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Road-Conditioned Diffusion for Traffic Trajectory Synthesis

Complete Journeys, Learned Route Preferences and GAF-Derived Generative Conditions

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Abstract

Synthesizing realistic traffic trajectory datasets requires modeling complete journeys that capture both feasible route choices and fine-grained, time-dependent vehicle dynamics. In this thesis, we propose a modular, road-conditioned diffusion framework to synthesize complete trajectories across five traffic participant classes on the Monaco SUMO Traffic (MoST) network. Our pipeline decomposes journey generation into four interpretable stages: (1) a learned behavioral policy that selects routes from a catalog of topology-valid alternatives; (2) a class-conditioned prior that samples travel duration; (3) a variable-length temporal diffusion model that generates normalized route progress at a native physical resolution of 0.25 s; and (4) a piecewise-linear road decoder that reconstructs timestamped planar coordinates. The framework is trained on an expanded corpus of 29,895 continuous vehicle trip legs derived from native activity-based SUMO simulations.

Beyond baseline spatiotemporal generation, we systematically investigate whether trajectory-image representations originally designed for participant classification can enhance conditional motion synthesis. Specifically, we integrate two complementary visual encodings derived from raw trajectories: a spatial Hilbert-curve encoding mapped to Gramian Angular Summation/Difference Fields and Markov Transition Fields (GASF/GADF/MTF), and a corresponding velocity-profile encoding. Pre-trained ResNet-18 classifiers on these spatial and velocity views achieve macro F1 scores of 76.47% and 90.34% on the development set, respectively. To leverage these representations for generation without requiring future trajectory observations at test time, we jointly adapt the CNN encoder tails alongside the temporal denoiser and train a conditional latent diffusion prior to sample representation codes given only route, duration, and class inputs.

We conduct a comprehensive evaluation across eight conditioning configurations, two random seeds, eight unseen test episodes, and thirty downstream data-augmentation training runs. Controlled ablation studies demonstrate that modeling motion on a native 0.25 s physical grid reduces acceleration Wasserstein distance (W_1) by 81–82% compared to a fixed-length relative-time grid, while our learned route preference policy reduces road-edge usage divergence by 13.6% over shortest-path selection. In the main conditioning comparison, however, jointly conditioning on both spatial and velocity GAF codes slightly degrades physical

acceleration fidelity, increasing W_1 distance by 2.44% relative to the class-, route-, and duration-conditioned diffusion baseline. Conversely, in downstream time-series classification tasks, augmenting observed data with trajectories conditioned solely on spatial GAF representations achieves the highest macro F1 of 96.95% (outperforming baseline diffusion augmentation at 96.26% and observed-only training at 94.41%), though this gain is primarily concentrated in one of the two evaluation origin–destination cohorts.

Overall, this thesis contributes an end-to-end road-conditioned trajectory synthesis pipeline and a rigorous empirical study of visual representation conditioning. Our findings highlight a critical divergence between discriminative separability, physical motion fidelity, and downstream data utility: while GAF encodings effectively capture participant-specific macro-structures, their benefits for trajectory synthesis depend heavily on temporal resolution constraints and the specific evaluation objective.

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Chapter 1

Introduction

Synthesizing realistic traffic trajectories requires modeling both spatial routing and temporal movement dynamics. In a real-world road network, multiple feasible paths can connect the same origin and destination (OD). Even along a fixed route, vehicles exhibit diverse temporal behaviors—cruising steadily, dwelling at transit stops, accelerating through intersections, or decelerating in response to traffic conditions. Consequently, generating a geometrically connected polyline solves only half of the trajectory synthesis problem. A practically useful generative model must simultaneously respect road-network topology and capture the distinct kinematic distributions and behavioral variations across different traffic participant classes.

The primary output of this work is a synthetic trajectory dataset in the standard `(time, id, x, y, type)` format, where each identifier corresponds to a complete, continuous journey from origin to destination. In our formulation, the requested participant class governs both road-access permissions and learned movement preferences. Rather than treating synthesis as deterministic point-to-point regression that attempts to recover a single ground-truth future trace, we formulate it as conditional distribution learning: repeated draws for the same request yield diverse, plausible journey realizations. This probabilistic perspective naturally separates our evaluation into distinct dimensions: road-network compliance, physical motion fidelity, within-request diversity, request coverage, and downstream training utility.

This study bridges two active research directions: conditional diffusion models for trajectory generation [4, 6, 7] and image-based time-series representations for traffic participant recognition [1, 2]. Combining these paradigms raises a central research question: *Can spatial and kinematic image representations learned for participant classification improve conditional traffic trajectory synthesis?*

1.1 A complete journey with spatial and temporal choices

Let c be a participant class, R a permitted directed route and $P_R(s)$ its coordinate interpolation at normalized progress $s \in [0,1]$. A synthesized journey is

$$\mathcal{T} = \{(t_i, P_R(s_i), c)\}_{i=0}^{n}, \qquad t_0 = 0, \quad t_n = T, \quad 0 = s_0 \le s_1 \le \cdots \le s_n = 1. \tag{1.1}$$

The route, duration T and intervening progress vary across samples. Repeated progress values represent stopped intervals. Different rates of progress encode movement changes. Sequence length follows the sampled duration, so a long service is not represented solely by a short observation window.

The implemented system combines a learned route-preference policy, a training-side duration prior, temporal motion diffusion and a deterministic road decoder. A request supplies the map, endpoints, class and declared permissions; buses additionally retain public stop anchors. The policy samples among supported road alternatives. Diffusion then generates motion on a quarter-second physical-time grid, and the decoder returns coordinates. Each component has a specific responsibility: road adherence comes from construction and decoding, duration comes from the prior, and the learned denoiser controls temporal progression. Experiments keep those responsibilities identifiable.

Training on observed route–motion pairs and participant labels allows the denoiser to learn a conditional distribution of complete journeys. At inference, independent noise draws yield alternative movement realizations for the same request, whose class consistency and kinematic quality are assessed through independent physical and downstream evaluations.

SUMO supplies training observations and held-out reference realizations on the public Monaco SUMO Traffic network [3, 47]. The framework focuses on individual complete journeys conditioned on network attributes and participant information, with the evaluation scope discussed in Chapter 8.

1.2 GAF representations for conditional generation

The spatial representation maps ordered coordinates through a Hilbert sequence into GASF, GADF and MTF image channels [10, 12]. A separate speed representation applies image transforms to a sequence derived from the same positions and timestamps. The spatial classifier receives coordinates, while the velocity classifier receives the derived speed sequence. Both classifiers are trained to predict the same participant classes, providing complementary views of participant-related movement.

The value of the image transforms lies in their representation of pairwise sequence relationships: image encoders can extract global movement patterns that

complement a categorical class embedding. Spatial quantization and run compression emphasize route-scale organization, whereas the velocity view emphasizes the shape and scale of whole-trip movement. Comparing the two views therefore reveals which aspects of participant-related structure transfer most effectively to motion synthesis.

Recognition and conditional generation emphasize different properties. A feature can distinguish buses from motorcycles by identifying familiar road corridors. Once a route and class are fixed, that feature may offer little information about a plausible within-route speed profile. A classifier can also assign an extreme class-associated behavior a high score without it being representative. The evaluation measures recognition, motion quality and downstream utility separately.

The final method makes the GAF representations part of the generation training objective. Small spatial and speed codes condition every denoising stage, and the final CNN stages can adapt through motion reconstruction alongside the participant-classification objective. A training-only conditional latent diffusion prior supplies codes for new requests. This gives the model a possible channel for trajectory-specific information while preserving an out-of-sample interface: it never receives an image of the future trajectory being evaluated. The general encoder–conditional-decoder construction follows established diffusion representation ideas [53]; the thesis tests its use with spatial and speed encodings, road conditions and complete physical-time motion.

1.3 Simulation data and physical-time representation

The experiments use the public MoST scenario with SUMO to construct complete journeys. Activity-based demand, vehicle parameters and public services preserve associations between participant class and travel behavior. A controlled comparison of demand configurations motivates this choice: class-permitted random routes retain motion differences but weaken the spatial patterns used for participant recognition. Chapter 3 describes the learning corpus and its construction.

Complete journeys also require a suitable temporal representation. A fixed 257-point sequence gives a long bus trip a much coarser physical-time grid than a short journey. The native-time model retains quarter-second observations, repeated positions and variable lengths. A matched temporal-resolution pilot assesses its consequences for generated motion and computation. The pilot evaluates the trained generator through movement statistics and computational cost.

The learning population contains 29,895 complete vehicle legs from 24,073 source vehicles, covering bus, motorcycle, private car, bicycle and taxi/ride-hailing. Public services, activity context and vehicle permissions are retained. New simulator realizations provide reference trajectories for evaluation, with source-vehicle partitions

and shared service templates specified in the experimental protocol.

1.4 Research questions and contributions

The thesis addresses two questions. First, can a road-conditioned diffusion system produce complete journeys with reasonable route preferences and motion distributions, and what is gained by preserving physical time? Second, do spatial and speed GAF representations improve the generated data beyond class, route, duration and appropriate representation controls?

The first contribution is an executable complete-journey generator with a permitted road interface, learned route preferences and variable-length temporal diffusion. Its matched controls separate structural road support, sampled duration and learned motion. The second is a controlled study of GAF's role in that generator: recognition, terminal preferences, frozen-feature supervision and generation-trained conditions are tested as different mechanisms. The experiment compares frozen, single-view, dual-view, raw-series, wrong-trip and class-centroid conditions throughout the denoising process.

The evidence is built around actual synthetic trajectories. Five-class physical measures, fixed-order journey and GAF figures, diversity, coverage and an independent downstream augmentation task accompany the model comparisons. The final experiment uses eight fresh reference episodes organized as four OD cohorts with paired windows and two model seeds. Physical quality and downstream utility establish the effect of each conditioning representation.

The following chapters place the approach in the literature, establish the data and representations, describe the generation and GAF modules, and define the comparisons before presenting their results. Earlier findings retain their original populations and uncertainty. The discussion relates these results to representation design and conditional motion synthesis.

Chapter 2

Background and related work

2.1 Trajectories as geometric and temporal objects

A sampled trajectory is an ordered sequence $X = ((t_i, p_i))_{i=0}^{T-1}$ with $p_i \in \mathbb{R}^2$. The polyline through its points defines geometry. The assignment of time to those points defines motion. Even on a straight road these properties can be separated. A vehicle covering one hundred metres in ten seconds can travel at constant speed or spend half the window waiting. Its endpoints and total displacement are identical in both cases, but its sequence of increments differs. A trajectory synthesizer therefore needs both a spatial support and a distribution over movement on that support.

A directed lane graph supplies information that a background map cannot convey by itself. For example, a bridge may lie directly above a second street in plan view. The distance between projected centerlines can be zero even though a vehicle cannot turn between them. At a junction, the permitted successor also depends on direction and participant class. A small distance to the nearest road is useful evidence about position. A connected assignment of road states throughout the sequence establishes a stronger property. Both depend on the accuracy and version of the map.

Microscopic simulation offers a practical test bed. The MoST scenario represents Monaco and its surroundings in SUMO, including multimodal roads and mobility demand [3]. Simulation exports can be sampled frequently and can carry participant types without manual annotation. They inherit the modelling choices of the simulator. A generated trace resembling the export can therefore reproduce a simulated mobility pattern even when its performance on traffic measured in the field is unknown.

2.2 Diffusion models for trajectories

A denoising diffusion model learns to recover structure from samples corrupted at known noise levels [4]. The representation of the clean sample is a design choice. When the representation is a sequence of positions, denoising acts on a coordinate plane. In the present work it acts on the temporal sequence of distances travelled along a sampled path. Either representation can produce coordinates as output. Their intermediate states and their routes to enforcing feasibility differ.

DiffTraj applies diffusion to the joint spatial and temporal structure of GPS trajectories, providing a probabilistic model for journeys with multiple plausible realizations [6]. Road-constrained approaches incorporate network structure more explicitly: Diff-RNTraj combines segment information with continuous moving ratios in a vectorized representation [7], allowing denoising to model movement relative to road segments. Together, these approaches establish complementary strategies for representing the spatial support of trajectory distributions.

RNTrajGen further separates road-level movement from the generation of finer trajectories [8]. This hierarchical perspective motivates the route–motion decomposition used here: the temporal-resolution ablation compares learned and empirical movement on identical permitted paths, while the route-preference evaluation studies choices among catalog alternatives. The main representation comparison then holds the route policy fixed to assess its effect on movement. Differences in cities, sampling rates and task definitions make these published systems methodological context for the controlled experiments.

Scenario Diffusion jointly generates agents and their trajectories under map and scenario conditioning [9]. Such scene-level models address the interaction and simultaneous occupancy of multiple participants. The present formulation instead samples individual complete journeys, emphasizing long-range route choice, travel duration and within-route motion. This choice determines the evaluation unit and the road, movement and data-utility measurements used throughout the thesis.

2.3 Imaging a time series

Wang and Oates introduced Gramian angular fields and Markov transition fields as image representations of time series [10]. In GASF and GADF, angular transformations turn scalar observations into pairwise sum and difference relationships, so each pixel links two sequence positions. MTF instead encodes transition probabilities between discretized states, exposing how the sequence moves among value ranges. These structured pairwise arrays allow convolutional filters to recognize recurring patterns across temporal scales.

For a traffic trajectory with two spatial coordinates, Hilbert indexing converts

each location into a scalar while preserving observation order. Traversing a space-filling curve encourages nearby sequence indices to correspond to neighboring spatial cells, allowing a CNN to model temporal relations between locations through image patches. This locality is approximate: neighboring cells can be distant along the traversal, and quantization merges positional differences within a cell. The representation therefore trades spatial precision for a compact ordering suitable for time-series imaging.

Salvadori investigated trajectory images for traffic participant recognition, including encoding and classifier choices [1]. Ferreira et al. studied image-based classification on simulated traffic [2]. These studies motivate spatial image representations as a source of participant-related features. This thesis examines their use in conditional generation, using a published implementation to specify Hilbert indexing and sequence compression [12]. The classifiers used in the experiments are trained on the study corpus. Recognition results are evaluated under the stated data partitions and sampling horizons.

2.4 Evaluation under multiple objectives

Trajectory quality combines geometric consistency, motion fidelity, diversity and data utility. These properties are related but can respond differently to a modeling change: strict route decoding can preserve spatial feasibility even for repetitive movement profiles, whereas greater variation can increase both useful alternatives and implausible behavior. Evaluating each dimension under a common route and candidate budget provides an interpretable account of the generated distribution.

For distributional evaluation we use the first Wasserstein distance. In one dimension this is the area between two empirical cumulative distribution functions. Its units are those of the measured quantity. A difference of twenty metres in the travel-distance comparison has a direct interpretation as a reduction in distribution discrepancy; it is not the error of a predicted endpoint for a particular participant. For downstream evaluation we use macro-F1 on held-out participants. The averaging treats a rare class as a separate classification task instead of letting the most frequent class dominate.

DDIM sampling provides a deterministic sequence of reverse updates conditional on the initial noise [5]. Reusing paths, request order and random seeds reduces incidental changes in a comparison. Each temporal representation uses noise arrays matched to its sequence dimensions. The temporal-resolution pilot reports two fixed model seeds on one new SUMO episode. The final GAF-conditioning comparison has four OD cohorts with paired windows and two model seeds. Its cohort intervals are descriptive; repeated windows, model seeds and candidate draws are not additional independent transport domains. Earlier resampling studies retain their own conditional scope.

2.5 Learned representations and conditional synthesis

Representation learning connects encoders to diffusion through both auxiliary supervision and explicit conditioning. REPA uses pretrained feature representations as training targets for diffusion models [50], motivating the transfer of structured information from a recognition model to a generative objective. For trajectory synthesis, this raises the question of whether participant-discriminative features also describe the variation in movement remaining after route, duration and class are specified.

Diffusion Autoencoders combine an encoder with a conditional diffusion decoder and learn a latent generative model for sampling [53]. This architecture supplies the basis for using observed representations during training and sampled representations at inference. The traffic adaptation in this thesis combines spatial and velocity images, partial CNN adaptation, a route/duration-conditioned latent prior and a physical-time trajectory decoder; representation sensitivity and sampled-code diagnostics characterize the resulting interface.

Route choice is a further component rather than an incidental coordinate error. Recursive models describe path preferences through local decisions and goal-dependent structure [54]. Here, a masked next-edge policy learned from observed branching decisions scores a finite set of permitted routes. Behavioral cloning supplies local preferences that are normalized over the permitted route catalog. This distinction makes both its learned preference and its coverage limitations explicit.

2.6 What a trajectory generator learns

A route model and a motion model face different ambiguities. TrajGAIL learns sequential vehicle decisions by adversarial imitation on a road network [15]. Such a policy can choose among candidate links, while the rate at which a vehicle travels along a link remains a separate quantity. Diff-RNTraj jointly represents segment identity and movement within segments [7]. The two-stage design of RNTrajGen likewise separates road sequences from fine movement [8]. These distinctions motivate the temporal pilot comparator here. Transferring an empirical speed sequence to the very same path gives the baseline every geometric advantage of the proposed model. Because both candidates complete the prescribed route, the relevant improvement is in motion distributions rather than in reaching a different travel distance.

Map-conditioned forecasting has developed representations for both geometry and interaction. VectorNet encodes polylines, while LaneGCN propagates information through lane connectivity [16, 17]. Trajectron++ models heterogeneous agents and feasible dynamics, and Scene Transformer jointly predicts agent futures [18, 19];

Social GAN captures multimodal pedestrian interactions [20], whereas Motion Transformer combines intention localization with movement refinement [21]. These methods typically exploit observed histories and surrounding traffic to forecast a scene. Complete-journey synthesis instead begins from an OD request and participant information, requiring a distribution over route choice, duration and movement for the entire trip.

2.7 Guiding a trajectory while it is being generated

In sequential decision problems, Diffuser treats planning as denoising a trajectory, while Diffusion Policy generates action sequences conditioned on observations [22, 23]. MotionDiffuser is a closer comparison for map-based movement: it represents joint agent futures in a compressed space and guides samples using differentiable costs [24]. The Hilbert transform used here quantizes coordinates and changes sequence length. A small change to an increment can leave every encoded pixel unchanged until a cell boundary is crossed. A coordinate-gradient perturbation therefore needs an additional surrogate or a different interface. The final learned-condition experiment preserves the discrete transform and differentiates CNN parameters during training, without requiring derivatives through Hilbert indexing.

Classifier guidance uses a classifier gradient during the reverse process, while classifier-free guidance combines conditional and unconditional model estimates [25, 26]. Universal Guidance extends the range of external guidance functions used at inference [27]. These methods make different assumptions about the information available at noisy states. In particular, a class score trained on clean trajectories has a physical interpretation only after a provisional motion sequence has been decoded. Inverse-problem samplers face a related distinction. DDRM uses structure in linear measurement models, whereas diffusion posterior sampling approximates a measurement term for more general observations [28, 29]. Participant classifiers provide a discriminative preference over trajectories, motivating separate evaluation of the motion distributions induced by their scores.

Twisted Diffusion Sampling uses weighted particles to approximate conditional distributions [14]. Feynman–Kac steering makes explicit how intermediate rewards, resampling times and potentials affect reward-guided generation [30]. A terminal scorer can choose only among trajectories already produced. An intermediate scorer can change which partial solutions receive subsequent computation. That opportunity also permits premature loss of viable solutions. If an early estimate of the final class preference is noisy, resampling can discard good trajectories before their motions become recognizable. At a very late update, a surviving state may have too little stochastic evolution left to move appreciably away from a terminal candidate. These are reasons to distinguish terminal feedback from in-sampler guidance. The earlier fixed-pool controls isolate terminal preferences. The final experiment learns

a condition used throughout denoising, without a classifier-gradient or resampling perturbation at a selected timestep. Results from these mechanisms are reported separately.

Sampling budget itself can change conclusions. Improved DDPM studies learned reverse variances and sampling efficiency [31]; EDM separates choices of noise schedule, preconditioning and solver [32]; DPM-Solver constructs efficient numerical updates for diffusion sampling [33]. These works motivate explicit cost accounting. Equalizing network calls controls one cost, but a denoiser call on a long physical-time sequence is more expensive than one on 257 bins. The temporal comparison matches optimization steps and candidates and reports measured runtime for each representation. The earlier GAF selection studies reuse the same eight candidates. The final direct-draw comparison uses four samples per request and records latent-prior and motion-denoiser work separately.

2.8 Representation, fidelity and utility

External representations can enter diffusion training through feature alignment. REPA aligns projected denoiser states with clean-image encoder features [50], while centered kernel alignment provides a way to compare relationships between learned representations [49]. In trajectory generation, these ideas motivate complementary measurements of representation structure and delivered motion quality: feature similarity characterizes what the encoder retains, and physical evaluation measures how that information affects synthesized trajectories.

A diffusion model over time does not uniquely specify a trajectory task. TimeGrad uses an autoregressive probabilistic model for multivariate forecasting, while CSDI conditions on observed entries for imputation [34, 35]. Those problems have observed context that a synthesis request may lack. DiffWave shows that an entire ordered waveform can itself be the generated object [36]. For road motion the temporal target used here is more structured: nonnegative increments give progressively ordered positions on a chosen path. A plausible marginal distribution of increments can still give an implausible sequence if their temporal ordering is wrong.

Recognition offers another perspective on that loss of information. Temporal convolutional models offer an alternative to recurrent sequence encoders [37, 38]. InceptionTime applies an ensemble of deep convolutional networks to time-series classification; ROCKET and MiniRocket show that convolutional transforms with simple classifiers can also be competitive [39, 40, 41]. Unsupervised multivariate sequence representations can learn reusable embeddings without training an image classifier [42]. These alternatives motivate evaluating class-related structure through several representation families. Raw geometry and motion classifiers serve as information diagnostics in the data study. Independent physical measurements provide the primary generation outcomes, alongside class-recognition diagnostics.

Classifier two-sample tests ask whether a learner can distinguish populations [43]. Distributional precision and recall instead separate sample quality from coverage [44, 45]. A selector can increase concentration in a recognizable part of the data distribution while decreasing the range of plausible behaviors available to the user. For that reason we compare variation across candidates, time near rest and steady movement in addition to speed and acceleration. Esteban and colleagues use synthetic time series to train predictors evaluated on real data [46]. Downstream augmentation asks a further practical question whose result need not track either classifier confidence or a single distribution distance. The final experiment measures this use directly with fixed downstream training budgets and separate reference cohorts. Its observed-only comparison must account for the larger labelled corpus used upstream to train the generator; a GAF-specific claim additionally needs the baseline and raw-representation controls.

Chapter 3

Data and complete-trip representation

This chapter describes the simulation scenario, the complete-journey learning population and the representations used by the generator. Native activity-based demand provides training and development trajectories. Separate SUMO runs provide reference observations for evaluation requests. The diffusion model produces the synthetic trajectories compared with those references.

3.1 Simulation scenario and learning population

The corpus is generated with SUMO on the public Monaco SUMO Traffic (MoST) scenario [3, 47]. Scenario revision `b29b2f65` provides the road network, activity-based demand, parking, background traffic, 24 bus services and 181 stops. Vehicle types retain their scenario-defined physical and behavioral parameters [51]. Participant labels and road-permission classes are recorded separately; in particular, taxi/ride-hailing labels can belong to vehicles with passenger permissions.

Each observation follows `time,id,x,y,type`, with time in seconds and coordinates in local planar metres. Speed and acceleration for evaluation are derived from positions and timestamps consistently for simulated and generated trajectories. A supplementary comparison of archived exports and demand configurations is provided in Appendix A.14.

The main corpus uses a ten-hour observation interval, simulation seconds 14,400–50,400, at a 0.25-second step. Two training realizations and one development realization retain the same public map and demand families. Each training source vehicle is assigned a realization before its motion is inspected, so a second seed is not counted as another copy of the same independent source vehicle. Known identities retain their earlier partition; original-test source vehicles are excluded by stored identity metadata without reading their frozen trajectories.

A vehicle's daily schedule is segmented into complete, continuous trip legs using departure, activity-stop boundaries and final arrival. For private vehicles, consecutive activities delimit separate legs; for buses, intermediate service stops remain within the journey until its terminal arrival. A source vehicle may therefore contribute several completed legs, each corresponding to the generator's output unit. The five-class synthesis task covers vehicle journeys, while pedestrian walking stages are excluded.

Table 3.1. Expanded native-demand corpus. Activity legs and distinct source vehicle identities are separate units.

Class	Train legs	Train source vehicles	Dev. legs
Bus	685	685	155
Motorcycle	6547	5226	1591
Private car	18381	13880	4436
Bicycle	2744	2744	668
Taxi/ride-hailing	1538	1538	380
Total	29,895	24,073	7,230

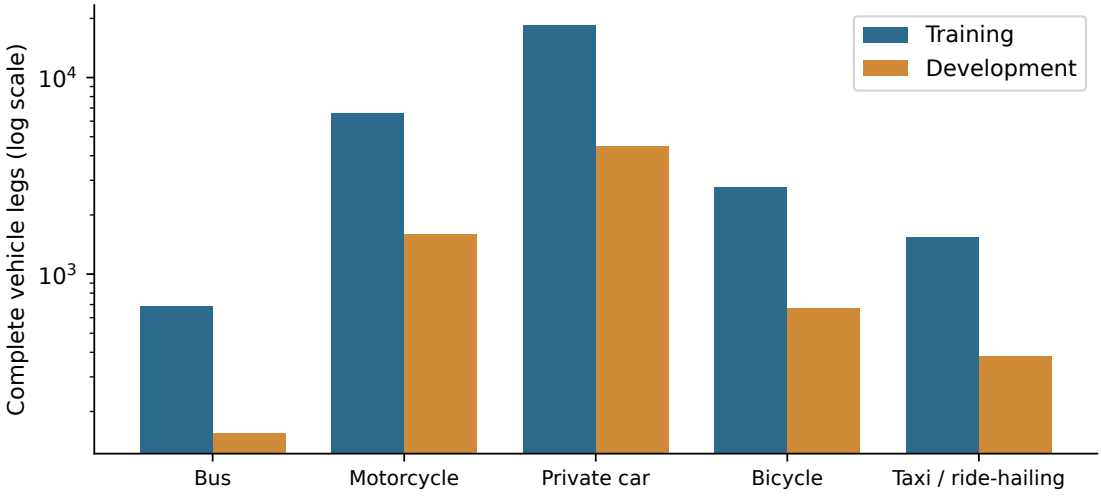

Figure 3.1. Expanded native-demand complete-leg counts. The logarithmic axis makes minority support visible while retaining the private-car majority. Legs, source vehicles and independently simulated traffic realizations are different observation units.

There are 29,895 training legs from 24,073 source vehicles and 7,230 development legs from 5,842 source vehicles. The recorded coordinate counts are approximately 65.1 million and 15.8 million. Median training duration is 549 seconds, the 95th

percentile is 1,152.75 seconds and the longest retained leg lasts 3,037.25 seconds. These counts describe the retained cohort, not a balanced set of independent scenes.

Continuity and event checks establish completed observations. Their numerical extent ends at the last recorded coordinate; no position is invented between that sample and a later event timestamp. Zero-displacement intervals and long trips remain in the data. Success of the out-of-sample map constructor is not a training admission requirement: an observed route can supply a valid supervised motion pair even if the separate map adapter cannot reconstruct every comparable request.

Short complete legs are retained through a constant mean-speed fallback when the image representation requires a longer speed sequence. In the expanded image cache, 2,454 training and 527 development examples use this subsecond fallback, ensuring exact alignment between the populations used by the image classifiers and the motion generator. Constant images, clipping and fallback frequencies are recorded as preprocessing statistics.

3.2 Decoupling Route Geometry, Progress, and Temporal Resolution

For observed positions $\mathbf{x}_i$, cumulative planar distance and normalized progress are

$$a_0 = 0, \qquad a_i = \sum_{j=1}^{i} \|\mathbf{x}_j - \mathbf{x}_{j-1}\|_2, \qquad s_i = a_i / a_n. \tag{3.1}$$

Repeated positions contribute zero progress while retaining elapsed time. All nonduplicate spatial vertices are stored in double precision for piecewise-linear coordinate reconstruction. A separate 129-point equal-arc summary conditions the neural network. The decoder retains the full-resolution polyline alongside this compact conditioning summary.

The native clock retains quarter-second samples and an exact fractional final interval when required at synthesis. The temporal-resolution pilot compares it with 257 equally spaced relative-time points. The same number of relative points gives a long trip a coarser physical resolution, which can remove movement details before learning begins. Reconstruction checks identify this representational limit; matched trained models test whether retaining finer detail actually improves generated motion.

The temporal pilot uses 925 training and 126 development vehicle legs, with two models for each clock. The expanded-corpus comparison is a separate study in which coverage and optimization exposure both increase.

3.3 Dual Trajectory-Image Representations

The spatial view starts only from ordered XY observations. Frozen coordinate bounds, the recursive Hilbert mapping and squeeze/mirror procedure produce the scalar sequence used for GASF, GADF and MTF. Hilbert indexing preserves observation order. The spatial classifier receives the coordinate-derived image. Position repetition and traversal can still affect its image through the sampling pattern.

The speed view derives one-second arc increments from the same XY/time observations and summarizes them at 128 relative sequence positions, with a fixed 35 m/s normalization scale. It predicts the same participant labels through a separate classifier. The two views are deterministic transformations of one observation, with different quantization and temporal losses. Their effects on synthesis are evaluated through the generative comparison. Chapter 5 specifies their exact construction and generative use.

3.4 Out-of-Sample Evaluation Episodes

The final request set contains eight fresh SUMO episodes organized into four non-bus OD-source vehicle cohorts and paired departure windows. Each manifest fixes 24 public bus services and 19 requests for each other class. Non-bus plans are complete single-leg requests without internal activity parking. Their source vehicles are disjoint from training and earlier evaluated non-bus cohorts. The public bus templates and map remain shared.

The generator sees endpoints, declared permissions and ordered public bus stop anchors. Its interior path is constructed from the network. The reference simulator uses its frozen public plans and background demand, while the generator samples from a network-only catalog. A different permitted route can be a valid outcome. Catalog membership of the reference route, geometric support and accepted physical references are separate coverage quantities.

All requested identities remain in the denominator even when simulation extraction, map construction or a particular metric lacks support. Coverage is evaluated against the original request set. New simulator realizations evaluate out-of-sample behavior on the shared road network. The detailed fit, sampling and uncertainty rules follow in Chapter 6.

Chapter 4

Road-conditioned synthesis of complete journeys

4.1 The conditional generation task

The output is a complete timed journey in the form `time,id,x,y,type`. A request supplies a road network M, an origin and destination, a participant class c, and the vehicle's declared road permissions. A scheduled bus additionally supplies its ordered public stop anchors. Intermediate reference coordinates, realised arrival time and future speeds are unavailable. The system samples a permitted route R, a duration T, and a motion realization along that route. A new random draw can change the route, the motion, or both.

Let $R(s)$ interpolate a constructed route at arc distance $s \in [0, L]$, and let $p(t)$ denote normalized progress. The complete system factors its sampling distribution as

$$\begin{aligned} R &\sim p_\rho(\,\cdot \mid M, \mathrm{OD}, c, \mathrm{stops}),\\ T &\sim p_{\mathrm{dur}}(\,\cdot \mid L, c),\\ h &\sim p_\eta(\,\cdot \mid R, T, c),\\ p &\sim p_\theta(\,\cdot \mid R, T, c, h), \qquad X_i = R\big(Lp(t_i)\big). \end{aligned} \tag{4.1}$$

The optional condition h represents the learned spatial and speed views described in Chapter 5; the baseline omits it. Its distribution is fitted using training trajectories and sampled for new requests. At synthesis, the sampled code conditions temporal diffusion directly, producing route progress for coordinate decoding.

The factorization defines a modular generative model in which route selection establishes spatial support, the duration prior determines the journey's temporal scale, and diffusion models its progress profile. Road-coordinate decoding combines these outputs into a complete timestamped trajectory. This decomposition

also supports controlled experiments: matching route and duration draws isolates changes in the generated motion profile.

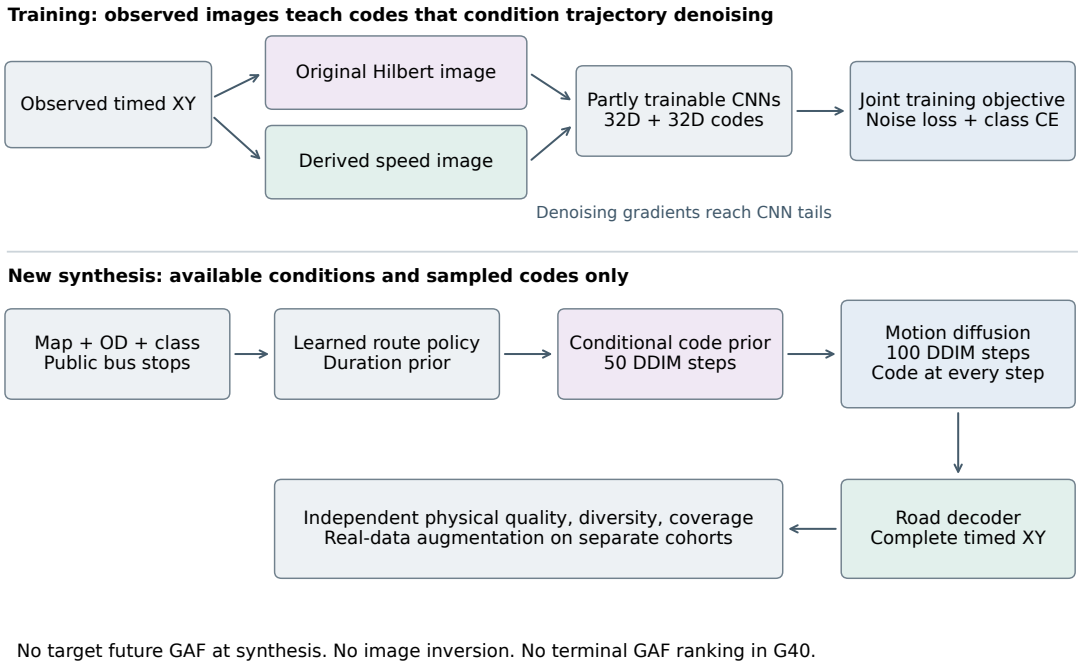

Figure 4.1. Implemented training and out-of-sample synthesis. Observed training trajectories produce the two image views and generative conditions; new synthesis samples conditions from a training-only prior. Route choice, duration, motion diffusion and coordinate decoding have distinct responsibilities. The learned condition enters every denoising block, coupling representation learning to complete-journey generation.

Training uses observed route–motion pairs, whereas test-time routes are constructed from the road network and request attributes. This distinction makes the learned temporal model usable for new journeys within the available road catalog [6, 7].

4.2 Learning route preferences within a permitted catalog

Road legality alone leaves a choice between several valid ways of travelling from an origin to a destination. The route module first constructs a finite catalog, then assigns learned probabilities to its members. Construction uses the directed

road graph and the declared SUMO vehicle permissions. Road access follows the declared vehicle permissions, including those assigned to taxi/ride-hailing vehicles. For buses, successive public stop anchors divide the route into ordered subproblems.

The constructor makes at most twelve deterministic proposals using shortest-path search with penalties that encourage alternatives. It retains at most four geometrically supported paths and discards candidates longer than 1.5 times the first valid route. Actual lane connections, internal junction geometry and endpoint positions are checked during construction. The resulting catalog defines the finite spatial support over which the learned policy assigns probabilities.

The policy is fitted by behavioural cloning of permitted branching decisions in training trips. At an edge e with next public goal g, it considers directed successors from which g is reachable. Each successor a has twenty static features: current and successor coordinates, goal coordinates and relative goal vectors, edge length, speed limit, lane count, turn and goal alignment, remaining shortest-path distances, excess path cost and a goal-arrival indicator. Their physical scales are fixed. These features describe network geometry and the current routing goal.

A multilayer perceptron with widths $20 \to 128 \to 64 \to 5$ produces a utility for each class. Selecting the requested class output and masking unsupported successors gives

$$\pi_\rho(a \mid e, g, c, M) = \frac{\exp u_{\rho,c}(f(e, a, g, M))}{\sum_{b \in \mathcal{A}(e,g,M)} \exp u_{\rho,c}(f(e, b, g, M))}. \tag{4.2}$$

Cross-entropy fitting samples a class, source vehicle, leg and branching decision uniformly at each level. The supported training subset contains 23,696 legs and 546,363 branching decisions; the corresponding development subset contains 5,748 legs and 132,341 decisions. Classification training uses transitions with multiple reachable successors. Two fixed neural fits are retained, alongside linear and shortest-cost controls in the route study.

For a catalog route r, the score is the sum of its branching-choice log probabilities, with the goal updated at ordered stop anchors. Normalizing these scores over the available catalog yields

$$p_\rho(r \mid M, \mathrm{OD}, c, \mathrm{stops}) = \frac{\exp\Big\{\sum_{j \in \mathcal{B}(r)} \log \pi_\rho(e_{j+1} \mid e_j, g_j, c, M)\Big\}}{\sum_{r' \in \mathcal{C}} \exp\Big\{\sum_{j \in \mathcal{B}(r')} \log \pi_\rho(e_{j+1} \mid e_j, g_j, c, M)\Big\}}. \tag{4.3}$$

The branching log probabilities are summed directly and normalized over the catalog, yielding a behavioral policy informed by recursive route-choice modeling [54]. An unsupported policy evaluation uses a recorded uniform fallback over the valid catalog.

Route quality is consequently assessed through route-use distributions, coverage and alternative-path diversity, together with the generated motion. Distributional

evaluation accommodates multiple permitted alternatives between the same endpoints. In the main conditioning comparison, the route policy and its random keys are fixed across all model variants. This pairing isolates changes in movement conditional on the selected route.

4.3 Representing motion on physical time

For a training trajectory, let $\ell_i = \|X_i - X_{i-1}\|_2$, $s_i = \sum_{j \le i} \ell_j$ and $p_i = s_i/L$, where $L = s_n > 0$. Repeated positions give repeated progress values and therefore retain stopped intervals. All nonduplicate spatial vertices are stored for exact piecewise-linear reconstruction; a separate set of 129 equally spaced arc positions provides the network's route description. The network thus receives a compact route summary while decoding retains the full polyline.

Two temporal representations are compared. The relative-time version uses 257 points uniformly distributed over $[0, T]$. The physical-clock version uses $0{,}0.25{,}0.50, \ldots, T$, appending the exact final time when it is not a multiple of 0.25 seconds. Its number of intervals varies with duration. The distinction matters because a fixed number of controls gives a long trip much coarser physical resolution than a short one. The temporal-resolution experiment measures how this representation choice affects generated movement.

With $\Delta t_i = t_i - t_{i-1}$, define the dimensionless relative speed and its transformed target as

$$r_i = \frac{p_i - p_{i-1}}{\Delta t_i / T}, \qquad u_i = \log(1 + r_i), \qquad z_{0,i} = \frac{u_i - \mu}{\sigma}. \tag{4.4}$$

Thus $r_i = 1$ is constant progress at the journey's average rate, and $r_i = 0$ represents a stop. The logarithm moderates the upper tail without removing zero motion. Training-only μ and σ assign equal mass to each trajectory and weight its intervals by $\Delta t_i/T$. Both clock versions use this target definition, with their own fitted normalization. Together, the two representations form a matched comparison on the current network.

4.4 Class-Conditioned Travel Duration Prior

Within each class, a ridge regression relates log travel time to centred log route length. For training journey j of class c,

$$\log T_j = a_c + b_c(\log L_j - \bar{\ell}_c) + e_j, \qquad T^* = \exp\{a_c + b_c(\log L - \bar{\ell}_c) + e^*\}, \tag{4.5}$$

where e^* is sampled uniformly from that class's training residuals. The slope has ridge penalty 10^{-3}; the intercept is unpenalized. Residual sampling retains variation beyond the conditional mean without imposing a Gaussian residual law. A

numerical lower bound gives $T = \max(T^*, 0.25, L/55)$, with metres and seconds as the physical units. This lower bound provides a numerical guard for duration sampling.

In the temporal-resolution pilot, both diffusion versions and the empirical comparator share the same prior and request-keyed draws. The main conditioning experiment instead holds the expanded training-side prior fixed across all eight model variants, with matched route and duration draws. Consequently their unselected duration distributions are paired by construction. The shared prior makes the motion mechanisms the source of differences in generated speed variation. Terminal feedback may select different durations from the shared pool; that is a selection effect and is reported as such.

4.5 A masked temporal denoiser

The base denoiser is a one-dimensional U-Net with channel widths 64, 96 and 128 and 891,969 parameters. The conditioning framework includes an additional 24,832-parameter condition adapter; the B variant bypasses it. The separately trained image encoders are accounted for in Chapter 5. Two downsampling stages, skip connections and nearest-neighbour upsampling connect the resolutions. Residual convolutional blocks use dilations 1 and 2, with dilations 1, 2, 4 and 8 in the bottleneck. Each block also receives a masked global feature summary, allowing information to pass beyond its local receptive field.

The conditioning vector adds a sinusoidal diffusion-step embedding, a learned class embedding, a convolutional route embedding and a five-component context embedding. The context contains normalized starting coordinates, log route length, log sampled duration and log mean travel speed. Route coordinates are expressed relative to the start and scaled by 1,500 metres. Three channels describe each interval's physical midpoint, relative midpoint and actual width. These allow the network to distinguish physical time from relative trip position. The denoiser learns stopping behavior from the motion targets, using route geometry and the sampled duration as context.

Sequences are padded to a multiple of four. The mask is reapplied through the network, pooling divides by the number of valid entries, and normalization acts across channels at each position rather than across padded time. The mask restricts computation and loss to the observed temporal support. Batches are sampled before length sorting; sorting and microbatching manage memory without removing long journeys.

Following noise-prediction diffusion [4], a uniformly sampled noise index k corrupts a target by

$$z_k = \sqrt{\bar{\alpha}_k} z_0 + \sqrt{1 - \bar{\alpha}_k}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{4.6}$$

on valid intervals. The 1,000-step schedule increases β_k linearly from 10^{-4} to 0.02, with $\bar{\alpha}_k = \prod_{j \le k}(1 - \beta_j)$. For an effective batch of B trajectories, the loss is

$$\mathcal{L} = \frac{1}{B} \sum_{b=1}^{B} \sum_{i} \frac{\Delta t_{b,i}}{T_b} \left[\epsilon_{b,i} - \epsilon_\theta(z_{b,k}, k, R_b, c_b, T_b, h_b)_i\right]^2 . \tag{4.7}$$

Padding has zero weight. Physical-time weighting within a trajectory prevents the fractional final interval from receiving a full interval's influence; averaging trajectories prevents long trips from dominating merely through their larger point counts.

4.6 Sampling and coordinate decoding

Generation uses 100 deterministic DDIM updates [5] between equally spaced indices of the training schedule. At a current index k, the predicted clean target and next state at $j < k$ are

$$\hat{z}_0 = (z_k - \sqrt{1 - \bar{\alpha}_k}\, \hat{\epsilon})/\sqrt{\bar{\alpha}_k}, \tag{4.8}$$

$$z_j = \sqrt{\bar{\alpha}_j}\, \hat{z}_0 + \sqrt{1 - \bar{\alpha}_j}\, \hat{\epsilon}. \tag{4.9}$$

The final update uses $\bar{\alpha}_j = 1$. Different initial Gaussian draws produce candidates; paired request and candidate identifiers reproduce duration draws across methods. Each temporal grid has its own noise-array dimensions.

After undoing standardization, the decoder clips u_i to [0,8] and constructs non-negative progress mass:

$$m_i = [\exp\{\mathrm{clip}(\mu + \sigma \hat{z}_{0,i}, 0, 8)\} - 1] \frac{\Delta t_i}{T}, \qquad \hat{p}_i = \frac{\sum_{j \le i} m_j}{\sum_j m_j}. \tag{4.10}$$

The initial progress is zero and the final progress is one. If total mass is below 10^{-12}, the decoder falls back to constant progress in physical time. Decoder diagnostics record negative-value truncation, upper clipping and constant-progress fallback. The normalization couples the whole journey: a local mass change can alter progress elsewhere. Endpoint completion follows from normalization, while the learned mass profile determines acceleration and stop timing.

Coordinates are interpolated along all retained vertices of the constructed map path. The motion model generates longitudinal progress along the selected route. For the relative-time model, progress is first interpolated onto the same quarter-second output clock; the exact fractional endpoint remains. Thus both methods deliver comparable timestamped coordinates. Permitted lanes, directed connections and source stop anchors are handled by the route constructor. This geometric constraint is shared with the empirical baseline and all GAF selectors.

The baseline generates complete journeys using class, route and duration alone. The GAF-conditioned variants retain the same target, physical-time grid, sampler

and decoder, while adding a sampled condition at every denoising step. Chapter 5 describes how that condition is learned and generated without future trajectory access. Earlier terminal-selection experiments remain separate controls on this synthesis system.

Chapter 5

GAF representations as generative conditions

5.1 Two representations of the same journey

Two image representations summarize each training journey. The spatial view uses ordered Hilbert coordinates and GAF channels following trajectory-recognition methods [1, 2, 12]. The speed view applies image transforms to the journey's derived speed sequence. Separate CNNs are trained on the same five participant classes; their encoders subsequently provide conditions for trajectory diffusion.

For a scalar sequence $a_1, \ldots, a_q$, let $u_i \in [0,1]$ be its normalized value and $\phi_i = \arccos u_i$. The two angular fields used here are

$$G_{ij}^{\text{sum}} = \cos(\phi_i + \phi_j), \qquad G_{ij}^{\text{diff}} = \sin(\phi_i - \phi_j). \tag{5.1}$$

A third channel is a Markov transition field. If b_i denotes a sequence value's bin and P_{ab} is the row-normalized count of adjacent transitions from bin a to bin b, then $H_{ij} = P_{b_i b_j}$. The angular construction follows time-series imaging [10]. For brevity, "GAF feedback" refers to the implemented three-channel GASF/GADF/MTF representation, including the transition channel.

5.2 Spatial Hilbert–GAF encoding

The spatial input consists only of the ordered coordinate samples. Frozen coordinate bounds map them into the unit square, with out-of-range values clipped and counted. The recursive Hilbert function assigns a scalar index to each point. Its precision parameter is 0.0009, corresponding to a 64×64 spatial grid. Hilbert indexing retains the temporal order of the observations.

The squeeze procedure compresses repeated-index runs, followed by mirror duplication when necessary, to obtain 224 values. This deterministic sequence compression prepares the input for image construction. If the procedure cannot reach the required length, the case is reported as an encoding failure. The scalar sequence is mapped to GASF, GADF and an eight-bin uniform MTF. The angular channels use per-sequence scaling to [0,1], while the frozen channel bounds convert the three fields to an unsigned eight-bit RGB image. Coordinate and channel parameters are estimated from training data and held fixed for evaluation and generated samples.

The spatial branch uses coordinate observations, whose sampling pattern also reflects movement. Different speeds change the number and order of sampled coordinates within quantized cells, and the run compression preserves some of that structure. Conversely, quantization and compression can map distinct motion profiles to the same image. On a fixed route, spatial feedback may therefore have little ability to rank the candidate pool even when it classifies trajectories from different routes well. Image hashes and the number of distinct spatial images in each pool make this limitation observable.

5.3 Speed encoding and its temporal limits

Speed is computed from the same coordinates and timestamps. Let $d(t)$ be the piecewise-linear interpolation of cumulative planar distance. At integer seconds, the speed summary is

$$v_j = d(j+1) - d(j), \qquad j = 0, \ldots, \lfloor T \rfloor - 1. \tag{5.2}$$

These one-second increments are expressed in metres per second and resampled to 128 equally spaced relative sequence positions. The complete-cohort encoder interpolates whatever complete one-second increments are available. For a valid leg shorter than one second, it repeats the mean arc speed over all 128 positions. This declared fallback retains 2,454 training and 527 development legs in the expanded cohort.

The angular normalization is $u_i = \mathrm{clip}(v_i/35{,}0{,}1)$, with a common absolute scale across classes. Values above 35 metres per second are counted. The MTF uses eight fixed bins on this normalized interval. The two angular channels are mapped from $[-1{,}1]$ to $[0{,}1]$; the resulting image is resized from 128×128 to 224×224. Both representations then use the same ImageNet channel normalization before entering a ResNet18 [11].

The speed image retains absolute speed scale but compresses temporal detail. One-second averaging and the 128-point summary emphasize whole-trip movement over quarter-second fluctuations. Its input consists of the resampled speed profile. The representation summarizes whole-trip speed structure.

5.4 From a generated journey to the two image views

Figure 5.1 makes the implemented representation concrete using an actual completed diffusion sample. The top row starts from its ordered Hilbert sequence; the lower row starts from speed computed from the same timestamped coordinates. Each row ends in the three channels that the corresponding class classifier receives. The image is computed from the generated journey.

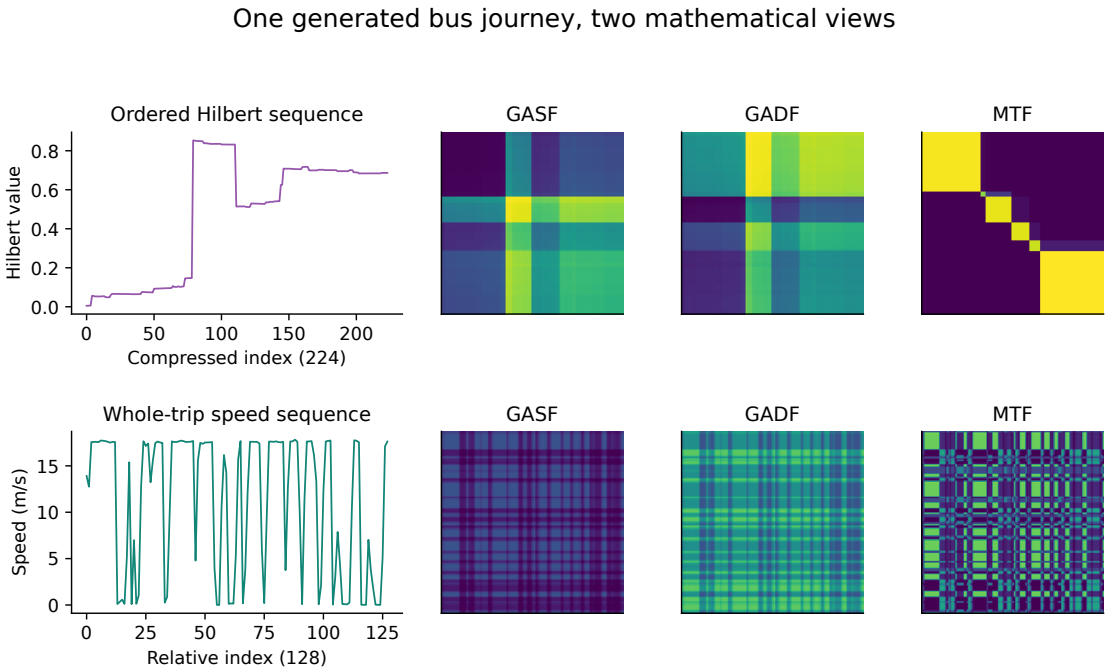

Figure 5.1. Exact representation channels of one preliminary candidate-selection bus journey, used to illustrate the unchanged encoding. The spatial row uses Hilbert run compression and frozen channel conversion; the speed row uses the 128-value absolute-speed summary and the implemented resize. Each field is shown with a common display range of [0,1] after its implemented conversion.

The two sequences have different coordinates on their horizontal axes. The spatial sequence index follows run compression, whereas the speed sequence index follows relative-time resampling. Their off-diagonal image patterns relate pairs of sequence positions. Visible image differences reveal which patterns each representation retains. The empirical question is whether the resulting representations provide useful constraints for generating motion. Chapter 7 links complete generated journeys and their output image views to independent population-level quality statistics. The output images visualize these measured trajectory properties.

5.5 Recognition and the limits of score-based feedback

The spatial and speed classifiers are separately trained on the same participant labels. Their different inputs remain separate: a speed sequence is never appended to the spatial classifier. Recognition measures participant-related information in recorded journeys. Its use for synthesis is evaluated through the conditioning experiments and independent motion metrics.

The earlier terminal experiments therefore compare matched candidates. Their scores are the floored log target-class probabilities,

$$s_v(X, c) = \log \max\{f_v(c \mid E_v(X)), 10^{-8}\}, \qquad v \in \{S, V\}. \tag{5.3}$$

Single-view policies use one score and the dual-view policy uses their equal-weight mean. The fixed candidate pool and common random keys prevent a larger sampling budget from being mistaken for a feedback advantage. The later route-and-duration control distinguishes global reweighting from reweighting that preserves those marginals. For that diagnostic, weighted distributions are the mechanism endpoint: an individual coupled random selection can differ even when the marginals are preserved exactly.

A second precursor fits the motion model against frozen classifier features. This tests the value of a fixed recognition representation as a motion-training target. The final experiment changes that interface: GAF-derived codes become sampled conditions of the generator, and the encoder's final stage is trained with the denoiser. The three interfaces are evaluated separately on their respective experimental populations.

5.6 Generation-trained spatial and speed conditions

Let $E_S(X)$ be the spatial image and $E_V(X)$ the speed image. For each active view, a CNN produces a pooled 512-dimensional feature $f_{\psi,v}(E_v(X))$. A separate bottleneck projects it through widths $512 \to 128 \to 32$, using SiLU and a final LayerNorm:

$$h_v = P_v\big(f_{\psi,v}(E_v(X))\big), \qquad h = [h_S; h_V] \in \mathbb{R}^{64}. \tag{5.4}$$

An inactive view contributes 32 zeros. The dimensionality of the denoiser interface consequently stays fixed across single- and dual-view conditions. Each view supplies a learned global condition.

In a joint variant, the final ResNet residual stage, class head and bottleneck are trainable. The earlier convolutional stages are frozen, as are all BatchNorm running statistics. Exact cached prefix activations reduce repeated image processing: their

pooled teacher features and cached spatial image bytes were checked against the existing training artifacts. The input encoding remains fixed.

The 64-dimensional code passes through a $64 \to 128 \to 128$ adapter and is added to the existing diffusion-step, class, route and context embedding. Every residual denoising block receives the resulting condition at every sampling step. The adapter's final layer is initialized to zero, reproducing the common starting denoiser exactly before adaptation. Adaptation therefore begins from the pretrained generator's output.

For active views $\mathcal{V}$, the joint fitting objective is

$$\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{noise}}(\theta, \psi, P) + \frac{0.1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \text{CE}(c, g_{\psi,v}(f_{\psi,v}(E_v(X)))) \,. \tag{5.5}$$

The first term is the physical-time-weighted noise-prediction loss in Equation 4.7. The class heads $g_{\psi,v}$ use the pooled CNN features, while the 32-dimensional bottlenecks supply the denoiser. The auxiliary term retains participant recognition as an explicit training task while denoising can adapt features that help reconstruct motion. The coefficient is fixed across the conditioning comparison.

Gradients update the CNN tail and projection weights. The discrete Hilbert, compression and byte-conversion operations remain fixed, so gradient updates act on encoder parameters and preserve the prescribed image construction. It also distinguishes the method from applying a classifier gradient to an intermediate noisy trajectory [25, 27].

Joint adaptation updates the encoder through both denoising and classification objectives. The frozen-classifier comparison measures their combined effect. Recognition scores reported for the original classifiers refer to their pre-adaptation checkpoints.

5.7 Conditional Latent Diffusion Prior for Test-Time Synthesis

Observed training trajectories supply the codes used in joint fitting. A conditional prior learns their distribution and supplies codes for out-of-sample synthesis.

After motion fitting, the final exponential-moving-average encoder is frozen and all training codes are extracted. Each active dimension is standardized by its training mean and standard deviation, with a standard-deviation floor of 10^{-3}. A conditional latent diffusion model then fits the distribution of these codes given class, the route's 129 two-dimensional points and the existing five-component route/duration context. It uses a 256-dimensional hidden state, four residual MLP blocks, the same 1,000-index linear noise schedule, and a fixed 5,000-update budget. The noise loss is evaluated over the active dimensions.

At synthesis, 50 DDIM updates sample a code from this prior; 100 separate DDIM updates generate motion conditional on that code. Route, duration and motion-noise keys are matched across the experimental model variants. The code prior and motion model use independent request-keyed random streams. The sampled code is held fixed through the motion denoising process. Inference uses the latent prior and temporal denoiser.

The encoder, conditional diffusion decoder and learned code prior follow the general construction of Diffusion Autoencoders [53]. Here the construction operates on road-conditioned variable-length motion and prescribed spatial and speed image views. The latent prior samples the condition, and the temporal decoder generates progress.

The code prior is checked through training-side perturbations and out-of-sample code-range statistics. Pilot settings and numerical diagnostics are reported in Appendix A.

5.8 Representation controls

Table 5.1 separates the main method from its controls. Every variant continues from the corresponding same-seed frozen-feature alignment baseline and uses the same motion target and sampling budget. A zero-initialized adapter and identical training streams make the starting point explicit.

The shuffled variant retains its assigned wrong-trip associations during code extraction, prior fitting and deployment. Its auxiliary labels remain unchanged. The shuffle tests instance pairing conditional on class and approximate trip scale, including their associated route and service correlations. The class-centroid variant learns projections of fixed training class means and uses a deterministic class-code lookup at sampling. The baseline and centroid model variants have no learned code prior. RAW provides an alternative representation control with its own encoder architecture and pretraining. Parameter counts and computation accompany its comparison.

Evaluation compares direct trajectory outputs under matched sampling budgets. Single-view and joint conditions measure the contribution of each representation; frozen, raw-series, shuffled and centroid conditions provide complementary controls. Physical distributions, diversity, coverage and downstream augmentation characterize the resulting synthetic data.

Table 5.1. conditioning configurations and their roles. B bypasses the code adapter. All other model variants use the same 64-dimensional interface; an inactive view is zero.

Variant	Condition source	Question isolated
B	No additional code	Class/route/duration diffusion baseline
F-SV	Frozen spatial and speed CNNs; learned projections	Is fixed recognition sufficient as a condition?
J-S	Joint spatial CNN tail and projection	What does the spatial view contribute?
J-V	Joint speed CNN tail and projection	What does the derived motion view contribute?
J-SV	Both jointly adapted views	Does combining views help beyond either alone?
J-RAW	Separate raw position and speed encoders	Is an image representation necessary for the observed gain?
J-SH	Same-class, approximately length/duration-matched wrong-trip images	Does trip-specific pairing matter?
C	Training class-mean CNN features	Is extra class information sufficient?

Chapter 6

Experimental protocol and independent evaluation

The main experiment evaluates complete trajectory synthesis under learned GAF-derived conditions. Temporal-resolution, route-choice and feature-alignment studies establish the modeling components, while a matched comparison measures the effects of representation choice and encoder adaptation under common training and sampling budgets.

6.1 Training Data and Preprocessing

All motion-model variants use the same expanded native-demand corpus: 29,895 training legs from 24,073 source vehicles, with the original physical-time grid and in-leg stops retained. The 7,230 development legs remain separate from fitting; final checkpoints and coefficients follow the common training protocol. Train-only quantities include coordinate and channel parameters, motion normalization, duration regression, class feature means and code-prior standardization.

The pre-trained spatial and velocity classifiers are self-trained ResNet18 models with ImageNet initialization, 50 epochs, batch 64, AdamW learning rate 2×10^{-4} and weight decay 10^{-4}. Their fixed final epoch, inverse-square-root class weights and unaugmented images match the recorded protocol. The RAW controls use the same labelled cohort, 50 epochs and final-checkpoint rule, but a separate one-dimensional architecture without ImageNet pretraining. The comparison matches labels and epochs, with architecture and computational costs reported for each representation.

During fitting, each motion target is paired with its observed route and duration. At test time, the generator receives the map-derived route catalog, participant class, road permissions and public bus stop anchors, then samples duration and latent conditions. Reconstruction and condition-perturbation diagnostics use

training examples and are evaluated separately from held-out trajectory synthesis.

6.2 Training Configuration and Matched Sampling

The eight model variants in Table 5.1 each have seeds 42 and 43. For a seed, all model variants start from the corresponding completed baseline B final EMA model from the frozen-feature alignment study, which was trained from initialization for 10,000 updates on the same motion corpus. The main conditioning experiment adds 5,000 motion updates to every variant. The 1,000-update pilots serve as implementation diagnostics; formal fits start from the common baseline.

Motion fitting uses effective batch 64, uniform class sampling followed by a training leg within class, FP32 arithmetic and physical-time-weighted noise loss. Selected examples are length-sorted into microbatches capped at 32 rows and 32,768 padded temporal values. AdamW uses motion learning rate 2×10^{-4}, encoder learning rate 5×10^{-5}, weight decay 10^{-4}, global gradient clipping at one and EMA update weight 0.001. Frozen prefixes and all BatchNorm running statistics remain unchanged. Every comparison uses final EMA weights. The final training step supplies each evaluated checkpoint.

Common initialization, valid-token totals, normalizers and final training random states are checked across the model variants for each seed. Code sampling uses a separate random stream. These checks verify the paired initialization and training inputs. Training and sampling cost, encoder parameters and peak allocated memory are reported separately from quality.

Each active code prior is fitted for 5,000 updates, with balanced batch 256, AdamW learning rate 2×10^{-4}, weight decay 10^{-4}, clipping at one and EMA weight 0.001. B uses no code; C uses a deterministic class lookup. The remaining six model variants require twelve learned priors across the two seeds. Code-prior fitting uses the training partition.

Each supported request receives four direct trajectory draws per variant and seed. Route, duration and initial motion-noise keys are common across model variants. Each learned-prior draw uses 50 latent DDIM updates and every motion draw uses 100 temporal DDIM updates. The first draw, K1, is the primary delivered-data comparison. Equal-weight distributions over all four draws and within-request diversity are secondary. The main comparison evaluates direct generated samples and records failed requests in its coverage results. Total computation includes both motion-denoiser calls and latent sampling for learned-prior variants.

6.3 Held-Out SUMO Evaluation Episodes

The final reference set comprises episodes 400201–400208 on the unchanged public MoST network. Four non-bus source-vehicle cohorts use selection seeds 40001–40004. Each cohort is simulated in two departure windows beginning at 18,000 and 32,400 seconds, with 600-second target departure spans and the frozen background-demand rules. A manifest contains 100 requests: the 24 public bus services and 19 source plans for each of the four other classes. All eight manifests are frozen before the first new outcome.

Non-bus source vehicles are disjoint from training and from the candidate-selection, route-preference and frozen-feature evaluation datasets. The two windows within a cohort deliberately reuse its underlying non-bus plans. Public bus service templates are shared across cohorts and with the learning domain. Four OD cohorts define the grouping for uncertainty descriptions; each has two departure windows.

SUMO exports are sampled at 0.25 seconds. Reference extraction applies its fixed continuity, arrival and service-event rules; catalog construction independently applies geometric support rules. Every scene retains the original 100-request denominator, and physical measures report the intersection appropriate to their support. Unsupported requests and rejected references are retained as coverage outcomes without substitute trajectories. Support intersections are computed during evaluation after generation. Exact reference-route membership characterizes catalog coverage alongside the distribution over permitted alternatives.

6.4 Physical quality beyond travel distance

The primary endpoints are native-resolution speed and acceleration distribution distances. An interval has speed $v_i = \|X_i - X_{i-1}\|/\Delta t_i$; successive speed differences are divided by $(\Delta t_i + \Delta t_{i-1})/2$ for acceleration. Coordinates and timestamps are processed identically for SUMO references and model output. The common coordinate-derived measurements make the physical comparison consistent across both sources.

Per-class distributions are compared using the first Wasserstein distance,

$$W_1(F, G) = \int_{-\infty}^{\infty} |F(x) - G(x)| \, \mathrm{d}x. \tag{6.1}$$

Requests have equal mass inside a class; candidates divide their request's mass; native observations are weighted by represented physical time. All five classes then receive equal weight. This weighting balances trajectory duration and class frequency. Each result includes its available class support. Distances are reported separately in their physical units.

The source FCD coordinates have centimetre-level reporting precision. Differentiation can amplify quantization on a quarter-second clock. Coarser-time movement summaries complement the native endpoints. Cumulative planar distance $d(t)$ is interpolated at integer seconds, with $v_j = d(j+1) - d(j)$ and $a_j = v_{j+1} - v_j$. Only complete one-second intervals contribute to these summaries; duration and planar distance retain the fractional endpoint. Speed and stopped-state summaries need one complete interval, acceleration and steady movement need two, and the four-second lag summary needs five. Each endpoint is evaluated over the journeys with sufficient temporal support, with its support count reported.

Table 6.1. Fixed per-journey motion summaries. Thresholds describe states rather than class-specific safety limits.

Summary	Definition
Duration; planar distance	Full timestamp span; sum of planar displacements
Mean; 95th-percentile speed	Mean and 0.95 quantile of one-second speed
Mean absolute acceleration	Mean of $\lvert a_j \rvert$ on the one-second clock
Stopped fraction; longest stop	Fraction and longest run with $v_j < 0.3$ m/s
Stop episodes per minute	Number of stopped runs per observed minute
Longest steady movement	Longest run with $v_{j+1} \geq 0.3$ m/s and $\lvert a_j \rvert < 0.5$ m/s^2
Four-second speed change	Mean of $\lvert v_{j+4} - v_j \rvert$

The shared route and duration draws control travel distance and temporal scale across the conditioning variants. Different movement distributions on those conditions remain possible. Distributional evaluation accommodates the several journeys that can satisfy the same request.

6.5 Coverage, diversity and statistical perspective

All classes, scenes, seeds and declared motion endpoints are retained. The primary grid has 128 model/scene cells and 640 class rows. Aggregate differences average classes, model seeds, windows and cohorts equally. A descriptive bootstrap resamples the four cohort means 2,000 times using a fixed seed. The intervals describe between-cohort variability within the shared simulator domain. The sixteen cell-level directions remain visible alongside the aggregates.

Request-level mean absolute discrepancies in physical summaries complement class-marginal distribution distances. The former preserve request correspondence, whereas the latter compare delivered populations. Together, these perspectives characterize agreement at the request and population levels.

Within-request diversity compares 129 relative-time coordinate positions and 257 relative-time progress values, as well as duration variation and effective route counts. Candidate-pair quantities are averaged within a request before averaging

requests. Diversity is interpreted jointly with physical quality to distinguish useful alternatives from implausible variability. A nearest same-class training-progress comparison measures similarity to a fixed library of observed movement profiles. Decoder clipping and constant-progress fallbacks are counted.

6.6 Independent synthetic-data augmentation

In the plots, "real" denotes observed SUMO journeys, not field-collected road traffic. The downstream task predicts the five participant classes from raw time series, using a network independent of the GAF classifiers and their weights. Its input has 257 relative-time positions with three channels: x, y and speed derived from coordinates. A 64-channel temporal convolution followed by four residual blocks with dilations 1, 2, 4 and 8 feeds concatenated mean/max pooling and a five-class head. The task measures downstream utility alongside the physical-quality endpoints.

The observed-data training subset contains 1,000 distinct source vehicles, balanced at 200 per class by a fixed identity hash. Training-only normalization comes from these rows. Synthetic sets contain 1,000 matched identities, also balanced by class, selected without quality scores from the first two OD cohorts and both generator seeds. B, F-SV, J-S, J-V, J-SV and J-RAW share the same selected request/draw identities. A resampling control bootstraps the same 1,000 observed rows; it is distinct from the earlier empirical motion-transfer generator.

There are fifteen regimes: observed-only; synthetic-only for six model model variants plus the resampling control; and observed-plus-source for those seven sources. Each has downstream seeds 4001 and 4002, giving thirty fits. All receive 2,000 AdamW updates, balanced batch 64, learning rate 2×10^{-4}, weight decay 10^{-4} and gradient clipping at one. A seed shares its initialization across regimes. The final checkpoint is fixed before loading evaluation trajectories.

The last two OD cohorts, each with both windows, supply the observed SUMO evaluation journeys. Non-bus source vehicles are disjoint from the synthetic-condition cohorts, and no evaluation source vehicle is in the 1,000 observed training vehicles. Shared bus templates remain a limitation. Pooled macro F1, class F1, episode results and vehicle-equal non-bus results are retained. Both downstream seeds are reported to show fitting variability.

The generator and its encoders used all 29,895 labelled training legs. The observed-only downstream learner uses 1,000 source vehicles. Results therefore reflect a larger upstream label budget for the generative models. B, RAW and resampling provide the corresponding augmentation controls.

6.7 Representation Sensitivity and Conditioning Diagnostics

Before formal generation, a fixed 500-training-vehicle audit checks original encoding reproduction and sensitivity to time warping, within-second perturbations and smoothing. It compares centered feature changes with the corresponding changes in physical movement. Four 1,000-update motion pilots check implementation, code use and prior behaviour. Their failures are retained; the single prior recovery uses the already declared 5,000-update formal prior budget and does not refit motion or inspect new reference quality.

Before any formal generation outcomes, a final-model condition audit is also fixed. It uses 25 training vehicles, five per class, at noise indices 50, 250, 500, 750 and 950. Assigned code, a same-class other-vehicle code, zero code and sampled-prior code are compared under identical route, duration and noisy motion input. All model variants and seeds are included. Zeroing a code retains the learned adapter and its biases. The noise indices sample different stages of the denoising process for analysis. The audit measures condition use and prior mismatch on training vehicles.

A further read-only audit uses the recorded code magnitudes and decoder counters from every completed out-of-sample draw. It reports fixed quantiles and exceedance of the observed training-code maximum. Exceedance of the empirical maximum describes code-range differences between training and generation. The recorded draws provide the data for this diagnostic.

The temporal-resolution pilot, source-demand study, route-choice study and earlier GAF controls retain their original populations and budgets. Each study contributes evidence about its own modeling component and evaluation population. Fixed example selection uses episode 400201, model seed 42 and the first lexicographic request per class supported by the reference and every variant. Candidate zero supplies the displayed sample. Both image views are recomputed from the fixed-rule outputs using the common image conversion.

Chapter 7

Results: synthesis, representation and data utility

This chapter evaluates complete-journey synthesis through route preferences, physical-time motion and downstream data utility. The temporal and routing controls establish the generation framework. The main comparison then measures the effect of spatial, speed and joint GAF conditions under matched route, duration and sampling budgets. Five-class numerical results are paired with complete generated trajectories and their image representations.

7.1 Evaluation population

The main comparison comprises eight model conditions and two fitted seeds, evaluated over eight out-of-sample SUMO episodes. It produces 50,176 candidates, or 6,272 per condition. The primary endpoint uses the first direct draw. Thirty downstream fits evaluate the utility of the generated data. Artifact verification and scalar recomputation are documented in the accompanying reproducibility records.

The 800 declared episode requests yield 784 supported generated requests and 755 physical/reference-route pairs. The realised reference route occurs in the finite catalog for 476 of those pairs. Two windows reuse each OD cohort and public bus services recur across cohorts. Each episode has 98 generated requests, while accepted physical pairs range from 90 to 97. The downstream task has 384 observed references under its own extraction support. Complete coverage is retained in Appendix A.

The temporal and routing controls use their original matched populations and budgets. Their results are reported separately from the main GAF-conditioning comparison.

7.2 Impact of Physical-Time Resolution on Motion Fidelity

The temporal-resolution pilot study compares a 257-point relative-time representation, a native 0.25-second representation and empirical transfer of a training motion profile onto a generated path. It uses 925 training legs, two fitted seeds and one new SUMO reference episode. Of 100 declared requests, map construction and reference extraction each support 97, with 94 in their intersection. Every full pool contains eight draws per supported request. The pilot isolates the temporal representation at a common update budget; runtime differs between clocks.

Native acceleration W_1 falls from 0.8397/0.8319 to 0.1492/0.1575 m/s^2 across seeds 42/43, reductions of 82.2/81.1%. Empirical transfer gives 0.3245/0.3588 m/s^2. The native model improves that endpoint in all five classes for both seeds. Native-speed distance also falls in the macro, from 0.6794/0.8238 to 0.6093/0.7437 m/s, but not in every class. One-second mean-absolute-acceleration distance decreases from approximately 0.113 to 0.0661/0.0560 m/s^2, providing a related observation at a coarser clock.

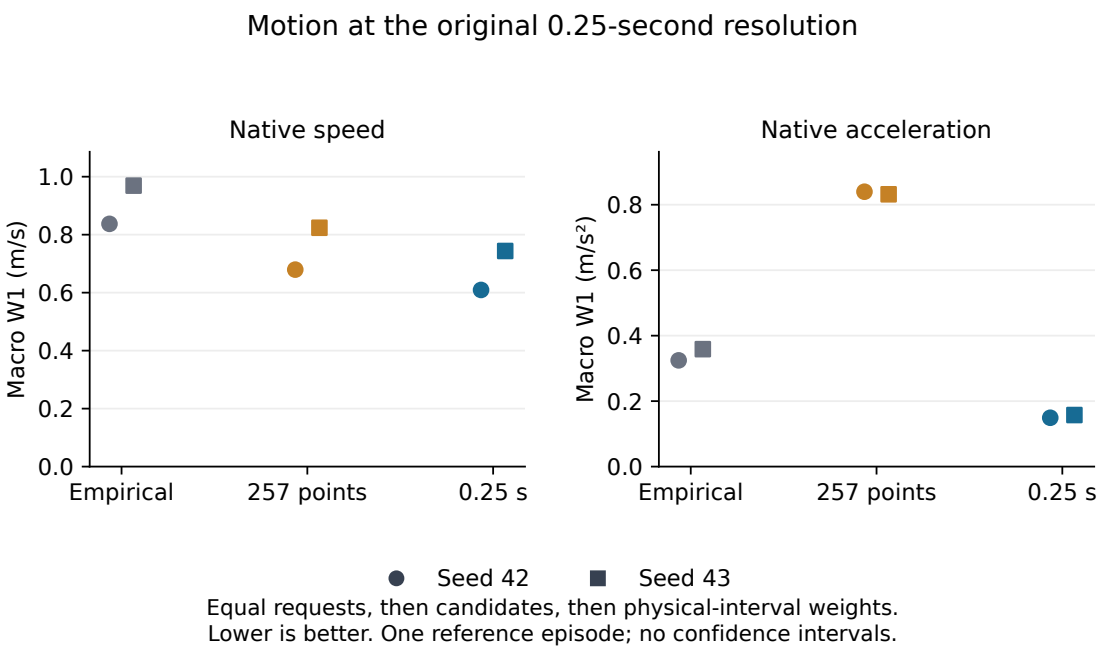

Figure 7.1. Matched temporal-resolution pilot: native-interval distributions for two diffusion clocks and empirical motion transfer. Both model seeds use the same request population and corresponding shared duration draws. The pilot uses one reference episode and motivates the clock used subsequently.

Longer-timescale structure remains difficult. Longest-stop distance worsens from

4.69/4.89 to 7.75/7.66 seconds; for buses it rises from 12.03/11.76 to 26.37/25.66 seconds. Acceleration and stop organization exhibit different responses to the temporal representation. Native fitting takes 1,520/1,529 seconds compared with 301/319 seconds for the relative-time fits, at the same 10,000 updates. The complete physical and class tables, including adverse outcomes, remain in Appendix A.

7.3 Evaluation of Learned Route Preferences

The route-preference experiment uses 23,696 accepted training legs and 5,748 development legs for next-edge learning. The class-equal development next-choice accuracy is approximately 96.8% for the two neural policies, compared with 94.9% for the shortest-cost rule. This accuracy measures observed choices at road-branching decisions.

Two out-of-sample SUMO episodes and two policy seeds compare distributions over a finite, geometrically constructed route catalog. Both episodes have 100 declared requests and 98 supported map conditions, with 97 and 92 physical/reference-route pairs. The exact realised SUMO route is in the catalog for 64 and 61 of those pairs. These coverage counts accompany the distributional evaluation of alternative permitted routes.

Averaged over the four cells, learned preference R reduces road-edge usage Jensen–Shannon divergence from 0.04000 for shortest-route choice to 0.03458, a 13.6% reduction, with the same direction in all four cells. Effective route count increases from 1.000 to 1.176. Weighted native-speed and acceleration distances decrease from 1.20327/0.17435 to 1.16728/0.17192 in their respective units. These measurements characterize route choice on the shared public network.

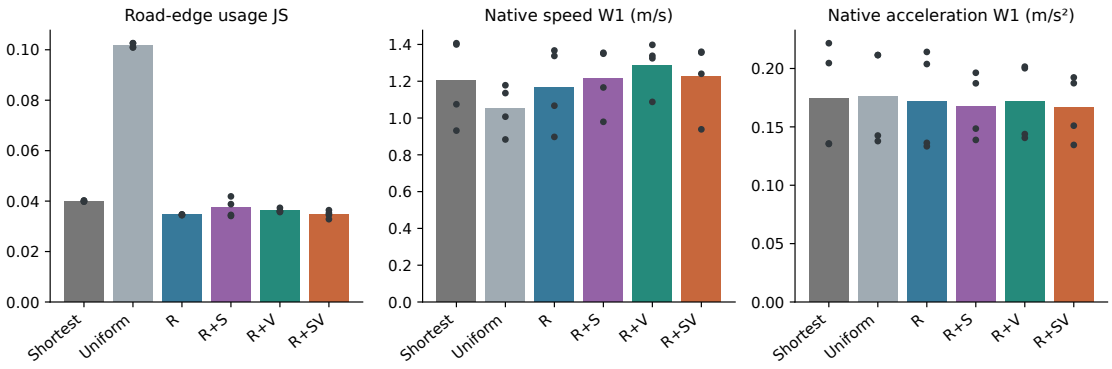

Figure 7.2. Learned route preference and route/motion distribution distances in the route-preference evaluation. Values summarize two episodes and two seeds; the underlying class and cell results remain archived. Lower edge-use divergence is closer to the observed distribution. The plotted points show each scene/seed cell.

Uniform route sampling retains a relevant tradeoff: its speed distance is lower at 1.05113 m/s and its effective route count is 3.454, but edge-use divergence is higher at 0.10177. The neural route policy balances empirical route preference against alternative-path diversity. The main conditioning variants use common R draws so that GAF effects can be attributed to conditional motion changes rather than different route selection.

7.4 Participant Classification Performance of Spatial and Velocity Views

The pre-trained GAF classifiers use the same 29,895 training legs and 7,230 development legs as the expanded motion corpus. Spatial macro F1 is 76.47% and speed macro F1 is 90.34%. Accuracies are 86.17% and 89.13%, respectively. Table 7.1 exposes class imbalance and the spatial classifier's taxi/ride-hailing weakness; overall accuracy alone would obscure that result. These scores characterize the classifiers used to initialize joint adaptation.

Table 7.1. Pre-adaptation pre-trained GAF classifiers on the same development population. F1 values are percentages. Support counts are legs, not independent simulator scenes.

Class	Development legs	Spatial F1	Velocity F1
Bus	155	95.65	100.00
Motorcycle	1591	73.57	78.90
Private car	4436	95.20	92.23
Bicycle	668	72.10	99.48
Taxi / ride-hailing	380	45.85	81.07
Macro F1	7,230	76.47	90.34

Spatial images classify private cars well but do not reliably separate the service-related taxi label. Speed images retain useful information about the same labels, without being added to the spatial classifier's input. The generative comparison evaluates how these class-related features affect motion synthesis. The source-demand comparison in Appendix A.14 and the matched earlier/current-classifier comparisons in the appendix provide the data context.

7.5 Final generative conditions: native motion quality

Table 7.2 gives absolute primary distances and their paired contrasts against B. The dual-view J-SV condition changes native speed distance from 1.437941 to

1.440104 m/s and acceleration distance from 0.176001 to 0.180289 m/s^2: increases of 0.150% and 2.437%, respectively. The acceleration increase occurs in all four cohort means. Only four of the sixteen scene/seed cells improve that endpoint. Equal weighting over all four generated draws also leaves both dual-view distances above baseline.

Table 7.2. Out-of-sample direct-draw quality on eight episodes, two model seeds and all five classes. Wins count scene/seed means; lower distances are better.

Variant	Speed W_1	Acceleration W_1	Speed wins	Accel. wins
B	1.43794	0.17600	–	–
F-SV	1.43543	0.17611	8/16	10/16
J-S	1.43909	0.17694	6/16	7/16
J-V	1.43740	0.17921	6/16	7/16
J-SV	1.44010	0.18029	8/16	4/16
J-RAW	1.44368	0.18068	7/16	4/16
J-SH	1.43449	0.17450	10/16	11/16
C	1.43850	0.17578	10/16	8/16

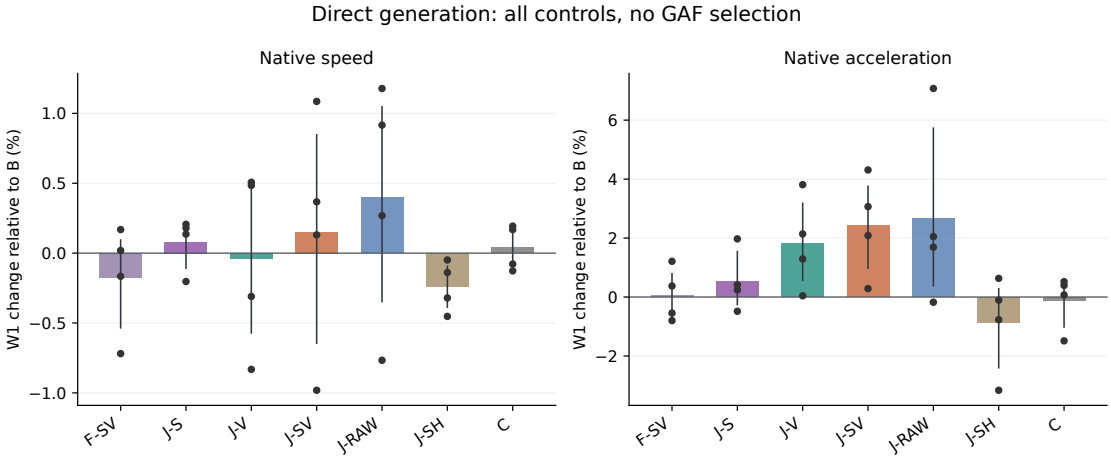

Figure 7.3. First-draw native motion distances relative to B. Bars are mean differences with equal weighting across classes, seeds, windows and test groups divided by the overall B mean; points are the four cohort differences with the same denominator. Whiskers are descriptive cohort-bootstrap intervals. Lower is better. Intervals summarize variability across the four same-network cohorts.

Spatial J-S alone changes speed by +0.080% and acceleration by +0.536%. Speed-only J-V changes them by −0.037% and +1.821%. Frozen F-SV has a small speed reduction and essentially unchanged acceleration. RAW also increases both primary distances. Across these interfaces, the differences are small and mostly

adverse.

Same-class wrong-trip J-SH reduces speed and acceleration distances by 0.240% and 0.851%, respectively; its descriptive acceleration interval spans zero. Class-centroid C remains close to B. Correctly paired joint conditions therefore offer no consistent advantage over these simpler controls. J-SH preserves class and approximate trip scale, including associated route and service structure.

7.5.1 All five classes and longer-timescale motion

Figure 7.4 shows the class-specific responses. J-SV reduces private-car acceleration distance by 3.28%, while increasing it for buses, motorcycles, bicycles and taxi/ride-hailing by 1.60%, 1.97%, 5.68% and 5.26%. Spatial-only J-S has its largest acceleration increase for private cars; speed-only J-V increases bicycle and taxi distances. The effects depend on both representation and participant class.

The longer-timescale summaries show a similar tradeoff. J-SV reduces longest-steady-motion distance by 2.26% and stop-episode-frequency distance by 1.56%, while increasing longest-stop distance by 1.52%, one-second mean-absolute-acceleration distance by 3.32%, and four-second speed-change distance by 1.67%. The differences extend across native and coarser physical clocks.

Duration is identical across model variants by the paired sampling design. Its shared distance of 178.63 seconds is therefore a limitation of the common duration component, not a GAF effect. Generated planar-distance discrepancies are also almost unchanged; small differences arise because finite-time planar chords need not sum to the exact constructed route arc length. The appendix provides all ten physical summaries, request-paired discrepancies and four-draw results.

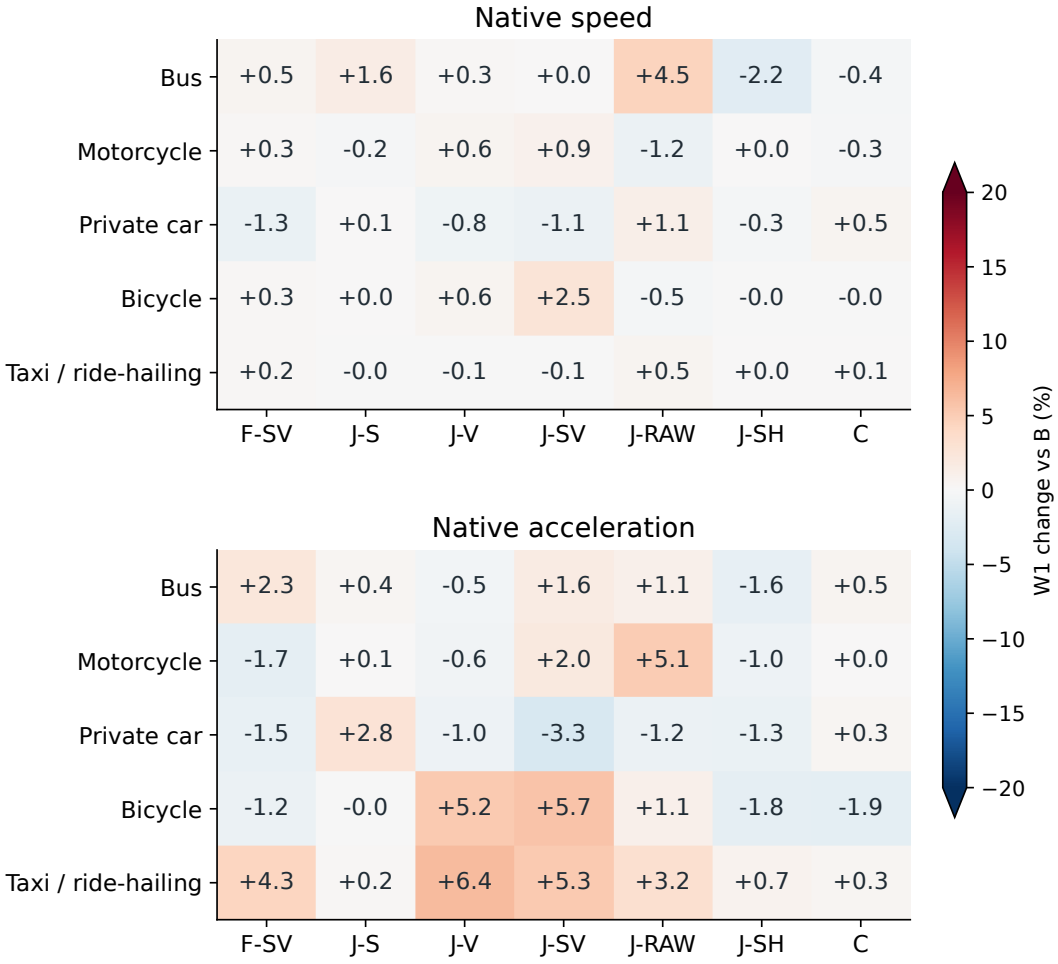

Figure 7.4. All five classes and seven condition contrasts against B: native speed and acceleration. Values are percentage changes in each class's distance, averaged over both model seeds and all eight episodes. Blue is lower distance. The fixed color range is shared; printed values are not clipped. Absolute values and both seeds are in the appendix.

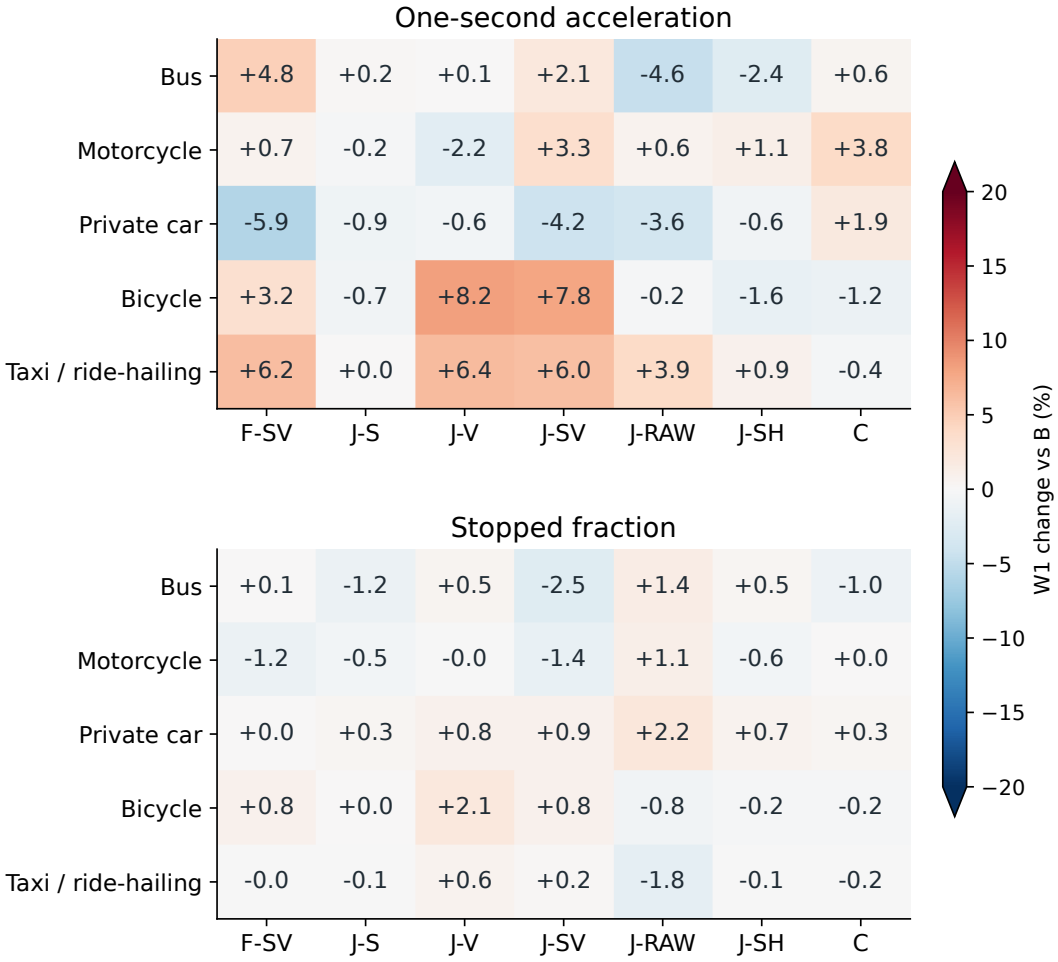

Figure 7.5. All-class one-second mean absolute acceleration and stopped-fraction distribution distances. These coarser-clock summaries complement the native endpoints. Per-endpoint support is retained in the evidence files.

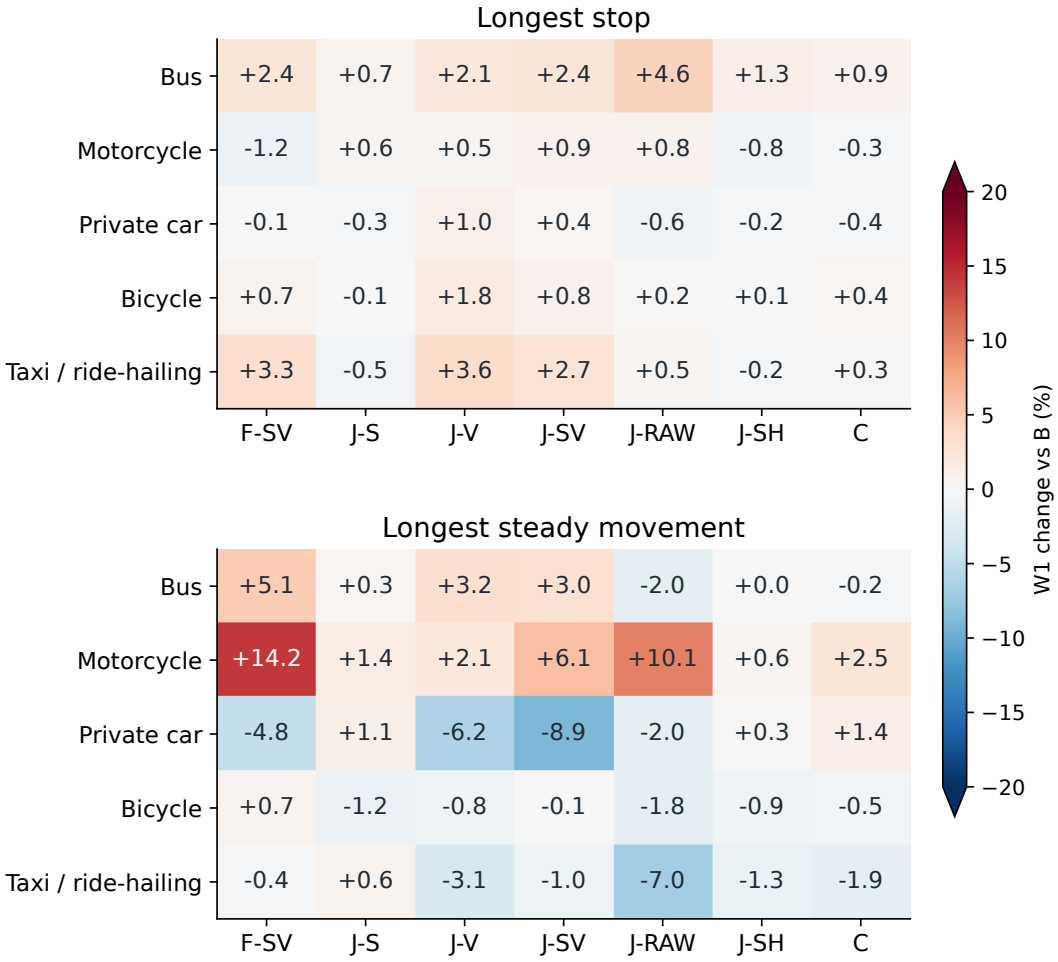

Figure 7.6. All-class longest-stop and longest-steady-movement distances. Different aspects of stop organization can change in opposite directions. These panels retain the same seven contrasts and five classes as the primary comparison.

7.6 Complete generated trajectories and their GAF views

The next ten figures show one fixed-order case for every class. The rule is episode 400201, model seed 42, first lexicographic jointly supported request per class, candidate zero. It was fixed before rendering. The displayed subset comprises the reference and B, J-S, J-V, J-SV and J-RAW; all eight model variants remain in the numerical comparison. Maps show the reference, B and J-SV, while the time curves show all displayed model model variants.

Map colors encode coordinate-derived one-second speed in metres per second, with a common scale across the reference and model variants in each case. The one-second speeds are interpolated onto the trajectory timestamps for spatial display; a circle marks the start and a cross the end. Curve comparisons use the whole elapsed-time axis. Their acceleration distributions use complete one-second increments for readability and do not replace the quarter-second primary metric. The examples compare timing and speed structure alongside the population statistics; individual stop sequences vary across stochastic realizations.

Each companion image figure uses GASF/GADF/MTF channels computed from those same output coordinates and timestamps. The rows are observed SUMO, B, J-S, J-V, J-SV and J-RAW. These output representations accompany the physical-time comparisons.

7.6.1 Bus: shared service geometry and variable dwell organization

The bus reference lasts 2,328 seconds, compared with 2,560.27 seconds for the generated journeys. The shared transit corridor produces closely related spatial images, while the speed curves reveal how the models distribute moving and stationary periods over the longer sampled duration. Although the generated progress curves largely overlap, the image panels retain visible differences in whole-trip movement organization. This distinction helps explain the class-level results: recognizable bus geometry can coexist with differences in acceleration and dwell timing, making the physical-time profiles essential to interpreting the spatial encoding.

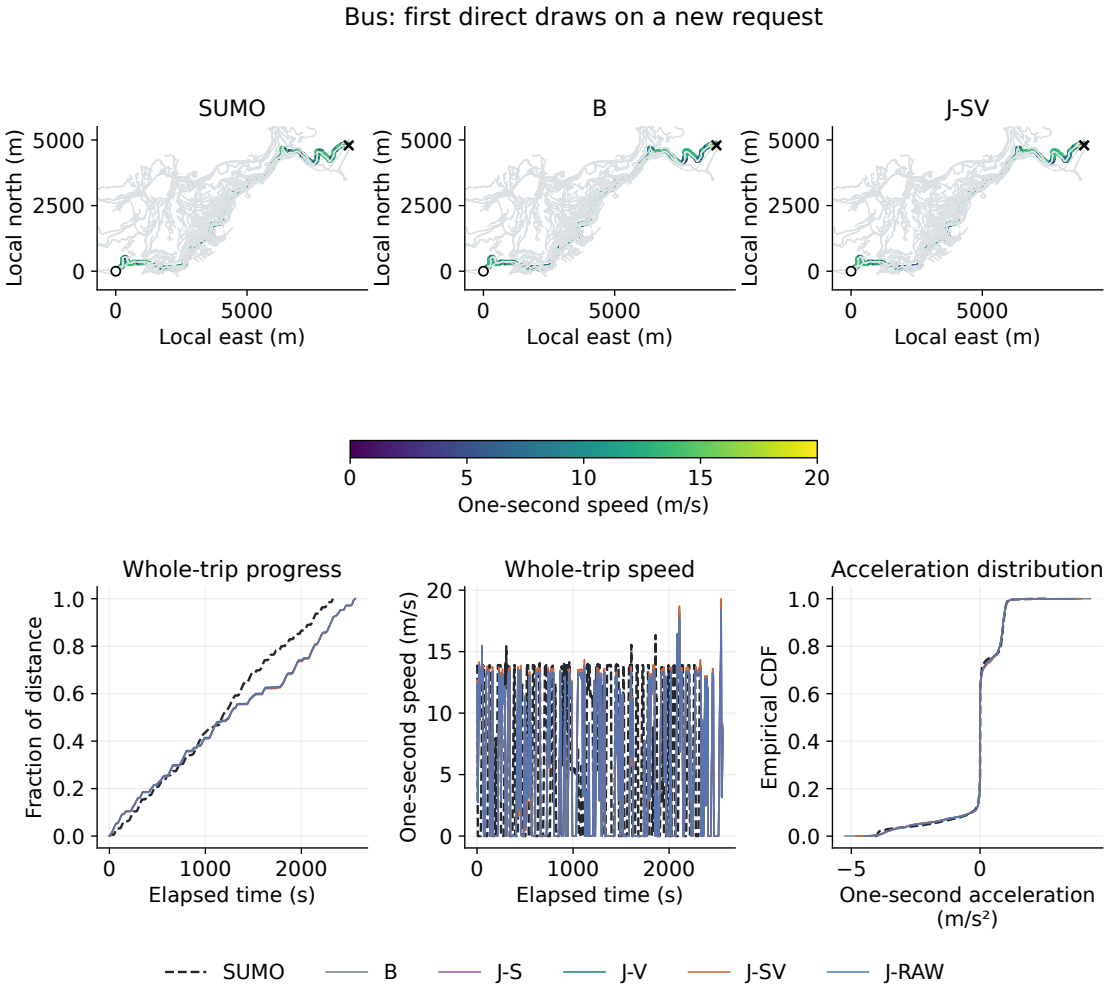

Figure 7.7. Fixed bus case: complete road geometry, cumulative progress, one-second speed and acceleration distribution. Route adherence belongs to the shared decoder; the condition variants are compared under common sampled route and duration.

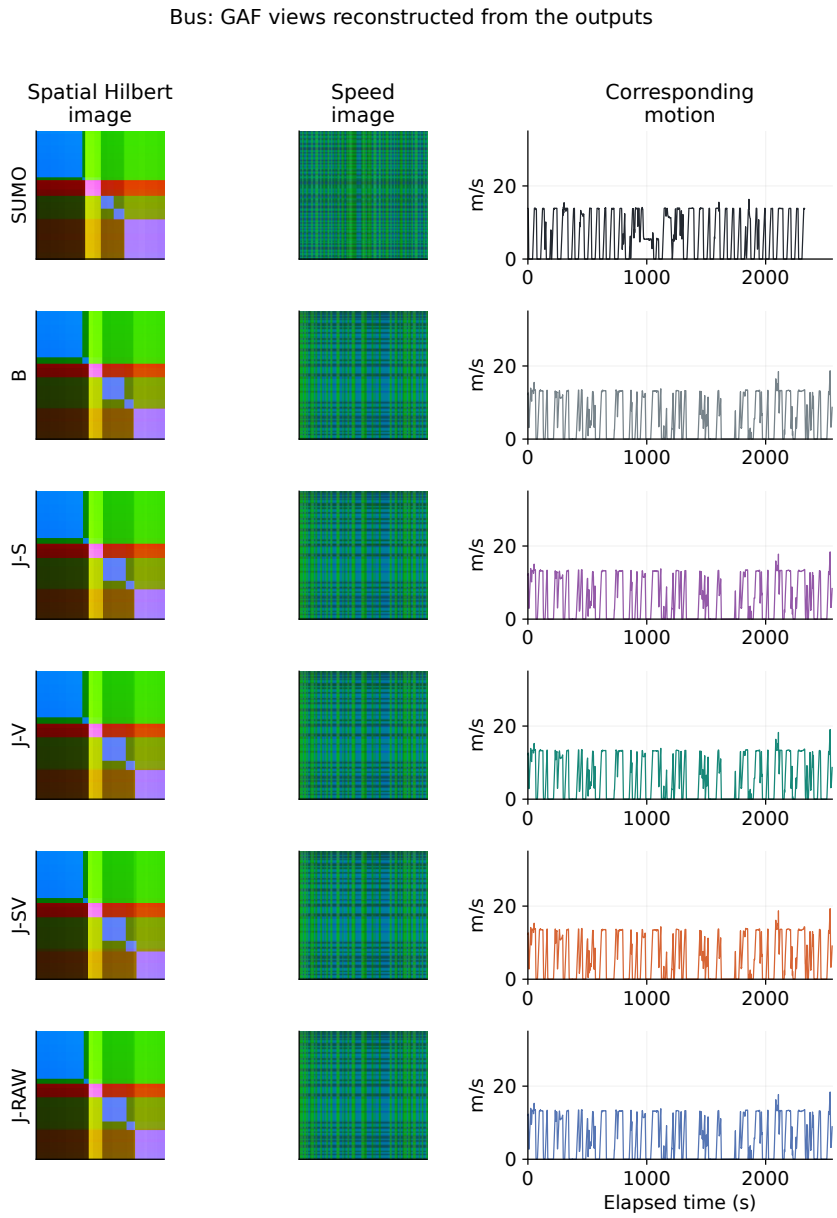

Figure 7.8. Spatial and speed image views reconstructed from the fixed bus reference and generated outputs. The corresponding full-trip speed traces expose timing differences that can be difficult to infer from the spatial image alone.

7.6.2 Motorcycle: route alternatives and the time axis

The motorcycle reference lasts 266.75 seconds, whereas the generated journeys last 355.31 seconds and follow a different permitted route. Reading the map and time curves together distinguishes this route-choice variation from changes in movement along the generated path. The spatial images encode the alternative geometry, while the velocity images summarize the resulting speed pattern over the complete journey. Across conditioning variants, the generated curves remain similar, consistent with the small aggregate changes in motion quality. Thus the example connects two sources of variation that are evaluated separately: selection among feasible routes and temporal behavior conditional on the selected route.

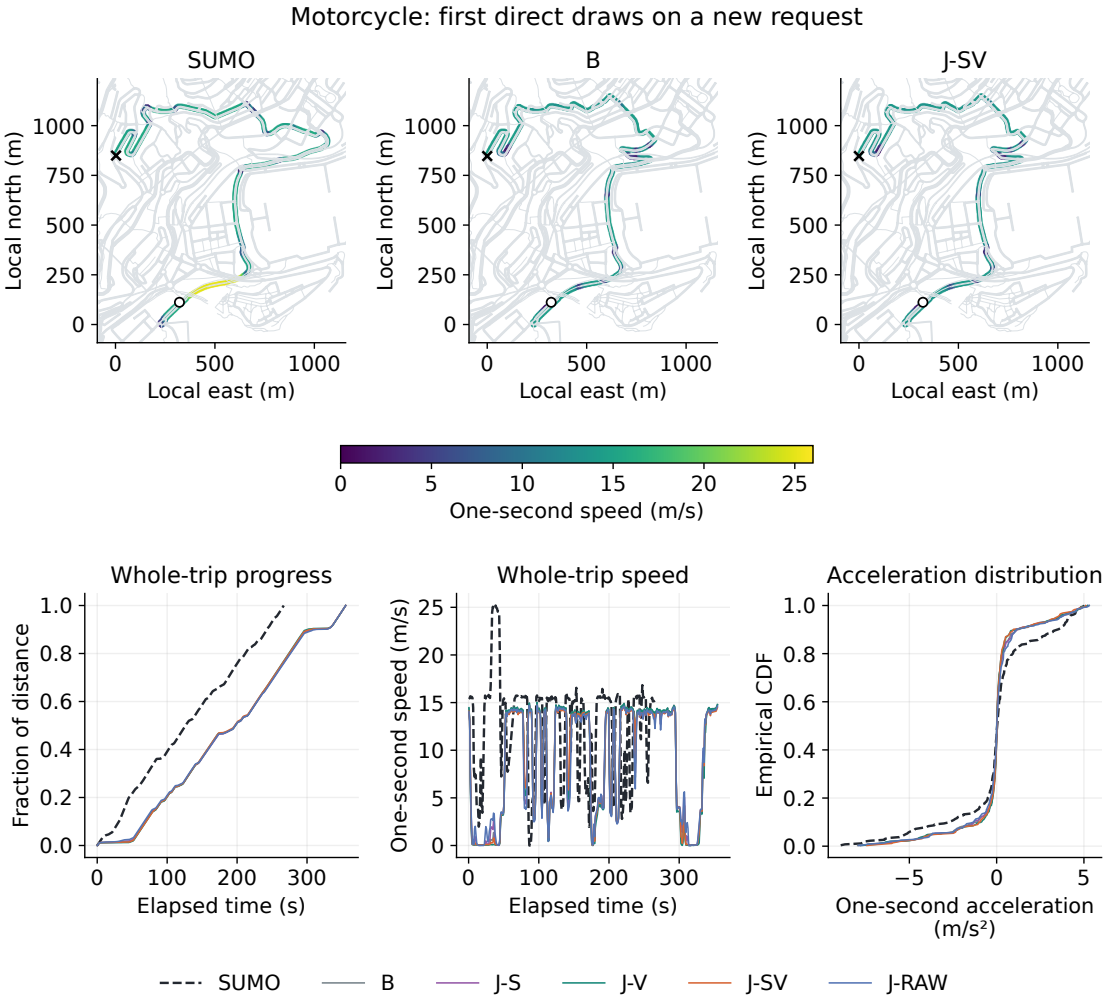

Figure 7.9. Fixed motorcycle case. Full time axes retain the difference between reference and generated duration; no time alignment is used to make the curves agree.

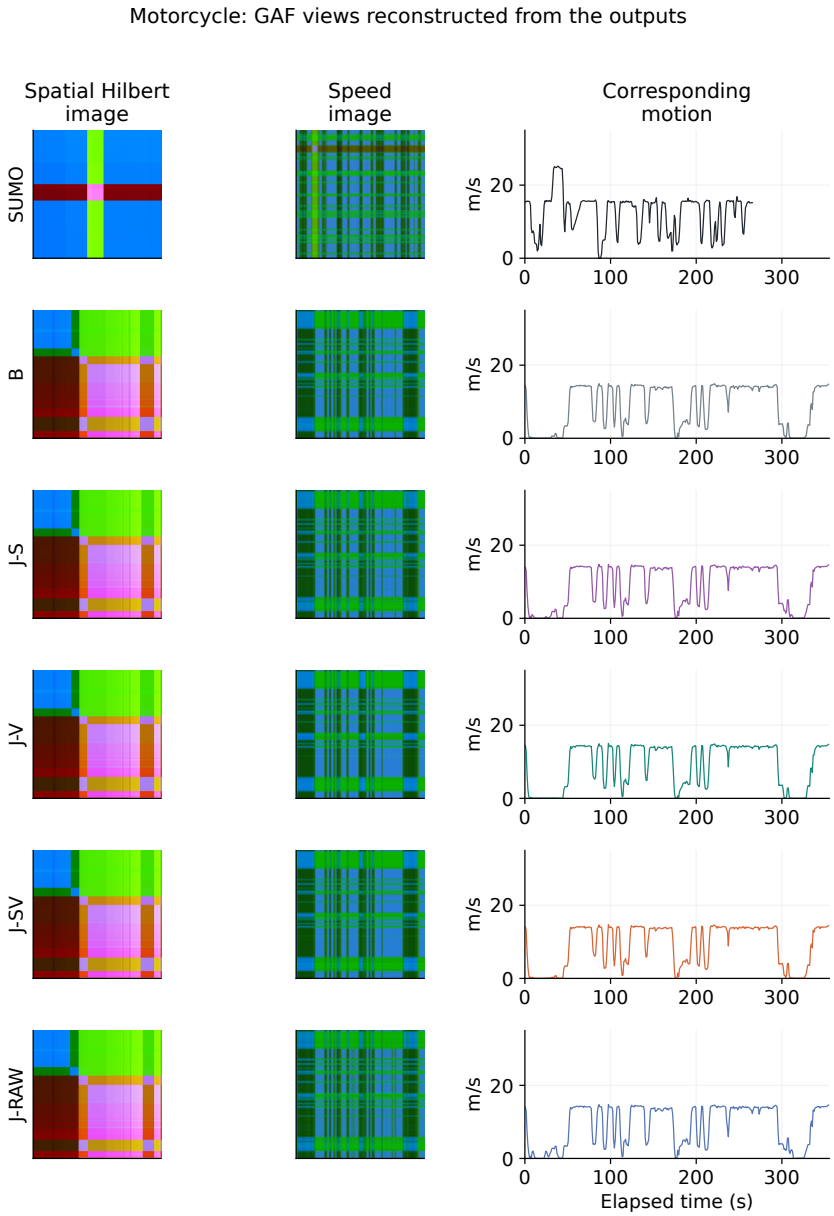

Figure 7.10. Fixed motorcycle output representations. Spatial and speed images encode different aspects of the same complete journey.

7.6.3 Private car: marginal acceleration and stop timing

The private-car reference lasts 631.75 seconds, compared with 763.90 seconds for the generated journeys. Similar acceleration CDFs accompany speed peaks and stationary intervals that occur at different elapsed times, illustrating how agreement

in a marginal distribution can coexist with differences in temporal organization. The GAF panels summarize these patterns through pairwise sequence relations, complementing the explicit speed curves. This case is consistent with the class-level acceleration improvement for joint conditioning while also showing why request-level timing summaries remain useful alongside distribution distances.

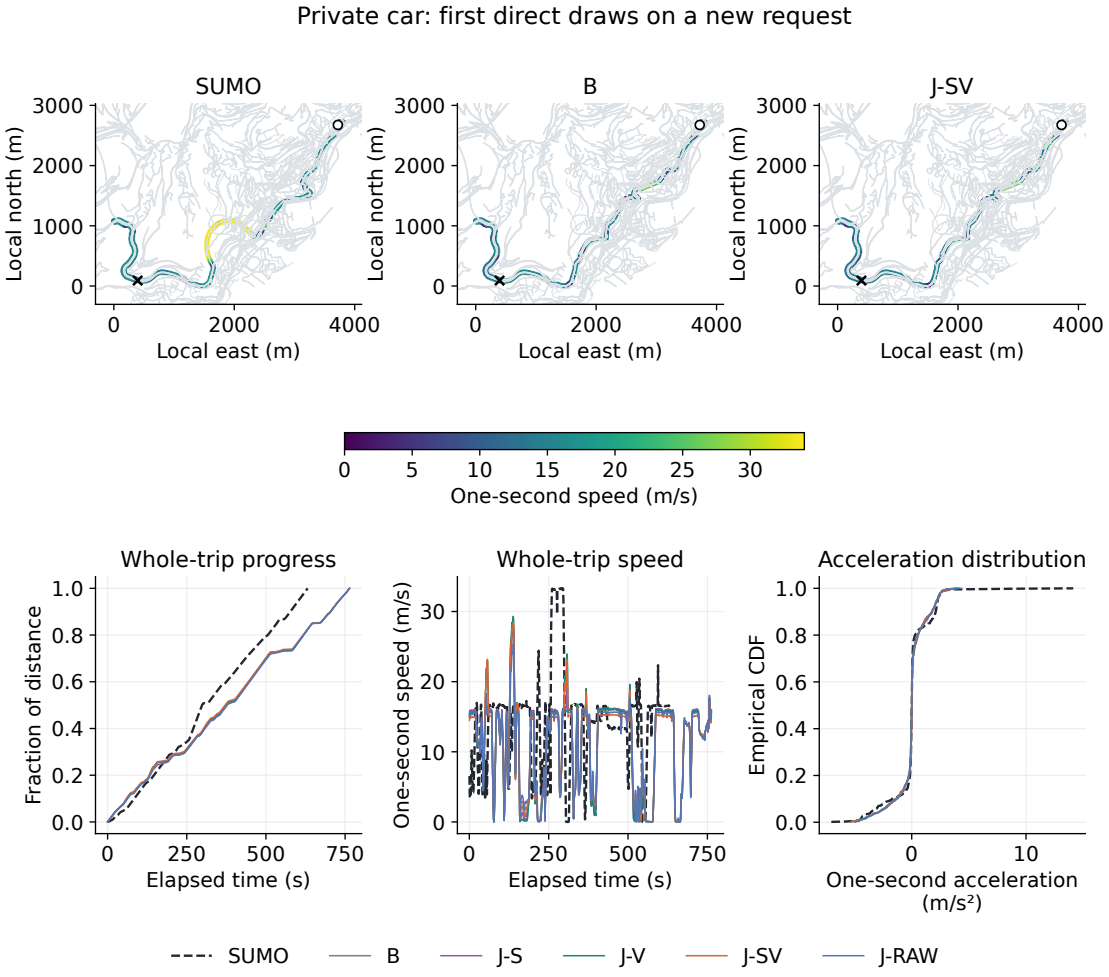

Figure 7.11. Fixed private-car case. The empirical acceleration CDF summarizes the movement distribution, while the full speed trace shows its temporal organization.

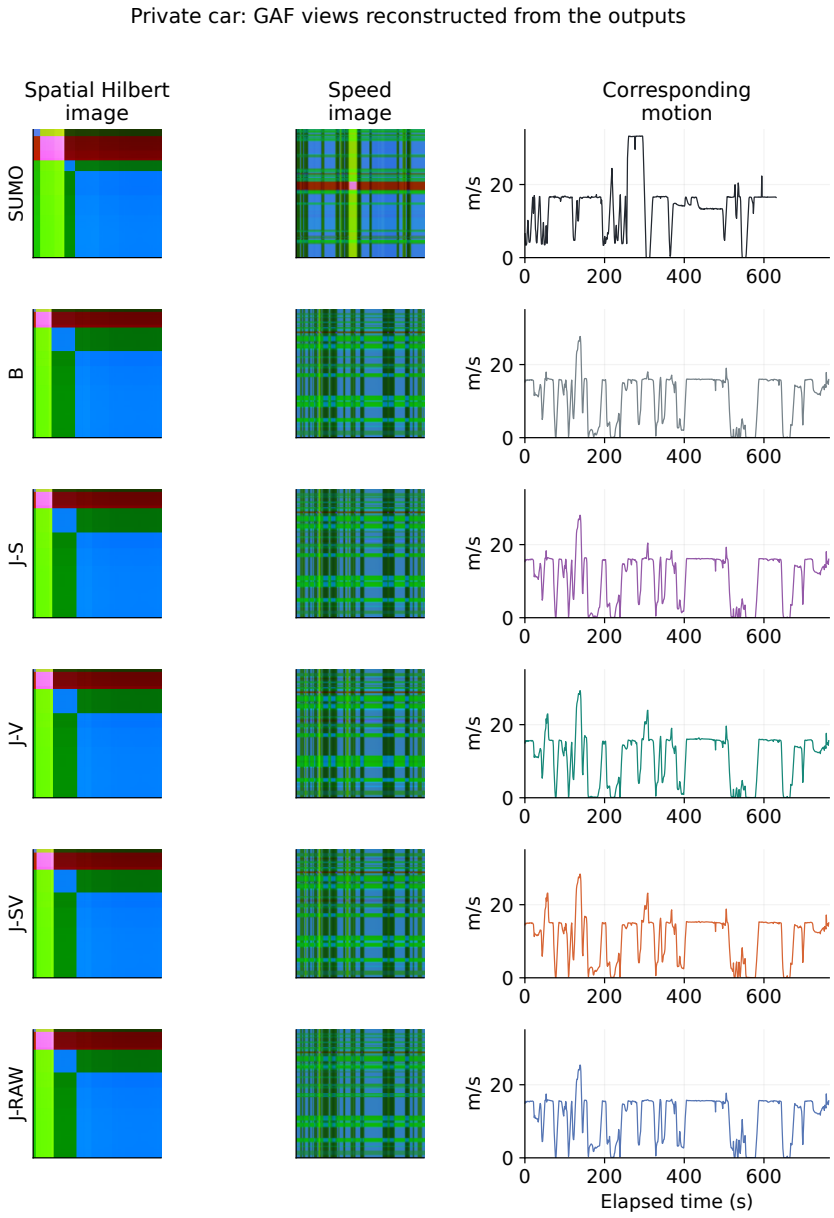

Figure 7.12. Fixed private-car GAF views and their source speed curves. The same frozen image conversion is applied to the reference and every generated row.

7.6.4 Bicycle: a visible shared travel-time limitation

The bicycle reference lasts 778.50 seconds, whereas the generated journeys last 499.15 seconds. Covering the selected path within this shorter duration raises the generated speed level across all conditioning variants, producing the steep progress curves visible in the figure. Both the velocity images and their source curves reflect this shared shift, while the spatial images retain the route's broader organization. Because route and duration are matched across variants, the example locates the dominant discrepancy in the common duration component. The adverse class-level acceleration changes further show that recognizable image structure and accurate physical movement assess different aspects of the generated journey.

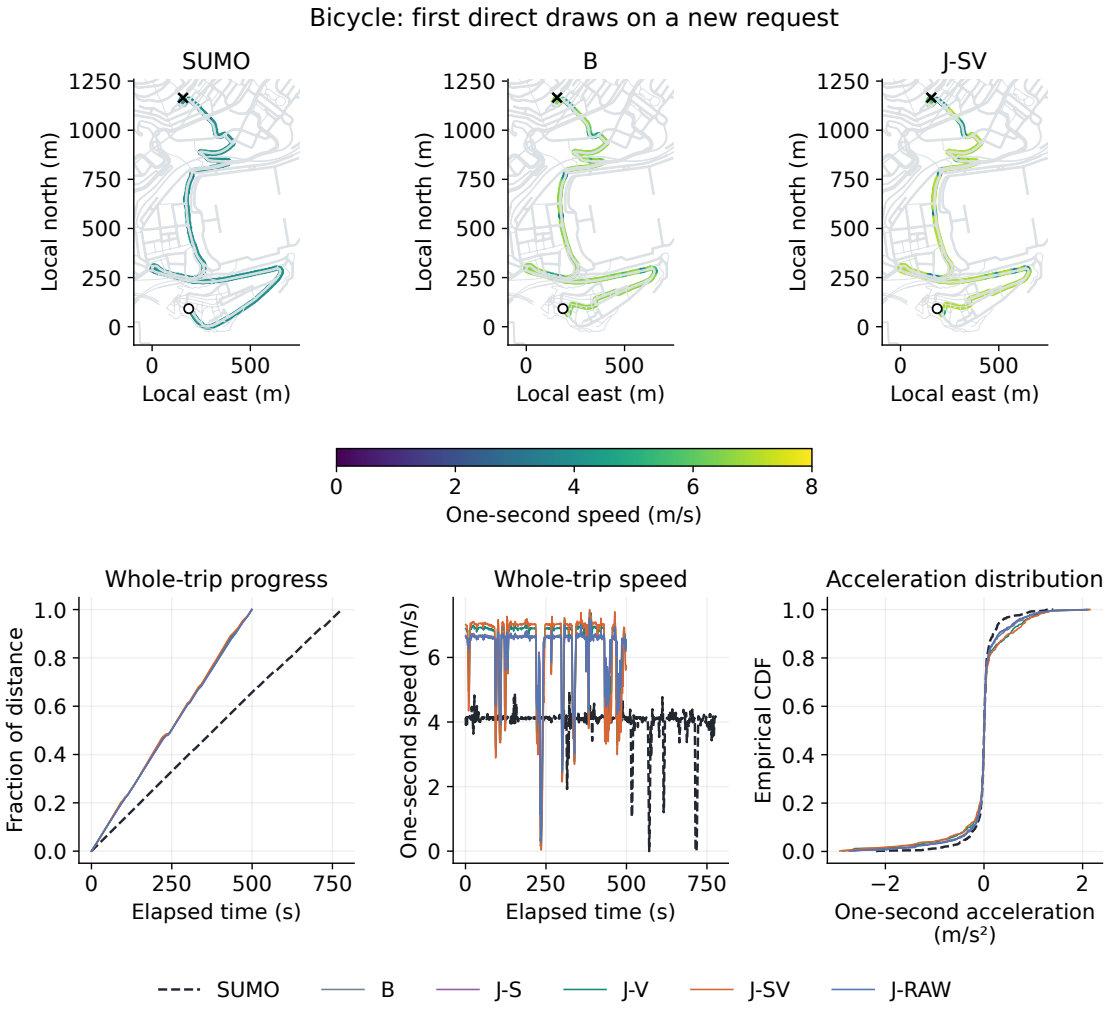

Figure 7.13. Fixed bicycle case. The shorter common generated duration and higher speed level remain visible. The case illustrates the effect of the common sampled duration.

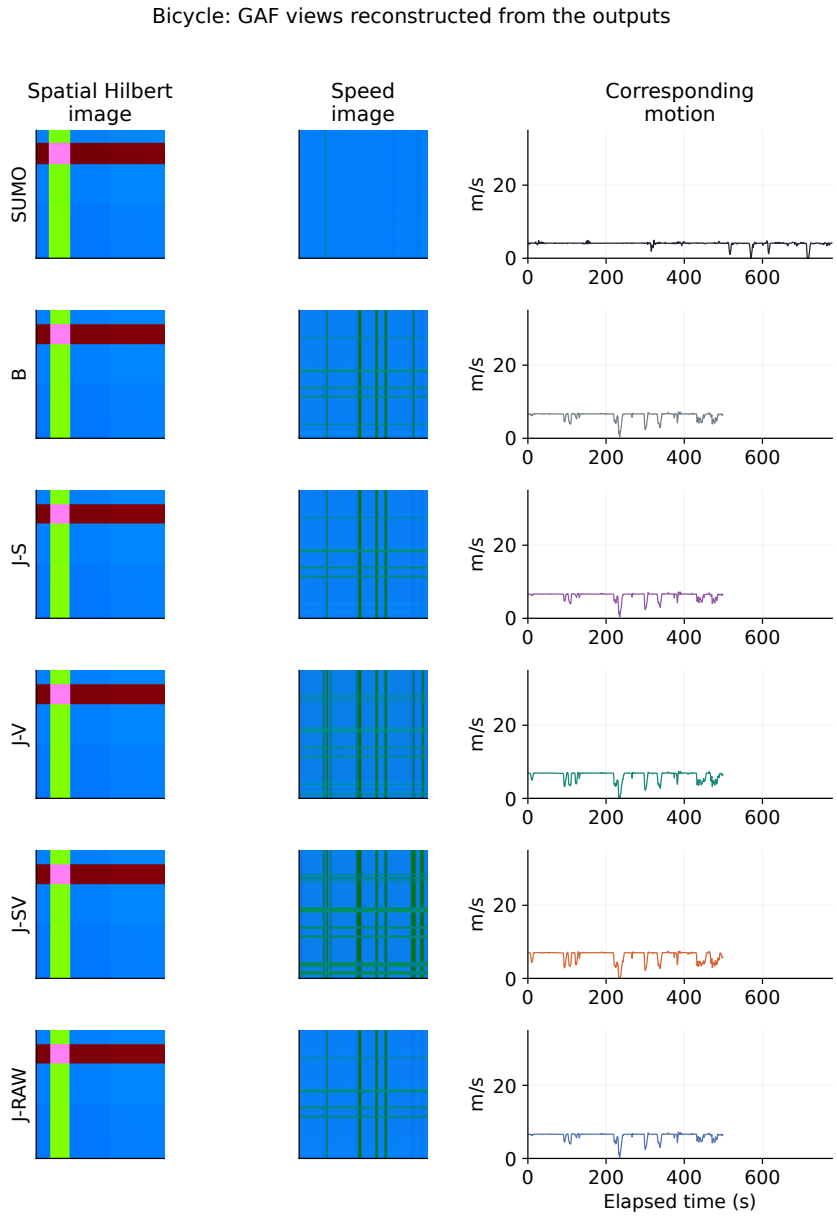

Figure 7.14. Fixed bicycle spatial and speed images. An image can preserve class-related structure while the associated physical-time profile remains mismatched.

7.6.5 Taxi and ride-hailing: service labels and geometry

The taxi/ride-hailing reference lasts 293 seconds and the generated journeys 345.18 seconds. Different permitted route choices produce distinct spatial patterns, while

modest changes in the generated speed plateaus are visible across conditioning variants. Taxi and ride-hailing labels describe a service category whose vehicles can share road access and motion characteristics with private cars; consequently, route geometry alone provides a weaker discriminative signal for this class. The spatial classification results and the mixed motion-quality response are consistent with that overlap, making the combined map, image and physical-time views particularly informative for interpreting this example.

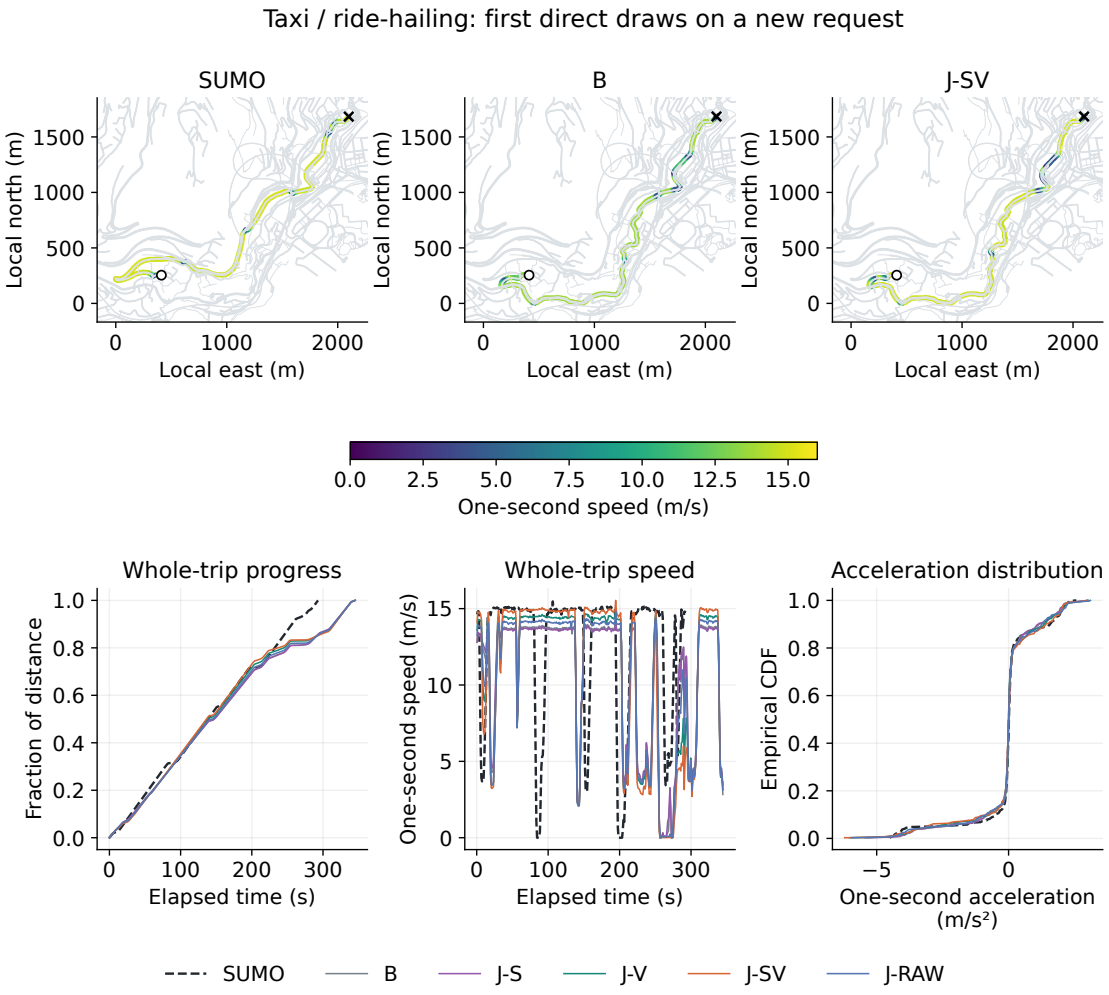

Figure 7.15. Fixed taxi/ride-hailing case: a permitted route alternative and its generated motion. Route-use distributions characterize alternative paths, while motion distributions characterize travel along them.

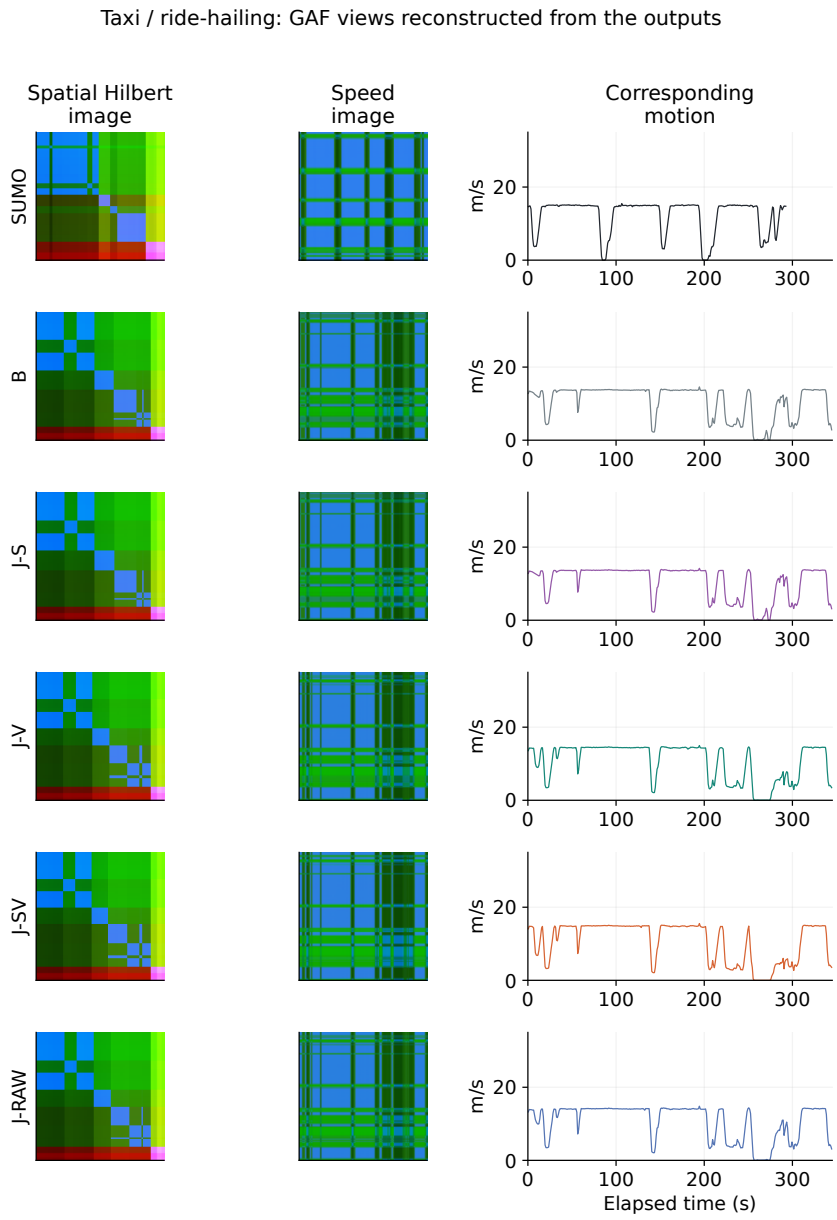

Figure 7.16. Fixed taxi/ride-hailing output images. The five class-specific pairs are retained in their predetermined order, including cases with clear timing discrepancies.

7.7 Candidate Diversity and GAF-Based Selection Across Vehicle Classes

The joint-conditioning results can be interpreted alongside the earlier candidate-selection experiment, where GAF scores act on a fixed set of generated journeys. The following panels retain the archived morning-episode outputs from the expanded native model with seed 42. Each class uses its first lexicographically ordered request supported by the reference, map and all four selection policies. Thus the visual comparison covers the same five participant classes under a common display rule.

Each trajectory panel links the road geometry to elapsed-time progress and speed. Map colors show one-second speed on a shared scale within each case, revealing where slow movement and cruising occur along the route. The eight candidate curves show the movement variation available for a fixed request, while the highlighted outputs identify Uniform, spatial, velocity and joint selection. The companion GAF panels show the corresponding spatial and velocity images together with their source movement profiles. These figures expose how different selection criteria use the available candidate variation and complement the main experiment, in which sampled representation codes influence denoising itself.

7.7.1 Bus: Candidate Movement and Image Structure

The bus case illustrates substantial variation in stop organization within a shared service corridor. The selected speed curves differ in cruising level and the placement of repeated stationary intervals, although their cumulative progress reaches the endpoint at the common sampled duration. The independent reference spreads movement over a longer elapsed-time axis. Reading the GAF panels beside these curves links spatial service structure to whole-trip motion, while the complete pool below shows the alternatives from which each policy selects.

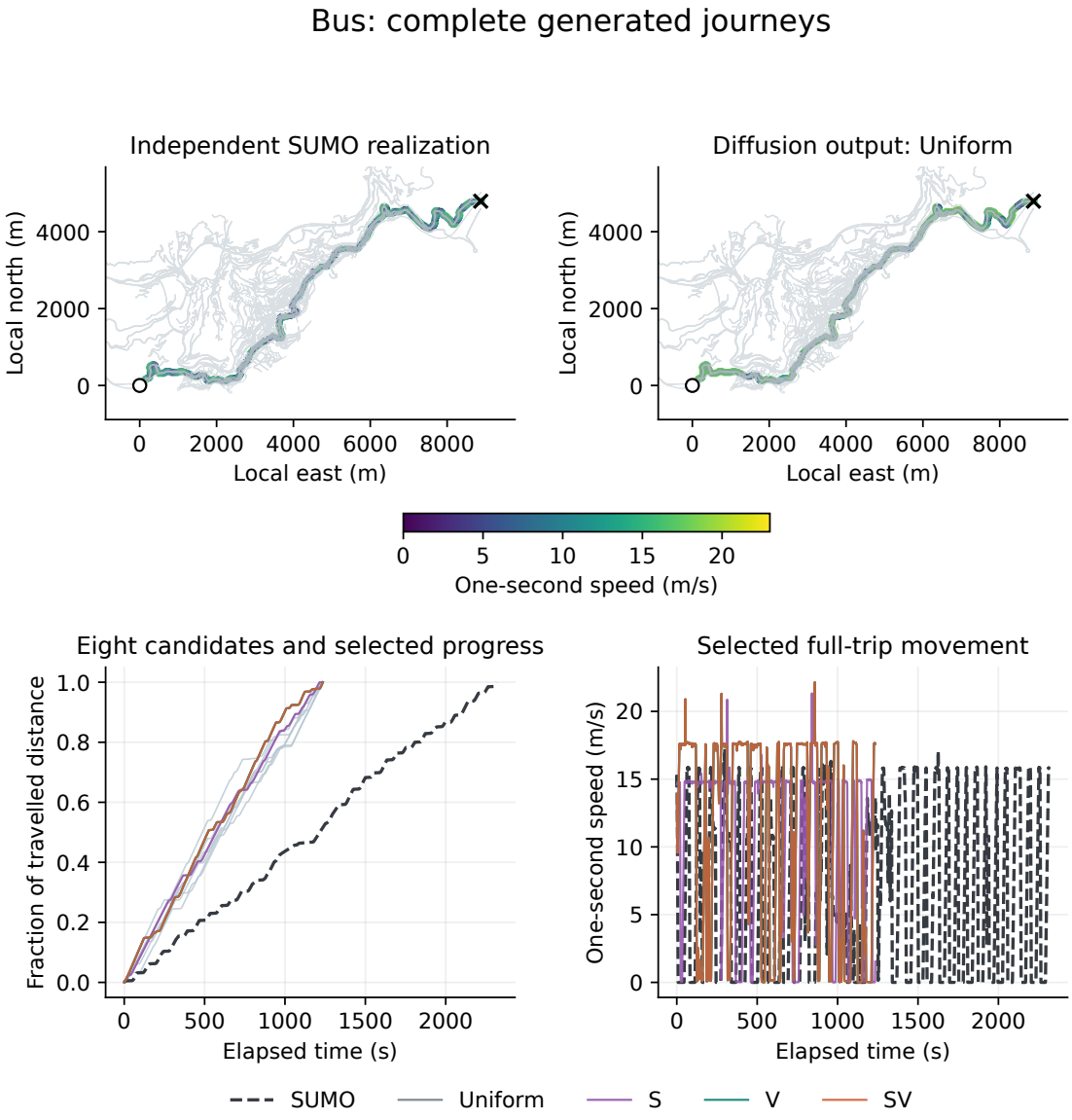

Figure 7.17. Bus: complete reference and generated journeys in the candidate-selection experiment. Pale curves retain the candidate pool; highlighted curves identify the four recorded selections.

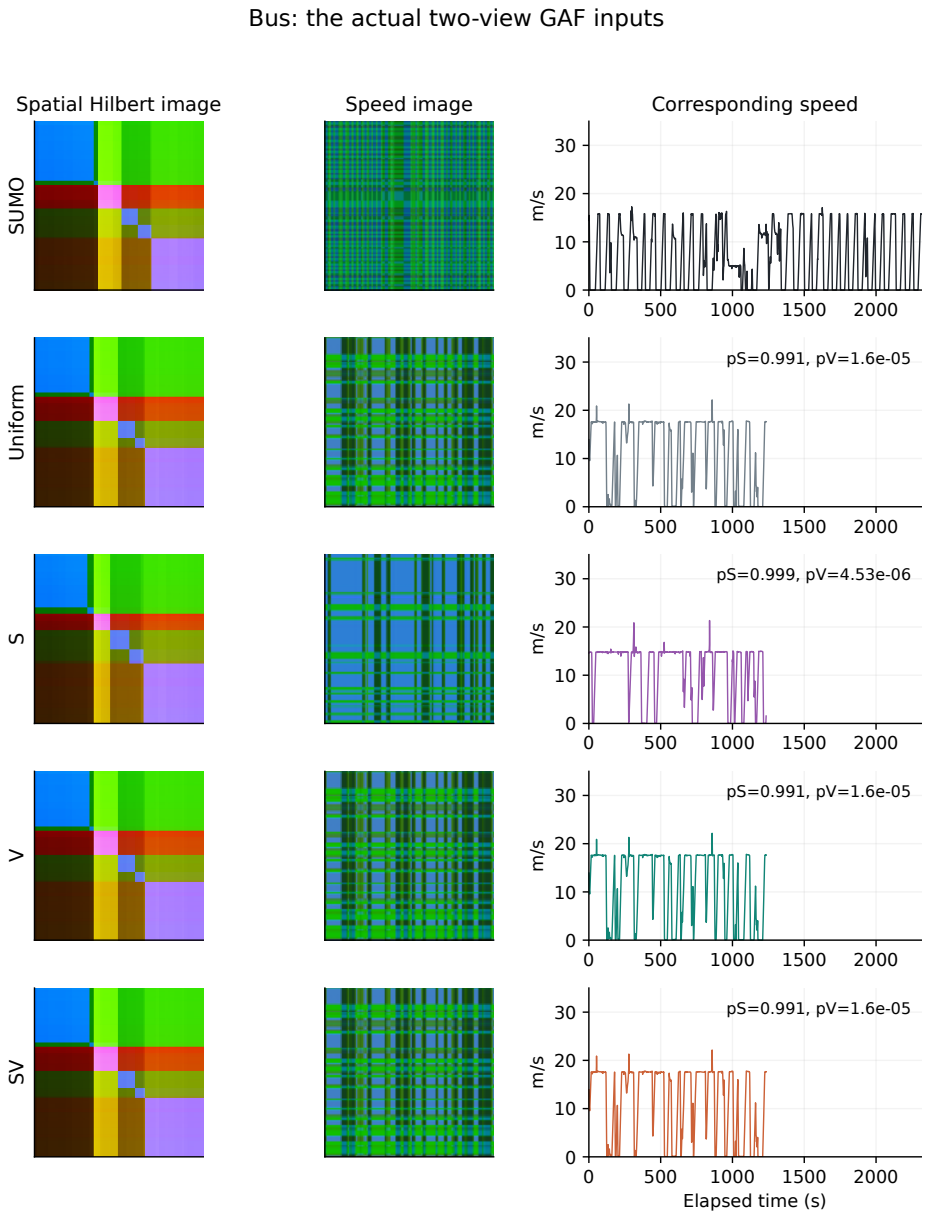

Figure 7.18. Bus: spatial and velocity GAF representations of the reference and recorded selections, paired with their source movement curves. Classifier scores describe the selection criterion.

7.7.2 Motorcycle: Candidate Movement and Image Structure

The motorcycle candidates share their road geometry but distribute stationary intervals differently along the journey. The highlighted selections retain different stop sequences, with visibly similar cruising plateaus over much of the trip. Their progress curves reveal cumulative timing differences that are less apparent in the map overlays. The paired image panels show how these movement patterns enter the spatial and velocity classifiers, connecting image-based selection to the actual delivered trajectory.

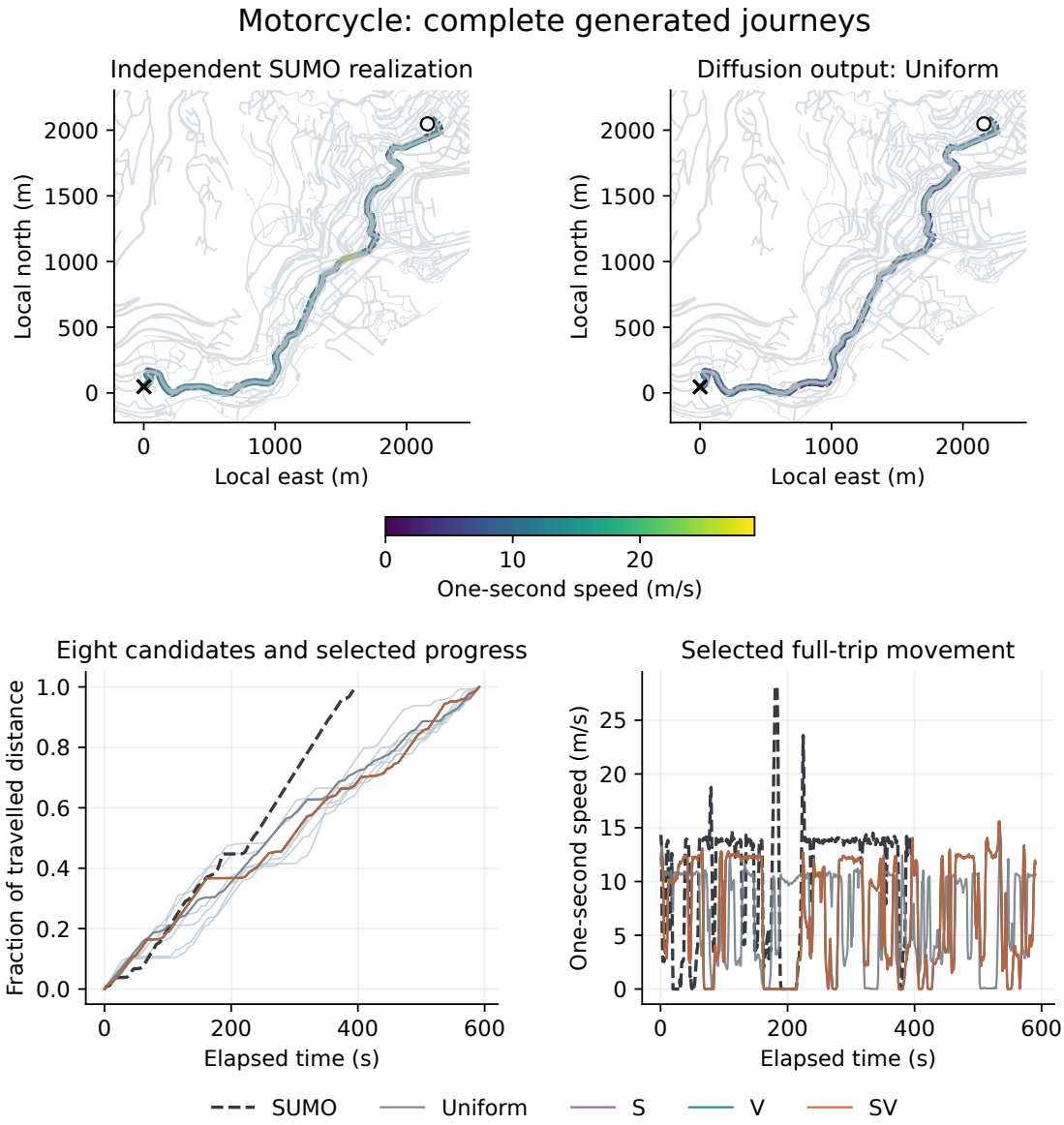

Figure 7.19. Motorcycle: complete reference and generated journeys in the candidate-selection experiment. Pale curves retain the candidate pool; highlighted curves identify the four recorded selections.

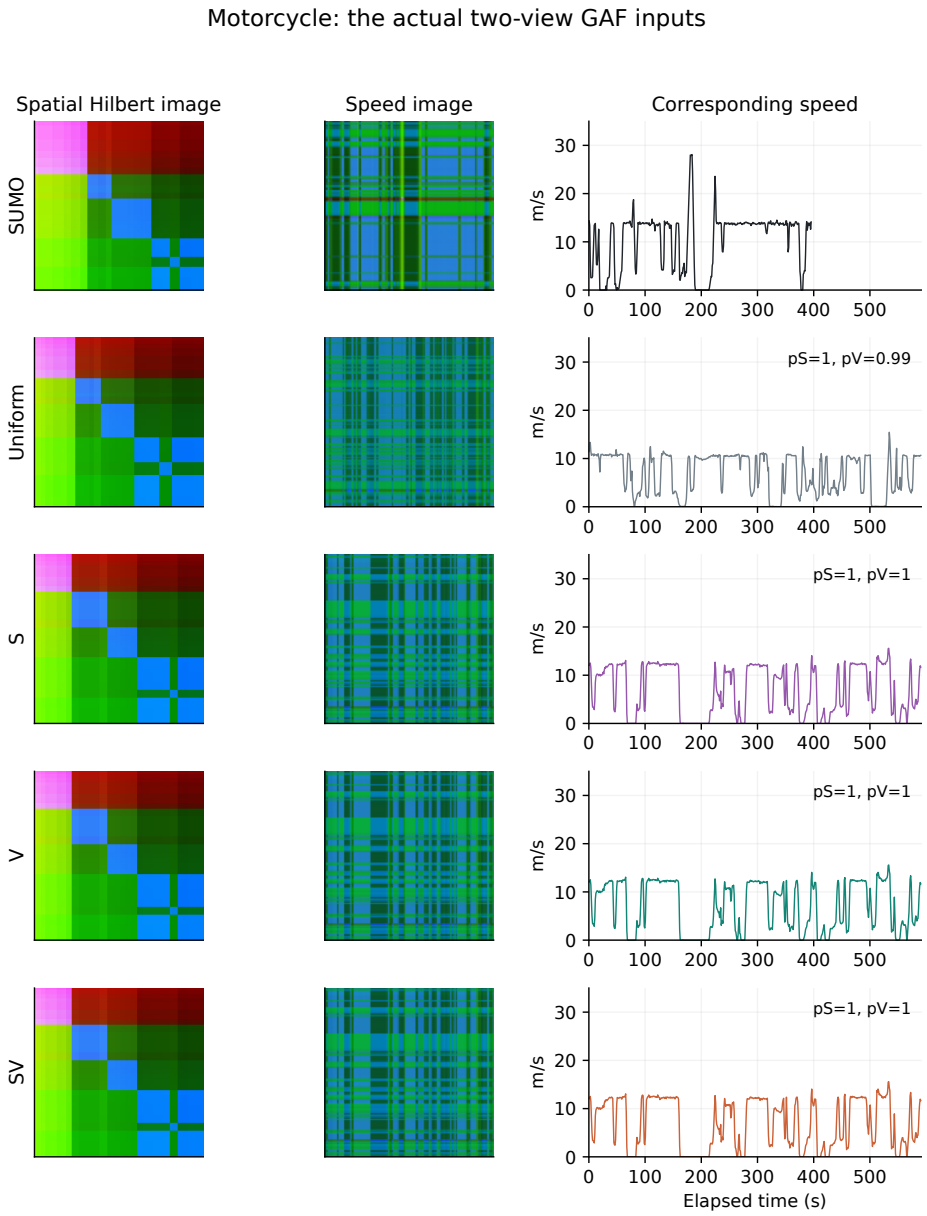

Figure 7.20. Motorcycle: spatial and velocity GAF representations of the reference and recorded selections, paired with their source movement curves. Classifier scores describe the selection criterion.

7.7.3 Private Car: Candidate Movement and Image Structure

The private-car selections largely overlap in the displayed speed and progress panels, despite visible variation among the pale candidate curves. This makes the case informative about agreement between selection criteria: different policies can retain the same output or motions with nearly identical profiles. The independent reference has distinct speed peaks and stationary periods. Comparing the corresponding GAF rows therefore helps distinguish agreement among selectors from agreement with observed movement.

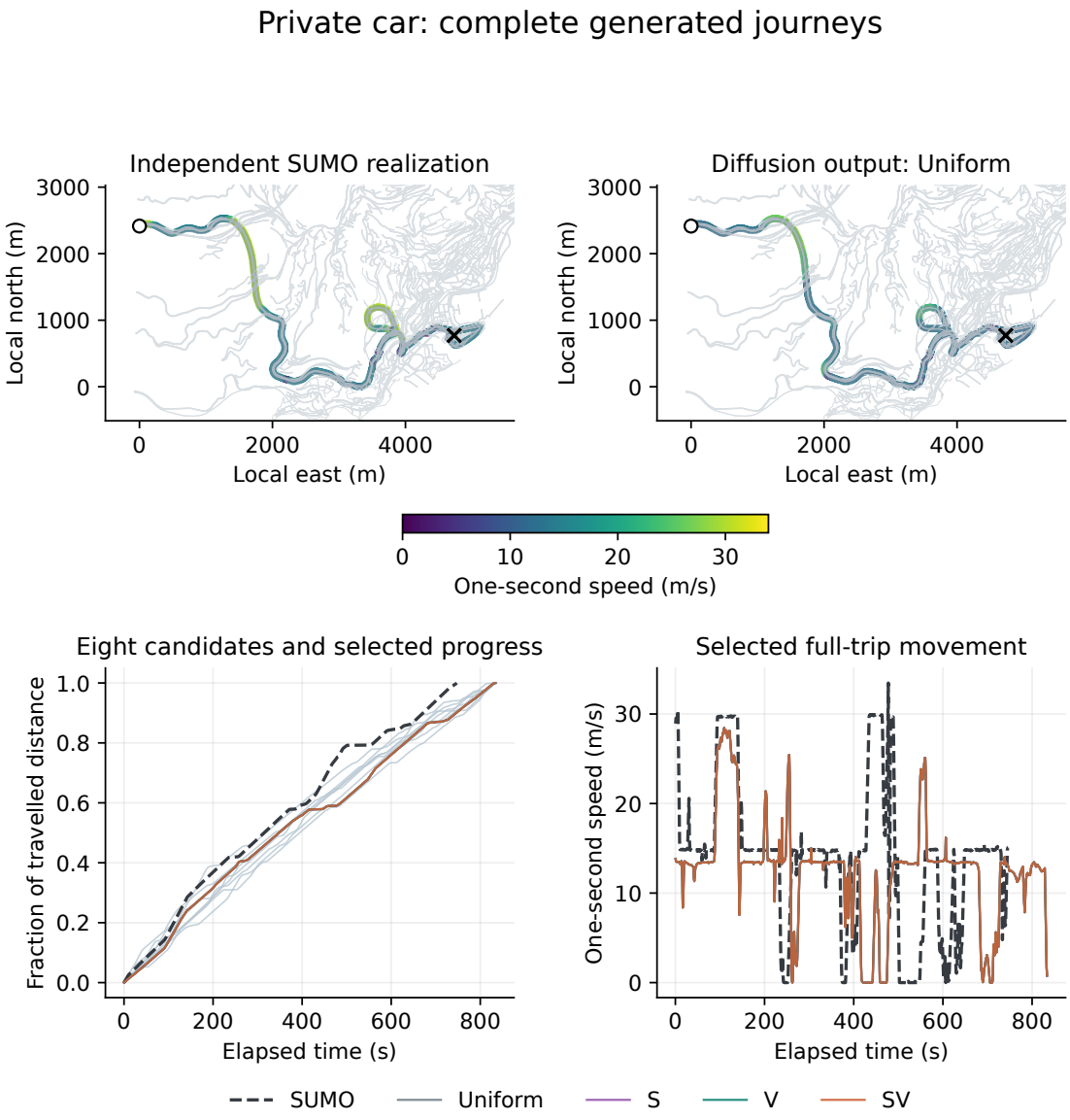

Figure 7.21. Private Car: complete reference and generated journeys in the candidate-selection experiment. Pale curves retain the candidate pool; highlighted curves identify the four recorded selections.

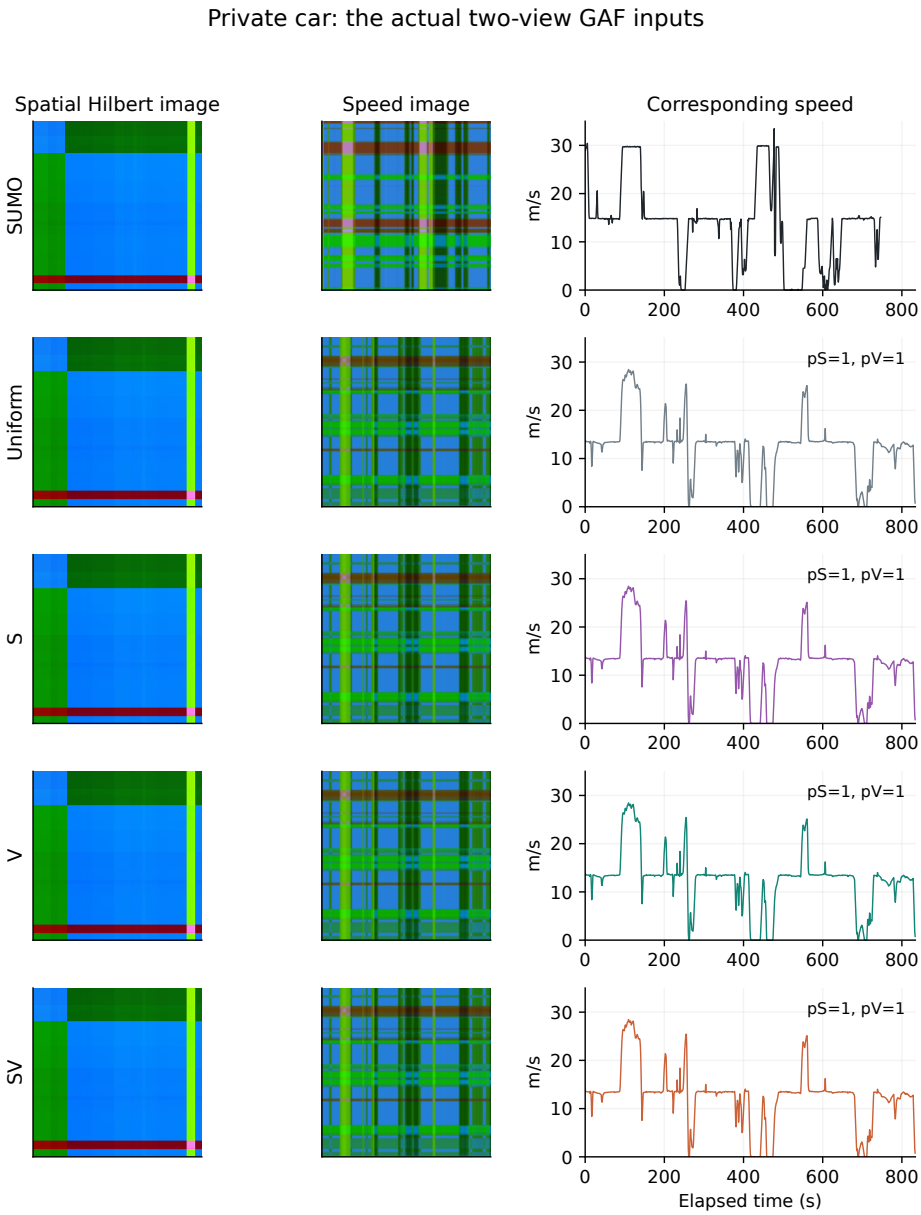

Figure 7.22. Private Car: spatial and velocity GAF representations of the reference and recorded selections, paired with their source movement curves. Classifier scores describe the selection criterion.

7.7.4 Bicycle: Candidate Movement and Image Structure

The bicycle reference progresses faster than the displayed generated selections, whose speed profiles remain at a lower level over most of the journey. Spatial, velocity and joint selection alter the positions and depths of brief slowdowns while retaining that common scale. This contrasts with the main-conditioning bicycle example, where the generated journey is shorter and faster than its reference. Together, the two fixed requests illustrate why class-level conclusions require the distributional metrics alongside individual trajectories.

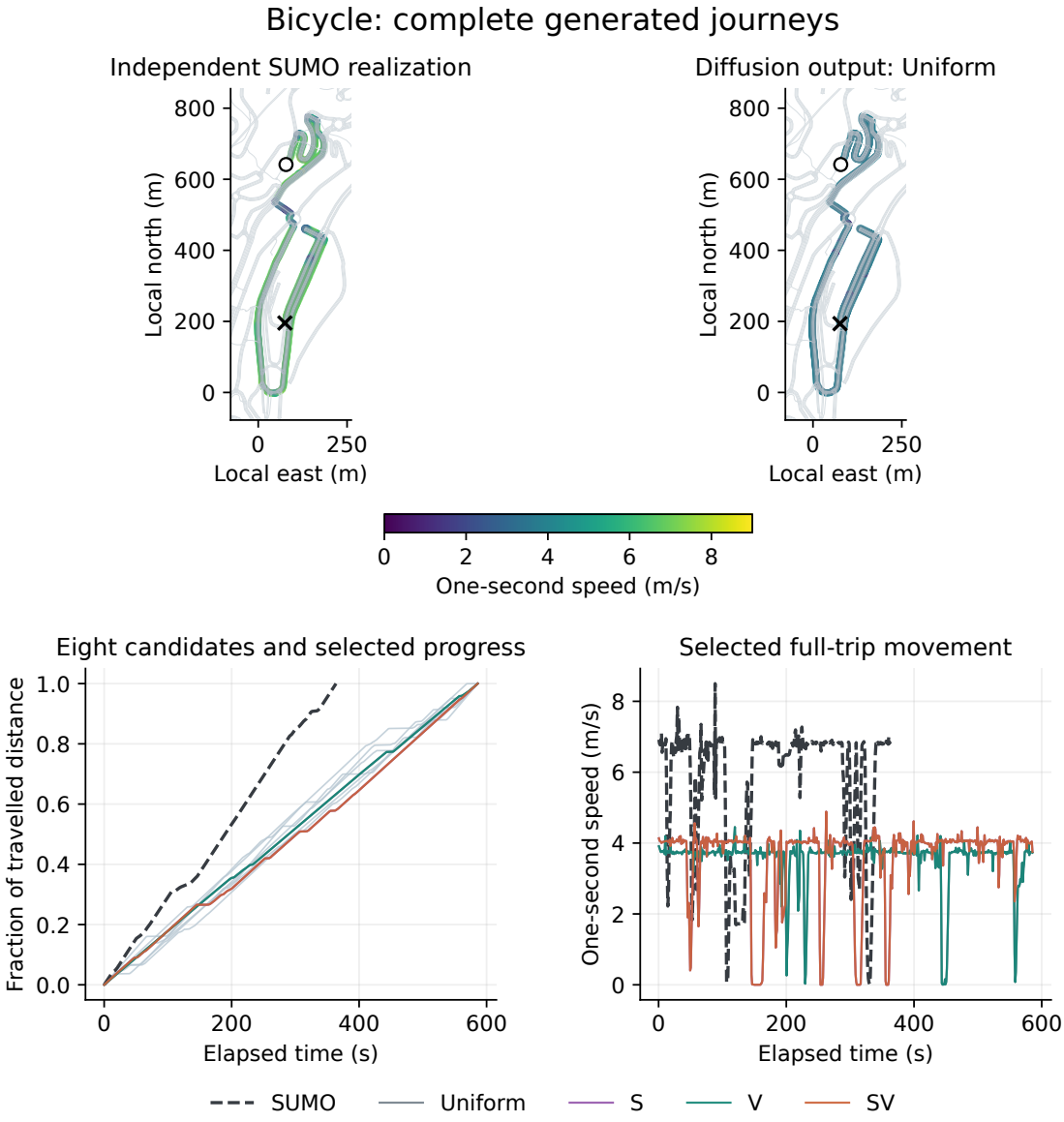

Figure 7.23. Bicycle: complete reference and generated journeys in the candidate-selection experiment. Pale curves retain the candidate pool; highlighted curves identify the four recorded selections.

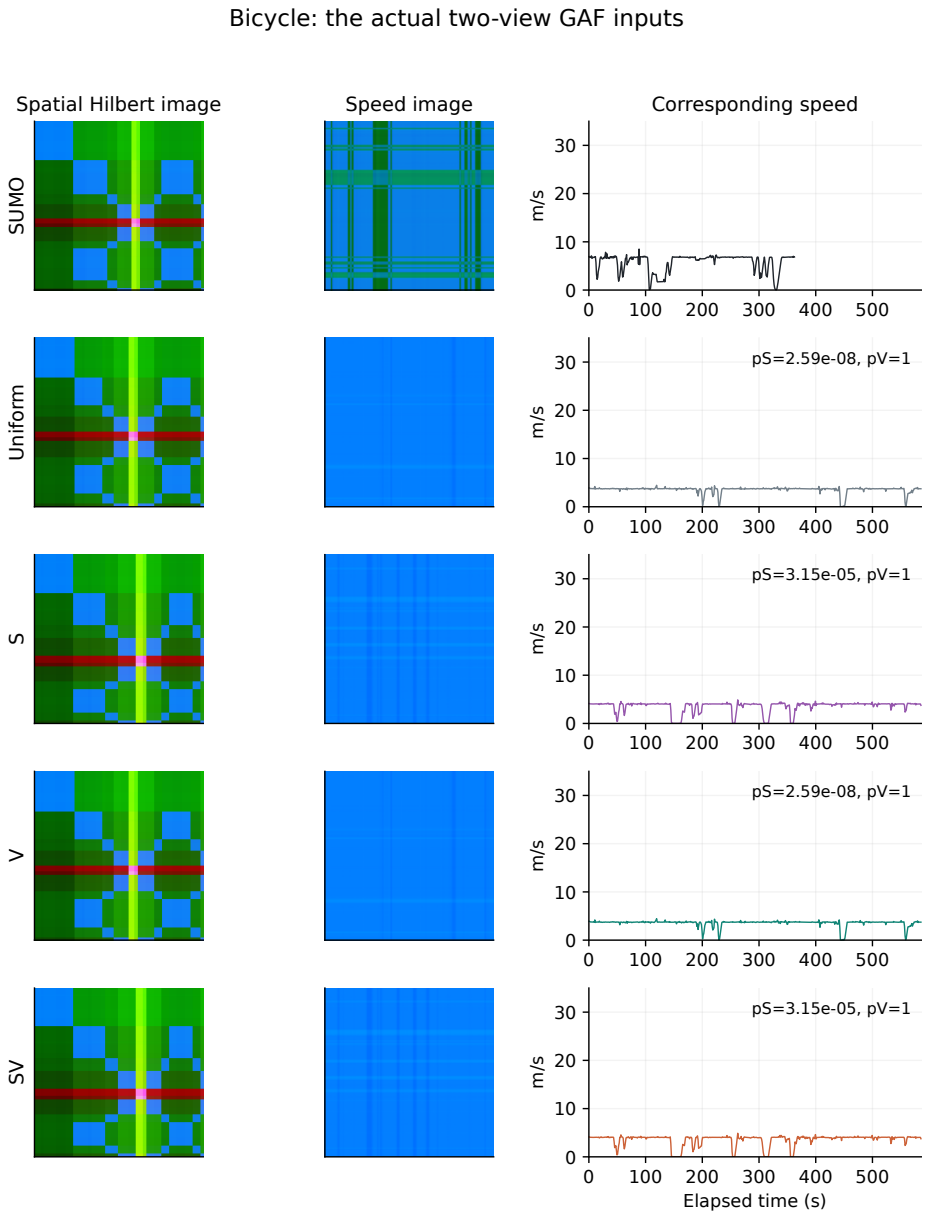

Figure 7.24. Bicycle: spatial and velocity GAF representations of the reference and recorded selections, paired with their source movement curves. Classifier scores describe the selection criterion.

7.7.5 Taxi and Ride-Hailing: Candidate Movement and Image Structure

The taxi/ride-hailing pool displays clear differences among the selected speed profiles. Joint selection favors a higher cruising plateau, whereas spatial selection retains a lower-speed pattern; their cumulative progress curves consequently separate during the trip. The maps remain closely aligned because the candidates traverse the same requested route. The GAF panels make this temporal choice visible in image space and show how class scores can select different movement styles from one road-constrained pool.

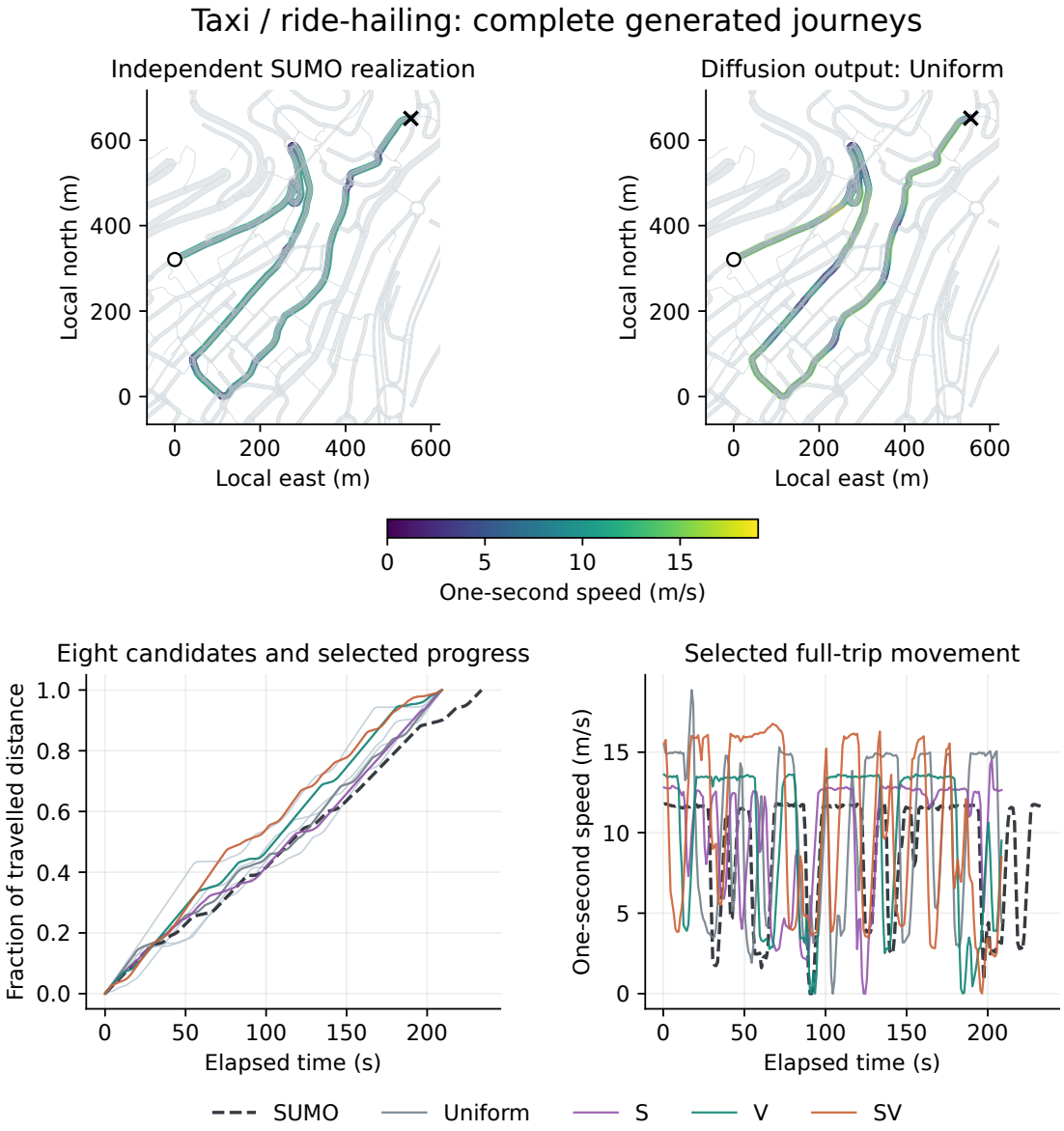

Figure 7.25. Taxi and Ride-Hailing: complete reference and generated journeys in the candidate-selection experiment. Pale curves retain the candidate pool; highlighted curves identify the four recorded selections.

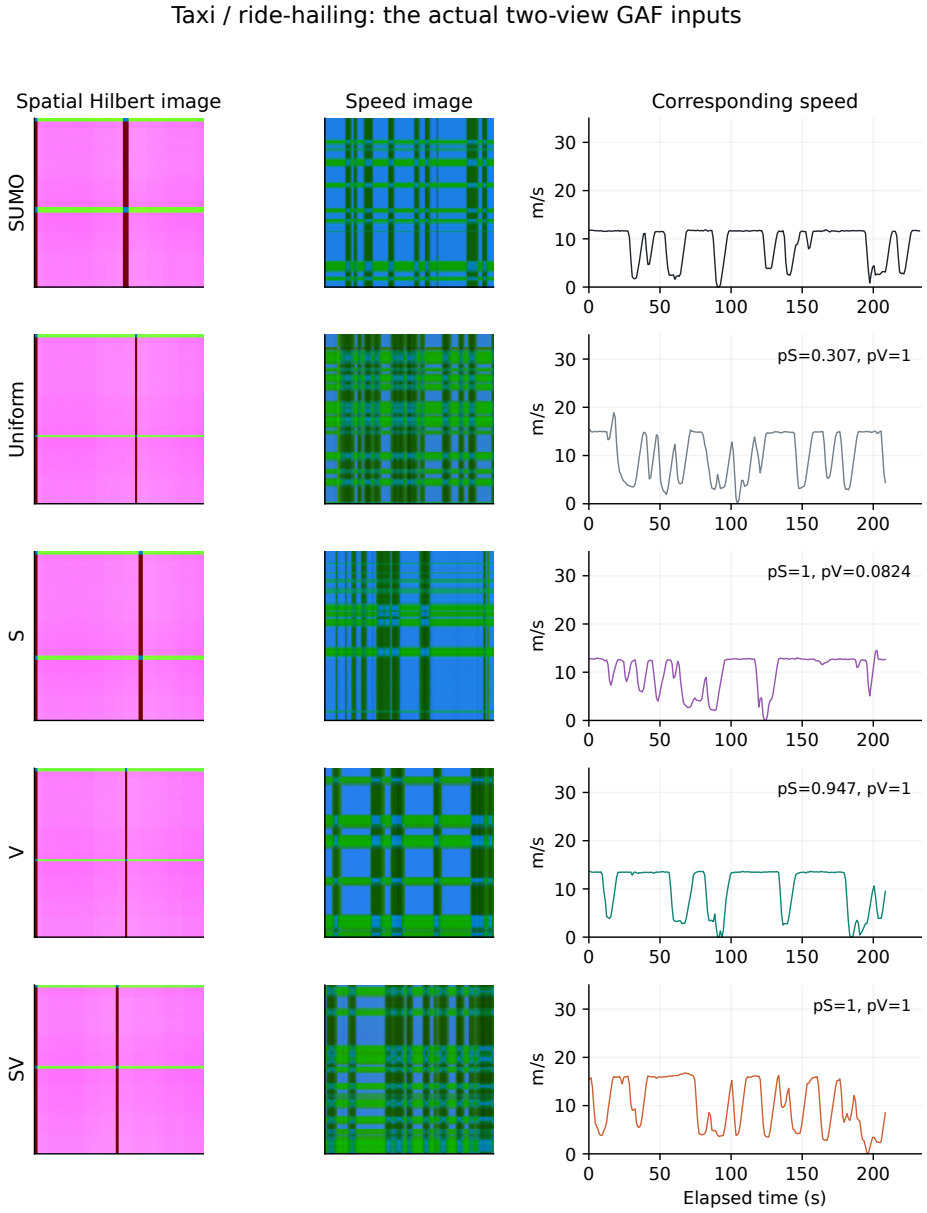

Figure 7.26. Taxi and Ride-Hailing: spatial and velocity GAF representations of the reference and recorded selections, paired with their source movement curves. Classifier scores describe the selection criterion.

7.7.6 Inspecting the Complete Candidate Pool

The bus pool makes the selection mechanism explicit: every candidate has its own speed sequence and associated spatial and velocity images, while the policy tags locate the retained outputs. A common road request can therefore produce multiple temporal realizations, and different GAF criteria can favor the same realization or different ones. This candidate-level view connects the generated dataset to the selection analysis and clarifies the contrast with learned latent conditioning.

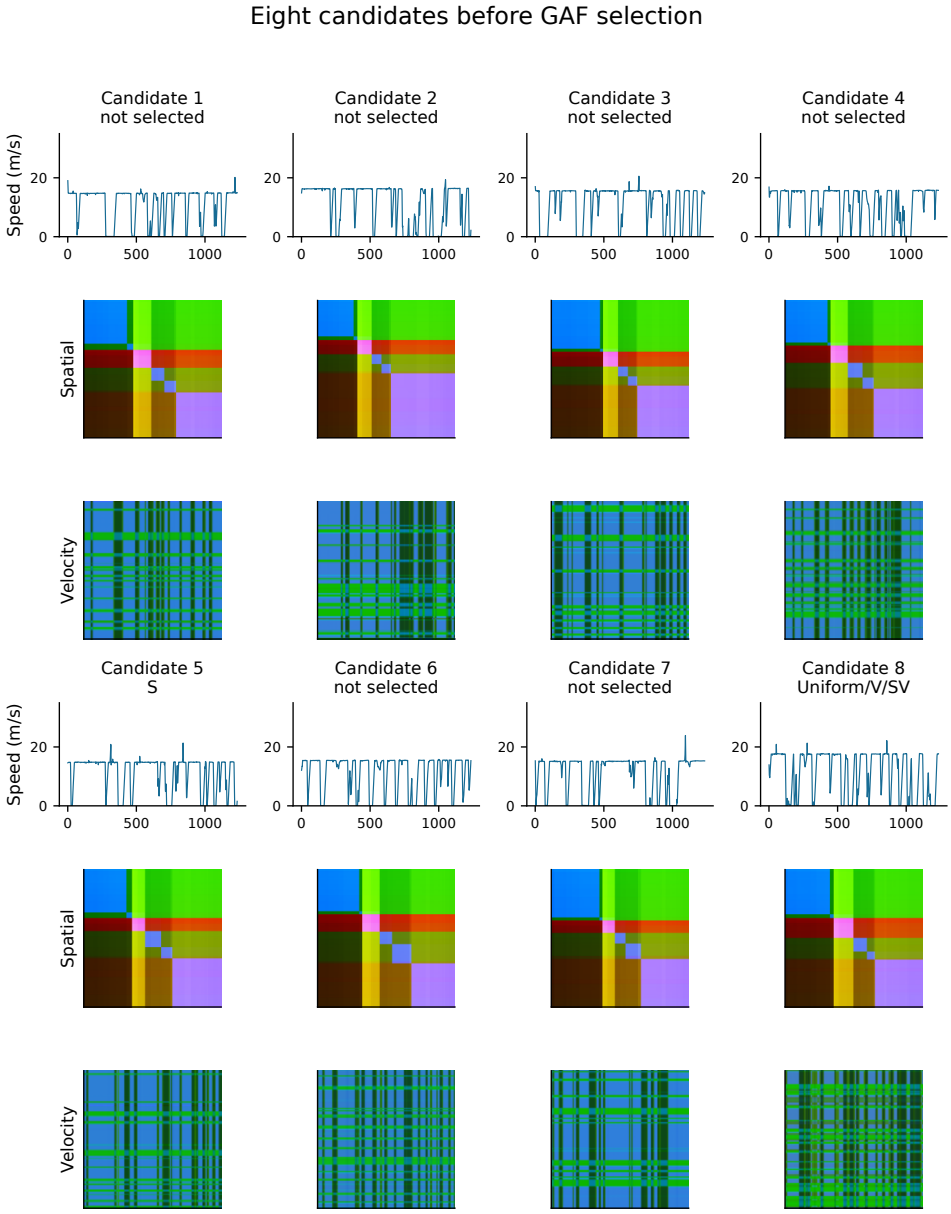

Figure 7.27. Complete bus candidate pool from the candidate-selection experiment. Every generated speed profile is paired with its two image representations; tags identify the recorded policy choices.

7.8 Synthetic-data augmentation

Table 7.3 retains every downstream regime, including synthetic-only training. Observed-only training achieves 94.41% mean macro F1. Adding unscored B samples increases it to 96.26%, whereas resampling the same observed set gives 94.17%. Thus the baseline synthesized data have a practical supplementation benefit for this particular raw-series recognition task and upstream information budget.

Spatial J-S augmentation reaches 96.95%, 0.699 percentage points above B. The direction is positive for both downstream training seeds. J-RAW reaches 96.66%, leaving a much smaller spatial-versus-RAW contrast of 0.294 points. The two-seed non-bus vehicle-equal F1 is 96.54% for J-S and 95.41% for B, so the pooled signal is not solely a bus-class effect. The physical and downstream measurements therefore yield different model rankings.

Table 7.3. All fifteen downstream regimes. Percent macro F1 on real simulated references from the last two OD cohorts. Each fit has the same optimization budget.

Training data	F1, seed 4001	F1, seed 4002	Mean F1
real	94.39	94.42	94.41
only-empirical	93.06	93.63	93.34
only-B	93.68	92.34	93.01
only-F-SV	93.44	91.83	92.63
only-J-S	92.90	92.88	92.89
only-J-V	92.35	92.87	92.61
only-J-SV	93.96	92.87	93.41
only-J-RAW	93.12	92.62	92.87
plus-empirical	95.18	93.15	94.17
plus-B	96.01	96.50	96.26
plus-F-SV	96.26	94.66	95.46
plus-J-S	96.83	97.08	96.95
plus-J-V	95.76	94.95	95.35
plus-J-SV	95.23	95.74	95.48
plus-J-RAW	96.80	96.52	96.66

The augmentation effect varies by evaluation cohort. Averaging episode macro F1 across the two windows and learner seeds, J-S minus B is -0.755 percentage points in cohort 3 and $+2.091$ points in cohort 4. Table A.12 gives these episode-equal summaries for all fifteen regimes. The pooled gain is concentrated in cohort 4.

Dual-view J-SV yields 95.48% macro F1 and speed-only J-V yields 95.35%, both below B augmentation. Synthetic-only regimes yield 92.61–93.41%, compared with

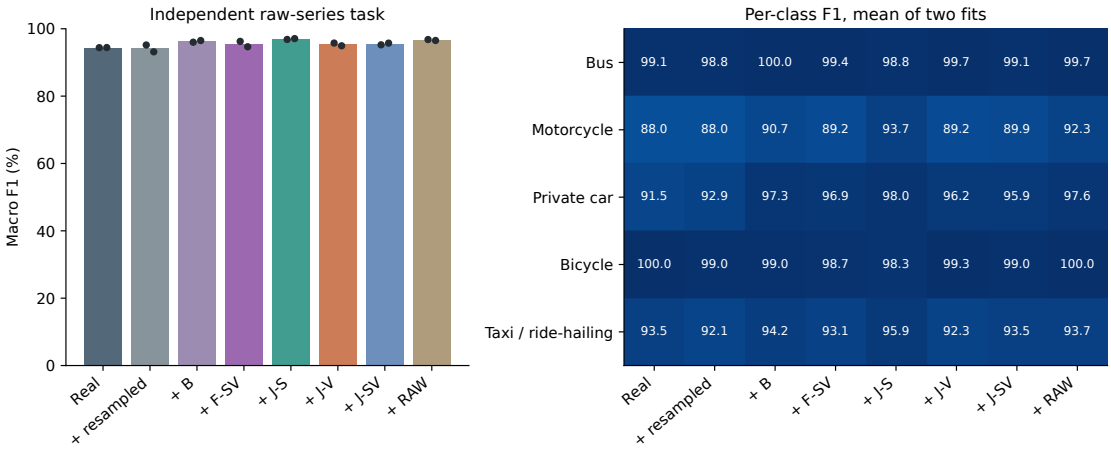

Figure 7.28. Independent augmentation on observed SUMO references. Left: pooled macro F1, with both learner seeds shown. Right: mean per-class F1. "Real" denotes simulator observations. All methods share downstream training budgets; the upstream generator uses the full labelled training corpus.

94.41% for observed-only training. Synthetic trajectories are most useful as supplementary training data in this experiment. Spatial conditioning provides the highest pooled augmentation score, whereas B retains the better aggregate physical endpoints.

7.9 Representation sensitivity and condition use

The pre-generation 500-vehicle sensitivity audit shows that whole-trip time warping changes both feature views. Within-second perturbations can substantially alter native acceleration while leaving the speed features almost unchanged, as expected from one-second averaging and relative-profile compression. Smoothing affects both views differently. The perturbations characterize the temporal detail retained by each view.

The final-model perturbation audit confirms that the code path is active. B is exactly invariant to the four perturbations; C is invariant to same-class exchange and its deterministic prior, as expected. Changing the code changes predictions in J-S, J-V and J-SV. At noise index 500, the J-SV mean prediction RMS changes by 0.012684 for a same-class exchange and 0.013908 for a sampled prior code relative to its assigned training code. The corresponding MSE values are shown in Table 7.4.

For J-SV, assigned training codes yield lower denoising MSE than all three perturbations at each diagnostic level. The absolute differences are small. This training-vehicle experiment identifies an active conditioning path and a gap between target-derived and sampled codes. J-S has smaller prediction changes, with

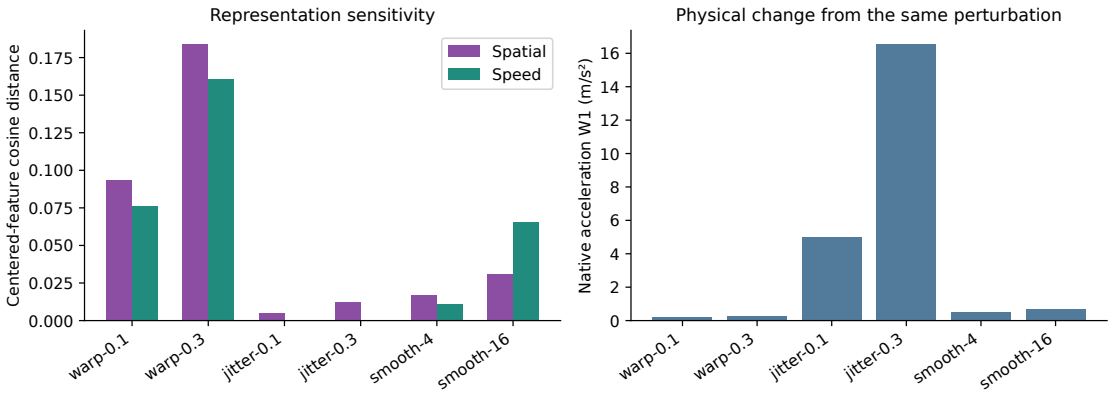

Figure 7.29. Training-side representation audit, fixed before test results. Feature changes and native acceleration changes are measured for the same perturbations. The panels report their respective units. The views respond differently across physical time scales.

Table 7.4. Condition use on training vehicles: J-SV noise-prediction MSE under four fixed perturbations. Means over 25 source vehicles and two model seeds. This is an in-sample mechanism check, not out-of-sample quality.

Noise index	Assigned	Same-class swap	Zero	Sampled prior
50	0.142174	0.142512	0.143795	0.142529
250	0.032396	0.032623	0.032764	0.032622
500	0.016933	0.017027	0.017050	0.017065
750	0.003504	0.003521	0.003573	0.003567
950	0.000201	0.000202	0.000208	0.000207

mixed MSE responses to same-class exchange. The complete perturbation tables are given in the appendix.

7.10 Diversity, correction and computation

Mean within-request relative-progress diversity ranges from 0.05575 to 0.05638, compared with 0.05626 for B. Thus the condition variants retain similar diversity under this measure. The near-shape rate is zero against the fixed library of at most 512 training progress profiles per class.

The decoder truncates negative log-motion values. The mean affected-bin fraction is 11.64% for B and 12.45% for J-SV, with all model variants between 11.52% and 12.45%. Upper clipping and constant-progress fallback occur zero times. These counters characterize the correction applied before coordinate decoding.

Table 7.5. Formal 5,000-update motion continuation on the available L20 setup. Times show seeds 42/43; peak allocated memory includes resident frozen-prefix caches. The common 10,000-update motion initialization, upstream classifier fitting, code-prior fitting and out-of-sample sampling are additional costs.

Variant	Trainable encoder	Minutes, 42 / 43	Peak allocated GiB, 42 / 43
B	0	22.90 / 24.02	0.61 / 0.61
F-SV	139,712	26.39 / 25.70	11.79 / 11.79
J-S	8,466,149	24.66 / 24.55	6.32 / 6.32
J-V	8,466,149	26.26 / 25.91	6.32 / 6.32
J-SV	16,932,298	27.46 / 26.60	12.02 / 12.02
J-RAW	5,922,250	26.18 / 25.74	1.23 / 1.23
J-SH	16,932,298	27.42 / 25.99	12.02 / 12.02
C	139,712	24.52 / 23.12	0.61 / 0.61

Each 5,000-update motion continuation takes 22.90–27.46 minutes on the assigned L20. The dependent formal queue closes in approximately 4.06 hours, including its reference, generation and utility stages. Summed generation-cell time divided by output count is approximately 0.451 seconds per B candidate and 0.486–0.499 seconds for the learned-prior model variants, including per-cell model loading and export. Each variant uses 627,200 motion-network forwards; a learned-prior variant additionally uses 313,600 prior forwards.

Chapter 8

Discussion and conclusions

8.1 Complete journeys as the unit of synthesis

The synthesis system models a journey through route, duration and progress. Given a road network, an OD request, a participant class and public service constraints, it returns timestamped coordinates. Learned route preferences describe which feasible alternatives participants use; the duration model determines the physical time available; temporal diffusion generates movement along the selected path. This factorization connects route choice and motion within one usable synthetic-data pipeline.

Multiple routes can satisfy the same request. Evaluation therefore compares route-use distributions and movement statistics alongside individual examples. Learned route preferences reduce edge-use divergence relative to shortest-route selection within the available catalog. Uniform route sampling produces greater route diversity and lower speed discrepancy in that experiment, but substantially higher edge-use divergence. These results expose a practical tradeoff between route variety and agreement with observed preferences.

Physical timing is equally consequential. The temporal pilot reduces native acceleration discrepancy by 81–82% when progress is modeled on the quarter-second clock, while longest-stop discrepancy increases. The native representation is subsequently used for complete-journey generation. The five fixed examples illustrate the remaining effect of duration: different motion models can produce similar progress patterns because they share the sampled route and travel time. In the bicycle example, the shorter sampled duration shifts all generated speed profiles upward. This common error originates upstream of the GAF motion condition.

8.2 From trajectory recognition to generative conditioning

The GAF study builds on spatial trajectory classification [1, 2, 12] through an explicit generative interface. Ordered Hilbert images retain spatial organization, while speed images summarize movement over the journey. Both classifiers are trained in this project. Native-demand data preserve the associations between participant class, route choice and service structure that were weakened by randomized class–route assignment. On the expanded development population, the spatial and speed classifiers reach 76.47% and 90.34% macro F1.

The generation model uses image-derived codes throughout denoising. Joint adaptation couples the encoders to the trajectory objective, and a learned prior supplies codes for out-of-sample requests. This follows the encoder–conditional decoder–prior construction of diffusion autoencoders [53], adapted here to road-supported variable-length motion. Code perturbations change denoising predictions, confirming an active connection between the representation and the generator.

The physical comparison measures the effect of that connection. Joint spatial–speed conditioning increases aggregate native acceleration distance by 2.44%, with increases in all four OD-cohort means. Single-view conditions and controls remain close to the baseline, and the private-car improvement under the joint condition accompanies adverse changes in the other four classes. Accordingly, the baseline provides the stronger aggregate physical result in this comparison. Similar diversity across conditions indicates that the differences arise without a large change in the measured within-request spread.

Downstream utility shows a different ordering. Spatial-only augmentation yields the highest pooled macro F1, 96.95%, compared with 96.26% for baseline augmentation and 96.66% for raw-series conditioning. Its episode-equal advantage over baseline augmentation is -0.755 percentage points in one evaluation OD cohort and $+2.091$ in the other. All synthetic-only regimes score below observed-only training. The generated data are therefore most useful as a supplement in the studied task, with a cohort-dependent benefit from spatial conditioning.

8.3 Interpreting the representation effect

The two GAF views are deterministic transformations of trajectory measurements. They organize existing information into a form accessible to image encoders. Class recognition can exploit broad speed scale, route identity and service regularity. The motion generator already receives class, route geometry, length and duration,

so part of this class-related information is shared with its existing inputs. The near-baseline class-centroid result and the small improvements from same-class wrong-trip codes are consistent with such overlap.

Temporal resolution provides a second explanation. Each view is compressed into a 32-dimensional global code. The speed image further summarizes one-second movement over 128 relative journey positions. The sensitivity experiment shows that within-second perturbations can change native acceleration substantially while barely changing speed-image features. Thus the representation emphasizes whole-trip structure more strongly than some local dynamics assessed by the physical metrics.

A third factor is the source of the code. During fitting, the encoder observes a training target; during test-time generation, the prior samples a code from the request conditions. Assigned J-SV codes produce slightly lower training-side denoising error than sampled, exchanged or zero codes. This gap identifies a difference between target-derived and sampled information. Together, information overlap, temporal compression and conditional sampling explain plausible constraints on this interface; the experiments measure their signatures rather than isolate a single cause.

8.4 Evaluation and scope

Trajectory synthesis requires several complementary measurements. Route statistics assess alternatives between endpoints; native and coarse-clock movement summaries assess dynamics; diversity characterizes variation; downstream classification measures data utility. Classifier confidence describes a separate property. Calibration research and proper scoring rules clarify the distinction between predictive probabilities and task-specific rewards [48, 52]. Even accurate class recognition can favor a typical class pattern while overlooking dynamics within that class. This motivates the physical and downstream evaluations used here.

The evidence concerns five participant classes on one SUMO road network with shared bus services. Four OD cohorts provide two departure windows each; the downstream experiment uses two of those cohorts. The temporal pilot has a smaller population and a common update budget rather than equal runtime. The downstream learner uses 1,000 observed source vehicles, whereas the generators use the larger labeled corpus. These design choices define the populations and information budgets of the reported comparisons.

Road adherence is enforced by the path decoder. The system models individual journeys without joint traffic interactions, lateral behavior or evolving signal-state conditioning. Generalization to field GPS data, new cities and unseen services remains outside the evaluated scope. The representation controls also have distinct meanings: joint adaptation combines denoising and class losses without a separate class-loss-only variant; raw-series encoders differ in architecture and pretraining;

same-class shuffling retains approximate trip scale. These conditions qualify the attribution of differences among interfaces.

The route policy operates over a finite catalog, and nearest-shape analysis uses a fixed training-profile library. These choices define the spatial support and similarity reference of the evaluation. Candidate-selection illustrations and joint-conditioning examples retain their respective model populations and request protocols.

8.5 Conclusion

This thesis delivers a road-conditioned diffusion framework for complete multi-class traffic trajectories. Native physical timing and learned route preferences connect movement generation to the road network and observed travel behavior. The resulting synthetic journeys support both physical evaluation and downstream data augmentation.

The GAF contribution is a systematic transfer of spatial and speed image representations from participant recognition into generative conditioning. Both views carry class information, their codes affect denoising, and spatial conditioning produces the highest pooled augmentation score in the studied task. Joint conditioning offers no aggregate physical advantage over the baseline. The resulting account connects representation, generation and data utility through measured outcomes, and identifies the role of temporal detail and already available conditions in that transfer.

Appendix A

Complete numerical results and fixed cases

A.1 Complete Physical Results and Class-Specific Coverage

The following tables retain all eight conditions, both fitted seeds and all five participant classes from the completed out-of-sample comparison. The K1 tables concern the first direct draw; the all-four comparison weights the four draws equally inside each request. The ten physical summaries appear in their original units, without a cross-unit composite or a choice of favourable endpoints. Class-marginal distributions and request-paired discrepancies address different questions. Both accommodate alternative realizations of each OD request.

Table A.1. Final coverage by reference episode. Counts are requests, not independent vehicle populations: paired windows reuse source plans and bus templates are shared. Exact means the realised reference route is in the catalog.

Episode	Requested	Generated	Physical pairs	Route pairs	Exact route
400201	100	98	96	96	61
400202	100	98	96	96	61
400203	100	98	95	95	61
400204	100	98	92	92	61
400205	100	98	97	97	63
400206	100	98	90	90	62
400207	100	98	96	96	54
400208	100	98	93	93	53

Table A.2. Native distribution distances for every fitted seed. K1 is the first direct draw; all four gives equal mass to four draws per request. Speed is in m/s and acceleration in m/s^2. Means cover eight episodes and five classes.

Variant / seed	K1 speed	K1 accel.	All four speed	All four accel.
B / 42	1.4726	0.1739	1.3227	0.1699
B / 43	1.4033	0.1781	1.2665	0.1721
F-SV / 42	1.4643	0.1747	1.3149	0.1691
F-SV / 43	1.4065	0.1775	1.2694	0.1729
J-S / 42	1.4713	0.1776	1.3215	0.1731
J-S / 43	1.4069	0.1763	1.2718	0.1706
J-V / 42	1.4663	0.1799	1.3191	0.1735
J-V / 43	1.4085	0.1785	1.2692	0.1737
J-SV / 42	1.4669	0.1796	1.3235	0.1708
J-SV / 43	1.4133	0.1810	1.2725	0.1750
J-RAW / 42	1.4691	0.1780	1.3297	0.1707
J-RAW / 43	1.4183	0.1833	1.2725	0.1753
J-SH / 42	1.4703	0.1729	1.3211	0.1679
J-SH / 43	1.3987	0.1761	1.2647	0.1717
C / 42	1.4716	0.1740	1.3239	0.1699
C / 43	1.4054	0.1775	1.2710	0.1708

Table A.3. Native speed Wasserstein distances (m/s) by class. Means of eight episodes and two fitted seeds, K1 direct draws. The five classes are displayed individually, without ranking or omission.

Variant	Bus	Motorcycle	Private car	Bicycle	Taxi
B	0.5230	1.9020	1.9939	0.6824	2.0884
F-SV	0.5257	1.9068	1.9684	0.6844	2.0918
J-S	0.5311	1.8983	1.9956	0.6825	2.0878
J-V	0.5243	1.9128	1.9773	0.6864	2.0863
J-SV	0.5231	1.9197	1.9720	0.6996	2.0862
J-RAW	0.5464	1.8783	2.0158	0.6792	2.0987
J-SH	0.5114	1.9024	1.9877	0.6822	2.0887
C	0.5210	1.8962	2.0034	0.6822	2.0897

Table A.4. Native acceleration Wasserstein distances (m/s^2) by class. Means of eight episodes and two fitted seeds, K1 direct draws. The five classes are displayed individually, without ranking or omission.

Variant	Bus	Motorcycle	Private car	Bicycle	Taxi
B	0.0762	0.3249	0.1403	0.1466	0.1920
F-SV	0.0780	0.3193	0.1382	0.1449	0.2003
J-S	0.0765	0.3252	0.1442	0.1465	0.1924
J-V	0.0758	0.3229	0.1388	0.1542	0.2042
J-SV	0.0775	0.3313	0.1357	0.1549	0.2021
J-RAW	0.0771	0.3416	0.1385	0.1481	0.1981
J-SH	0.0750	0.3218	0.1385	0.1440	0.1933
C	0.0766	0.3251	0.1408	0.1438	0.1926

Table A.5. Class-marginal Wasserstein distances, part 1. K1 direct draws, averaged equally over classes, model seeds, windows and OD cohorts. All model variants remain; lower is closer to the reference. Units belong to each column.

Variant	Duration (s)	Distance (m)	Mean speed (m/s)	Speed p95 (m/s)	Stopped fraction
B	178.6336	61.0779	1.4848	1.3994	0.0895
F-SV	178.6336	61.0655	1.4848	1.3609	0.0893
J-S	178.6336	61.0930	1.4848	1.3794	0.0892
J-V	178.6336	61.1096	1.4847	1.3803	0.0902
J-SV	178.6336	61.0818	1.4847	1.4118	0.0892
J-RAW	178.6336	61.0768	1.4848	1.4223	0.0894
J-SH	178.6336	61.1025	1.4848	1.3816	0.0894
C	178.6336	61.0932	1.4848	1.3829	0.0893

Table A.6. Class-marginal Wasserstein distances, part 2. K1 direct draws, averaged equally over classes, model seeds, windows and OD cohorts. All model variants remain; lower is closer to the reference. Units belong to each column.

Variant	Mean $\lvert a \rvert$ (m/s^2)	Longest stop (s)	Stops per min	Longest steady (s)	Four-second change (m/s)
B	0.0793	28.9814	0.2488	20.3933	0.2792
F-SV	0.0811	29.2932	0.2514	20.4734	0.2840
J-S	0.0792	29.0212	0.2490	20.4436	0.2792
J-V	0.0809	29.5223	0.2430	20.0124	0.2815
J-SV	0.0820	29.4228	0.2449	19.9324	0.2838
J-RAW	0.0796	29.3519	0.2563	20.0474	0.2768
J-SH	0.0795	28.9978	0.2487	20.3505	0.2792
C	0.0804	29.0463	0.2463	20.4462	0.2821

Table A.7. Request-paired mean absolute discrepancies, part 1. K1 direct draws, averaged equally over classes, model seeds, windows and OD cohorts. All model variants remain; lower is closer to the reference. Units belong to each column.

Variant	Duration (s)	Distance (m)	Mean speed (m/s)	Speed p95 (m/s)	Stopped fraction
B	284.7414	65.7098	2.7417	3.2119	0.1504
F-SV	284.7414	65.7063	2.7416	3.1937	0.1504
J-S	284.7414	65.7248	2.7417	3.2380	0.1501
J-V	284.7414	65.7426	2.7416	3.1704	0.1524
J-SV	284.7414	65.7058	2.7415	3.1551	0.1516
J-RAW	284.7414	65.7200	2.7416	3.1859	0.1512
J-SH	284.7414	65.7248	2.7417	3.2207	0.1509
C	284.7414	65.7217	2.7417	3.2293	0.1506

Table A.8. Request-paired mean absolute discrepancies, part 2. K1 direct draws, averaged equally over classes, model seeds, windows and OD cohorts. All model variants remain; lower is closer to the reference. Units belong to each column.

Variant	Mean $\lvert a \rvert$ (m/s^2)	Longest stop (s)	Stops per min	Longest steady (s)	Four-second change (m/s)
B	0.1614	41.2845	0.4349	36.2157	0.5419
F-SV	0.1629	41.6194	0.4364	36.1815	0.5464
J-S	0.1612	41.3469	0.4347	36.1413	0.5412
J-V	0.1635	42.0044	0.4328	36.3024	0.5491
J-SV	0.1661	41.9202	0.4340	36.0281	0.5577
J-RAW	0.1605	41.7909	0.4328	35.5502	0.5404
J-SH	0.1625	41.4936	0.4362	36.2181	0.5462
C	0.1625	41.4257	0.4331	36.1700	0.5447

Table A.9. Progress-shape diversity and decoder accounting. Means weight classes and scene/seed cells equally. Near-shape is the fraction within 10^{-4} of a fixed, at-most-512-per-class training library; it is not a privacy guarantee. Negative and upper are pre-correction decoder-bin fractions.

Variant	Pairwise shape	Nearest shape	Near-shape rate	Negative	Upper
B	0.0563	0.0163	0.0000	0.1164	0.0000
F-SV	0.0561	0.0163	0.0000	0.1179	0.0000
J-S	0.0563	0.0164	0.0000	0.1177	0.0000
J-V	0.0563	0.0163	0.0000	0.1217	0.0000
J-SV	0.0561	0.0163	0.0000	0.1245	0.0000
J-RAW	0.0557	0.0162	0.0000	0.1244	0.0000
J-SH	0.0563	0.0164	0.0000	0.1152	0.0000
C	0.0564	0.0164	0.0000	0.1172	0.0000

Per-cell support, all individual class results, coverage and provenance are available in *the accompanying reproducibility archive*. Missing endpoints retain their support counts. The nearest-shape diagnostic uses a fixed hash-ranked library of at most 512 training examples per class and normalized relative-time progress, not a search over every training coordinate sequence. The diagnostic characterizes similarity to this fixed reference library. Decoder clipping quantifies numerical correction of the generated representation.

A.2 Condition perturbations and utility cohorts

Table A.10. All-variant condition perturbations, part 1. Noise-prediction MSE on the fixed training-vehicle audit; means over source vehicles and seeds.

Variant	Index	Assigned	Swap	Zero	Prior
B	50	0.144968	0.144968	0.144968	0.144968
B	250	0.032536	0.032536	0.032536	0.032536
B	500	0.016942	0.016942	0.016942	0.016942
B	750	0.003552	0.003552	0.003552	0.003552
B	950	0.000204	0.000204	0.000204	0.000204
F-SV	50	0.143556	0.142638	0.143871	0.143524
F-SV	250	0.032525	0.032563	0.032634	0.032574
F-SV	500	0.017010	0.016893	0.016955	0.016949
F-SV	750	0.003534	0.003544	0.003547	0.003555
F-SV	950	0.000203	0.000201	0.000206	0.000205
J-S	50	0.143936	0.143285	0.144029	0.143928
J-S	250	0.032478	0.032490	0.032512	0.032471
J-S	500	0.016920	0.016913	0.016927	0.016909
J-S	750	0.003556	0.003553	0.003562	0.003551
J-S	950	0.000202	0.000200	0.000203	0.000202
J-V	50	0.142568	0.142921	0.144263	0.142860
J-V	250	0.032438	0.032623	0.032790	0.032582
J-V	500	0.016950	0.016998	0.017088	0.016956
J-V	750	0.003505	0.003520	0.003579	0.003556
J-V	950	0.000202	0.000203	0.000210	0.000202

Table A.11. All-variant condition perturbations, part 2. Noise-prediction MSE on the fixed training-vehicle audit; means over source vehicles and seeds.

Variant	Index	Assigned	Swap	Zero	Prior
J-SV	50	0.142174	0.142512	0.143795	0.142529
J-SV	250	0.032396	0.032623	0.032764	0.032622
J-SV	500	0.016933	0.017027	0.017050	0.017065
J-SV	750	0.003504	0.003521	0.003573	0.003567
J-SV	950	0.000201	0.000202	0.000208	0.000207
J-RAW	50	0.144829	0.145033	0.146072	0.144945
J-RAW	250	0.032524	0.032630	0.032750	0.032557
J-RAW	500	0.016868	0.017033	0.017061	0.016828
J-RAW	750	0.003544	0.003556	0.003565	0.003558
J-RAW	950	0.000203	0.000205	0.000208	0.000204
J-SH	50	0.143075	0.142956	0.143539	0.143130
J-SH	250	0.032535	0.032546	0.032579	0.032538
J-SH	500	0.016922	0.016915	0.016946	0.016917
J-SH	750	0.003560	0.003559	0.003566	0.003559
J-SH	950	0.000201	0.000202	0.000206	0.000201
C	50	0.145555	0.145555	0.145880	0.145555
C	250	0.032570	0.032570	0.032575	0.032570
C	500	0.016921	0.016921	0.016929	0.016921
C	750	0.003557	0.003557	0.003561	0.003557
C	950	0.000202	0.000202	0.000208	0.000202

Table A.12. Utility by evaluation OD cohort. Entries average episode macro F1 over both windows and both learner seeds (percent); they are not the pooled main score. All fifteen regimes are retained.

Training data	Cohort 3	Cohort 4
only-B	93.35	92.68
only-F-SV	93.63	91.69
only-J-RAW	92.81	92.95
only-J-S	92.85	92.93
only-J-SV	93.38	93.47
only-J-V	92.54	92.70
only-empirical	92.37	94.24
plus-B	97.50	95.06
plus-F-SV	97.31	93.73
plus-J-RAW	97.78	95.56
plus-J-S	96.75	97.15
plus-J-SV	96.70	94.30
plus-J-V	97.25	93.53
plus-empirical	95.17	93.23
real	95.15	93.72

A.3 Temporal-Resolution Pilot Study: all physical summaries

Tables A.13 and A.14 retain all ten predefined physical summaries for all fourteen completed evaluations. They supplement the representative measures in the main text without selecting columns by the direction of an effect. Every entry is a five-class macro Wasserstein distance, with lower values indicating closer distributions. Each quantity retains its own unit; the columns are not combined into an overall score.

Every row begins with the same 100 out-of-sample requests and uses the same 94 paired requests for these quality measures: 19 buses, 18 motorcycles, and 19 participants in each remaining class. Map construction supports 97 requests and reference extraction accepts 97, but their supported sets differ. Each full-pool row contains 776 generated trajectories, of which 752 belong to the paired requests (94×8). Each terminal-selection row contains 97 delivered trajectories, of which 94 have accepted references. These denominators distinguish generation coverage from quality support; none of the six unsupported requests is replaced. The eight selection rows reuse candidates, so their outputs are not additional independent generations.

In the tables, "257" and "Native" denote the relative and quarter-second diffusion models, and "Empirical" denotes whole-motion transfer from training journeys. N42 and N43 identify the native models; U, S, V and SV denote Uniform, spatial, speed and joint terminal selection. Seed numbers refer to fitted diffusion models and their matched empirical draws, not independent reference episodes. All columns have full five-class support in this completed cohort. The figures and tables remain descriptive for one new simulator episode.

Table A.13. Complete macro physical distances, part I: trip scale and one-second motion summaries. Every row has 94 paired requests out of the original 100.

Model / seed	Duration (s)	Distance (m)	Longest steady (s)	Four-second change (m/s)	Mean speed (m/s)
257 / 42	48.2841	4.6610	20.4234	0.2330	0.4601
257 / 43	46.3904	4.6817	19.2598	0.2199	0.6682
Native / 42	48.2841	4.6980	21.5398	0.2379	0.4602
Native / 43	46.3904	4.7070	20.4322	0.1935	0.6682
Empirical / 42	48.2841	4.6858	22.5400	0.5810	0.4599
Empirical / 43	46.3904	4.8039	20.7270	0.5953	0.6685
N42 / U	57.7782	4.5722	20.5561	0.2155	0.6082
N42 / S	58.4039	4.5503	25.4848	0.3205	0.3985
N42 / V	74.4163	4.5190	25.3082	0.2722	0.9320
N42 / SV	59.0127	4.6159	24.1906	0.2950	0.6988
N43 / U	51.5908	4.5515	21.6164	0.2416	0.3590
N43 / S	51.4898	4.6036	19.1246	0.1354	0.7402
N43 / V	77.6837	4.5177	30.7719	0.2088	0.9148
N43 / SV	68.0578	4.4888	23.0170	0.2141	0.8416

Table A.14. Complete macro physical distances, part II: acceleration, stopped motion and upper speed. Stopped-fraction distances use fractions, not percentage points. Every row has 94 paired requests out of the original 100.

Model / seed	Mean $\lvert a \rvert$ (m/s^2)	Stopped fraction	Speed p95 (m/s)	Longest stop (s)	Stop episodes per minute
257 / 42	0.1129	0.0278	1.8650	4.6934	0.1111
257 / 43	0.1131	0.0268	1.8252	4.8932	0.1184
Native / 42	0.0661	0.0179	1.9802	7.7539	0.1000
Native / 43	0.0560	0.0171	1.9495	7.6598	0.0939
Empirical / 42	0.1970	0.0156	2.7377	4.7270	0.1265
Empirical / 43	0.2059	0.0128	2.4870	4.1470	0.1236
N42 / U	0.0547	0.0221	1.7914	9.2801	0.1185
N42 / S	0.0870	0.0167	1.7201	7.6515	0.1248
N42 / V	0.0700	0.0203	1.8115	7.7146	0.1312
N42 / SV	0.0759	0.0169	1.6402	8.9170	0.1100
N43 / U	0.0693	0.0163	1.5543	8.1813	0.0880
N43 / S	0.0432	0.0159	1.4687	7.9047	0.1039
N43 / V	0.0563	0.0234	1.9131	9.6322	0.1208
N43 / SV	0.0561	0.0198	1.7139	10.7994	0.1169

Duration and distance retain the complete observed span. The remaining summaries use complete one-second intervals; for example, "Four-second change" denotes the mean absolute four-second lag difference in speed, not acceleration over a four-second sampling interval. Precise state thresholds and support rules are defined in Chapter 6. All unselected generators share the duration prior, explaining the identical duration distances within a seed. Selection can change the distribution of durations retained from that pool.

A.4 Temporal-Resolution Pilot Study: native-resolution distances by participant class

Table A.15 reports interval-distribution distances for the six full-pool generators. These are computed at the native output resolution, including fractional terminal intervals, and therefore address a different question from the one-second journey summaries. Requests have equal mass within a class; candidate mass is divided within a request, and speed or acceleration samples are then weighted by their physical interval length. Class-specific tables characterize how the aggregate response varies across participants and movement properties.

Table A.15. Class-wise native speed and acceleration W_1 for every full-pool generator. Each cell lists speed (m/s) above acceleration (m/s^2). The 94 paired requests comprise 19 buses, 18 motorcycles and 19 members of each other class; each row has 776 generated and 752 paired trajectories.

Model / seed	Bus	Motorcycle	Private car	Bicycle	Taxi / ride-hailing
	1.0800	0.5157	0.9632	0.2570	0.5812
257 / 42	0.6174	1.4987	0.6751	0.8113	0.5961
	0.8305	0.5742	1.8238	0.2460	0.6445
257 / 43	0.5955	1.5061	0.6588	0.7934	0.6055
	0.7540	0.6876	0.8473	0.2243	0.5334
Native / 42	0.0837	0.1833	0.1280	0.1632	0.1879
	0.6947	0.5444	1.6986	0.2193	0.5615
Native / 43	0.0746	0.2199	0.1250	0.1563	0.2117
	1.4309	0.7069	1.2088	0.2879	0.5521
Empirical / 42	0.4435	0.4065	0.1947	0.2379	0.3401
	1.2757	0.6626	2.0861	0.2634	0.5578
Empirical / 43	0.4483	0.4308	0.3134	0.2502	0.3513

A.5 Fixed-Duration Control Study: Complete Results

Tables A.16 and A.17 report all ten predefined physical summaries for the fixed-duration control study's two full pools and eight terminal selections. Table A.18 gives the corresponding native speed and acceleration distances for every class. N42 and N43 denote the same native-clock models used in the temporal-resolution pilot study; "all" denotes a full pool, and U, S, V and SV retain their earlier definitions. Every row has 94 paired requests out of the original 100. A full pool contains 776 generated trajectories and 752 paired trajectories; a selected set contains 97 and 94, respectively. The class counts remain 19 buses, 18 motorcycles and 19 members of each other class. All reported metrics have that support.

The fixed-duration control study fixes one training-prior duration per request and model seed before scoring the shared candidate pool. Its reference episode is the already analyzed the temporal-resolution pilot study episode, not a new blind sample. The following tables use the completed attempt2 summary and retain each metric's units; all distances are lower-is-better. Values are rounded to four decimal places for presentation.

Table A.16. Complete the fixed-duration control study macro physical distances, part I: trip scale and one-second motion summaries. Each row covers 94 paired requests out of 100.

Model / policy	Duration (s)	Distance (m)	Longest steady (s)	Four-second change (m/s)	Mean speed (m/s)
N42 / all	57.7782	4.5295	20.7243	0.2212	0.6080
N42 / U	57.7782	4.5722	20.5561	0.2155	0.6082
N42 / S	57.7782	4.4864	22.4883	0.3001	0.6079
N42 / V	57.7782	4.6340	23.5357	0.2675	0.6069
N42 / SV	57.7782	4.5187	22.7556	0.3144	0.6076
N43 / all	51.5908	4.4876	21.1753	0.1956	0.3586
N43 / U	51.5908	4.5515	21.6164	0.2416	0.3590
N43 / S	51.5908	4.4406	19.5263	0.2063	0.3586
N43 / V	51.5908	4.4136	23.1175	0.1998	0.3577
N43 / SV	51.5908	4.4905	22.0246	0.2114	0.3587

Table A.17. Complete the fixed-duration control study macro physical distances, part II: acceleration, stopped motion and upper speed. Stopped-fraction distances are fractions, not percentage points. Each row covers 94 paired requests out of 100.

Model / policy	Mean $\|a\|$ (m/s^2)	Stopped fraction	Speed p95 (m/s)	Longest stop (s)	Stop episodes per minute
N42 / all	0.0591	0.0180	1.9190	8.3425	0.1059
N42 / U	0.0547	0.0221	1.7914	9.2801	0.1185
N42 / S	0.0868	0.0201	2.0436	9.2865	0.1142
N42 / V	0.0710	0.0204	2.0278	7.1930	0.1125
N42 / SV	0.0896	0.0185	1.9874	8.7971	0.1079
N43 / all	0.0577	0.0157	1.7418	8.0993	0.0901
N43 / U	0.0693	0.0163	1.5543	8.1813	0.0880
N43 / S	0.0553	0.0172	1.7112	6.4591	0.0953
N43 / V	0.0562	0.0158	1.8483	7.6474	0.0861
N43 / SV	0.0604	0.0165	2.1182	7.2901	0.0883

Table A.18. Complete class-wise the fixed-duration control study native speed and acceleration W_1. Each cell lists speed (m/s) above acceleration (m/s^2), using the same 94-request paired population as the macro tables. Both full pools and all terminal policies are included.

Model / policy	Bus	Motorcycle	Private car	Bicycle	Taxi / ride-hailing
	0.7412	0.7595	0.7589	0.3107	0.9118
N42 / all	0.0778	0.1877	0.1208	0.1434	0.1945
	0.7129	0.9390	0.7597	0.3934	0.8594
N42 / U	0.0745	0.2073	0.1217	0.0872	0.1316
	0.7579	1.1450	0.8358	0.4140	0.9080
N42 / S	0.0890	0.1356	0.1257	0.1852	0.1502
	0.8729	0.7472	0.7720	0.4152	0.9175
N42 / V	0.0797	0.1405	0.1270	0.1019	0.1128
	0.9038	1.1317	0.6814	0.4000	0.8677
N42 / SV	0.0927	0.1348	0.1235	0.2062	0.1578
	0.6245	0.7224	0.6906	0.2492	0.4653
N43 / all	0.0800	0.2065	0.1103	0.1778	0.1939
	0.5770	0.6701	0.6441	0.2998	0.7232
N43 / U	0.1049	0.1968	0.1223	0.1678	0.2067
	0.6618	0.7591	0.6005	0.2941	0.3924
N43 / S	0.0742	0.2525	0.1146	0.1688	0.2150
	0.5973	0.7557	0.5623	0.2950	0.4191
N43 / V	0.0878	0.2119	0.1125	0.1432	0.1057
	0.6200	0.9857	0.7289	0.2790	0.3258
N43 / SV	0.0828	0.1880	0.1112	0.1660	0.1578

A.6 Five the temporal-resolution pilot study cases selected before visual inspection

For the temporal-resolution pilot study model seed 42, the case for each class is the lexicographically first request with an accepted reference and outputs from all four terminal policies. This rule is based on request order and common support, not physical quality, teacher confidence or visual appearance. The five plots illustrate the same declared conditions and the consequences of selecting different candidates; they are not five independent tests of a method advantage.

Each figure shows coordinate geometry, relative-time progress and a physical-time speed curve. Speed is derived from cumulative planar distance over complete one-second intervals, with the final partial second omitted from that panel. The one-second curves emphasize longer movement patterns, complementing the native quarter-second metrics. Different selected durations are allowed in the temporal-resolution pilot study. The shared route produces near-coincident map traces, while temporal curves distinguish movement along it.

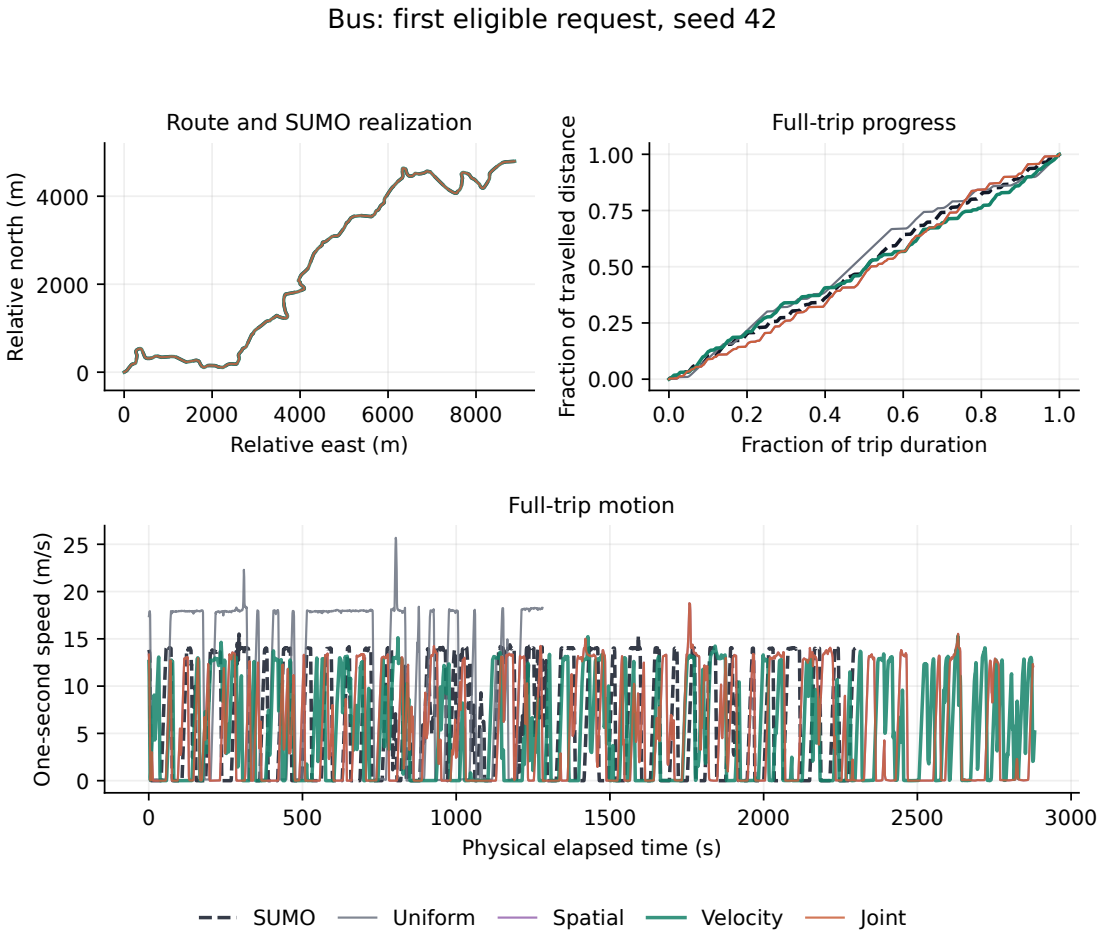

Figure A.1. Bus. Fixed seed-42 case: the first request by identifier order with a reference and all four policy outputs. Speed summarizes complete one-second intervals and omits the final partial second; it is not the native quarter-second signal.

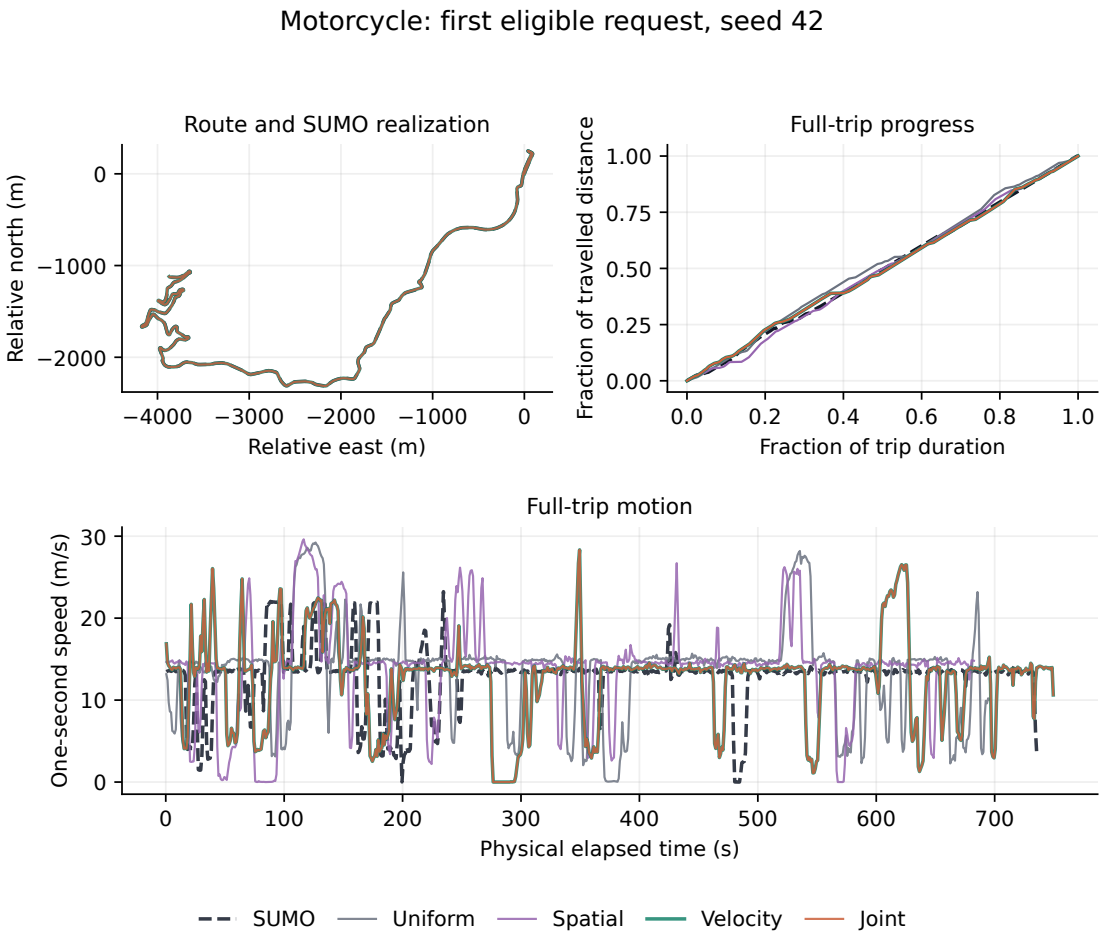

Figure A.2. Motorcycle. Fixed seed-42 case: the first request by identifier order with a reference and all four policy outputs. Speed summarizes complete one-second intervals and omits the final partial second; it is not the native quarter-second signal.

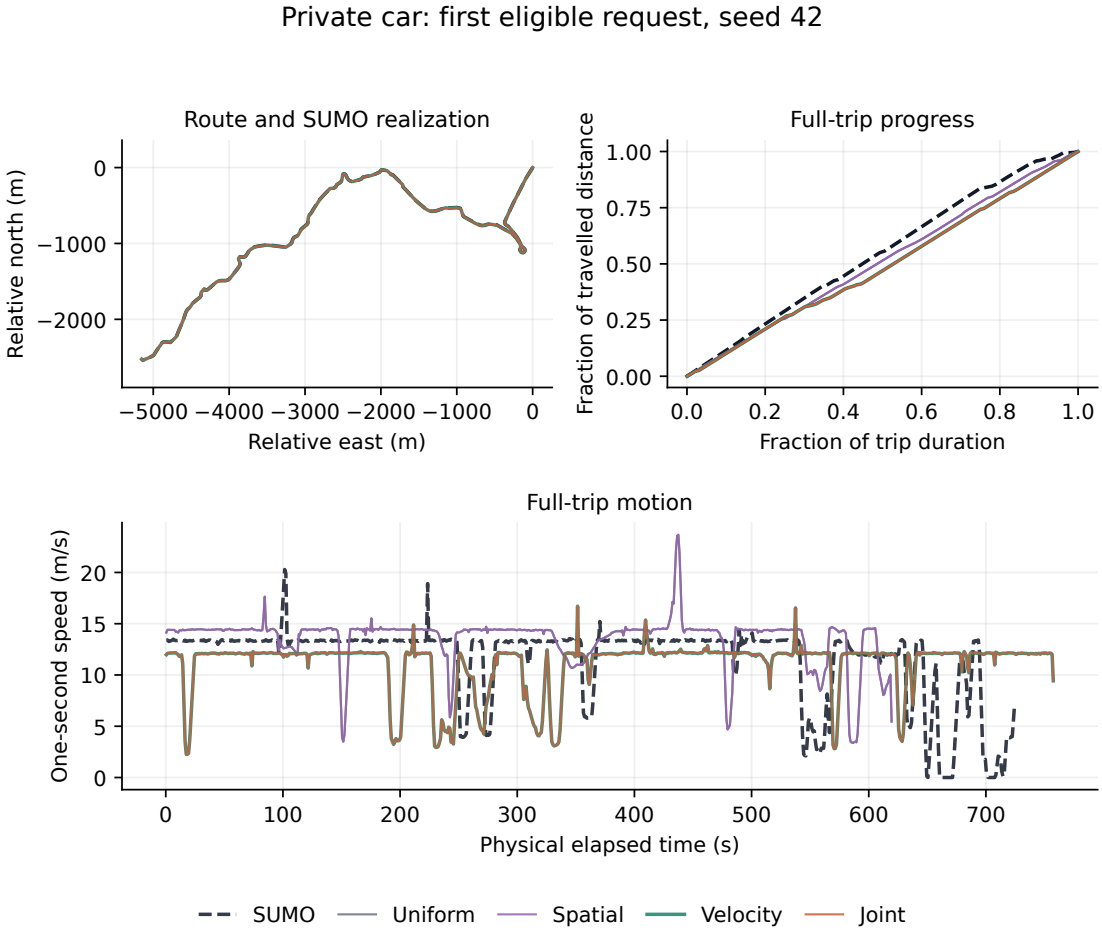

Figure A.3. Private car. Fixed seed-42 case: the first request by identifier order with a reference and all four policy outputs. Speed summarizes complete one-second intervals and omits the final partial second; it is not the native quarter-second signal.

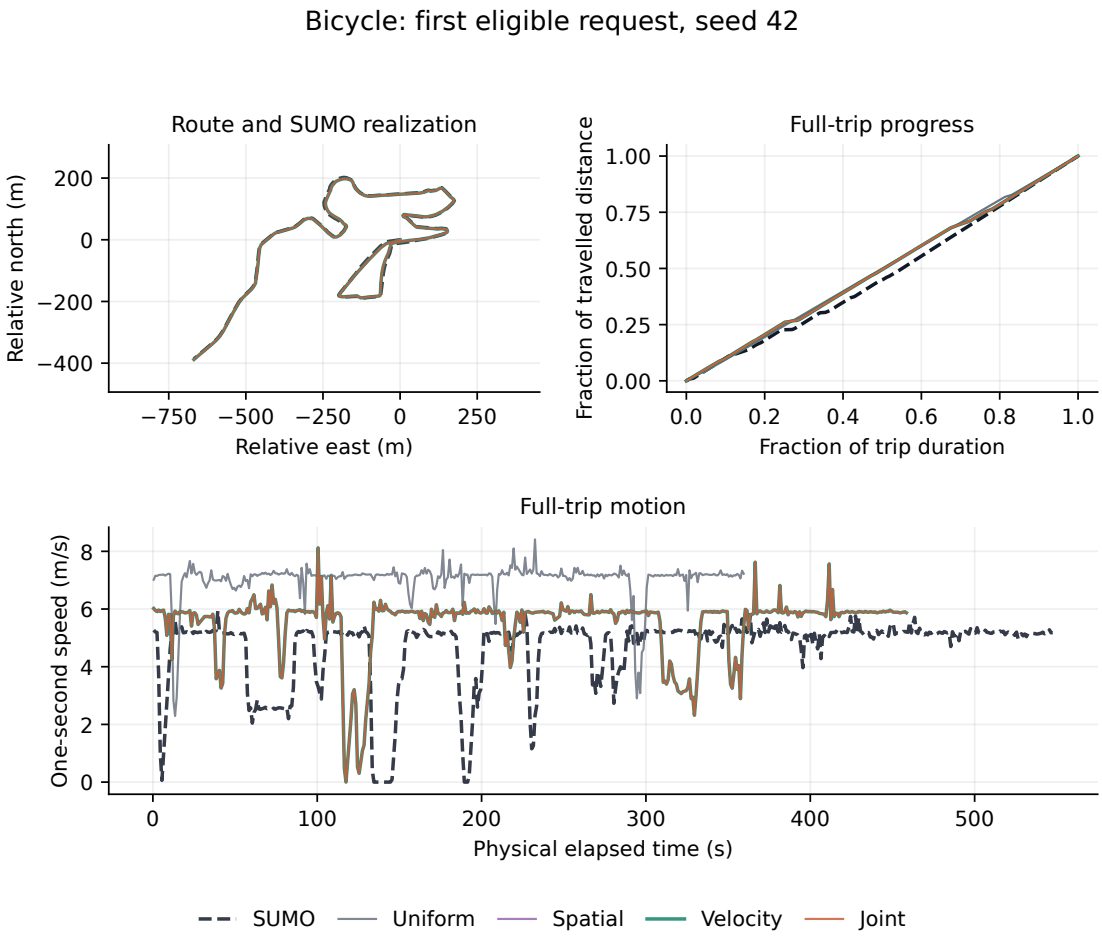

Figure A.4. Bicycle. Fixed seed-42 case: the first request by identifier order with a reference and all four policy outputs. Speed summarizes complete one-second intervals and omits the final partial second; it is not the native quarter-second signal.

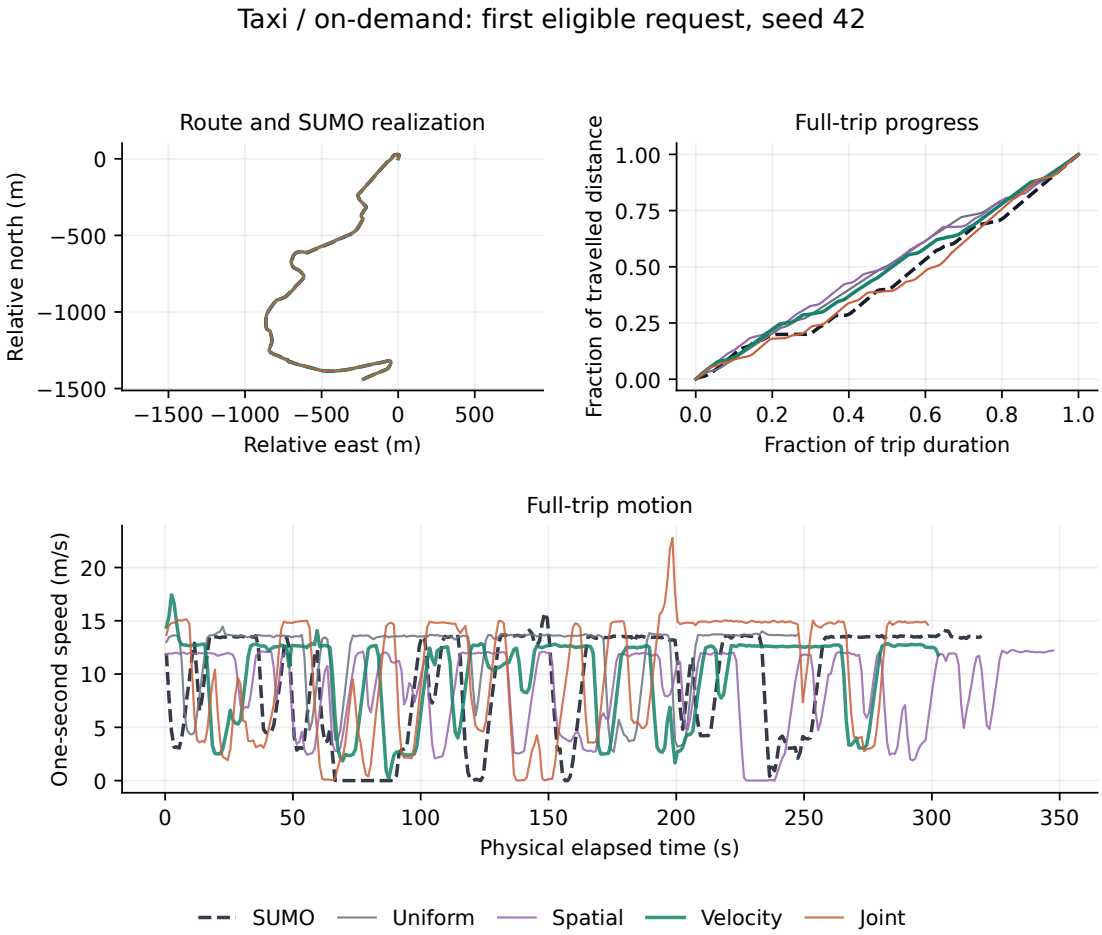

Figure A.5. Taxi or ride-hailing vehicle. Fixed seed-42 case: the first request by identifier order with a reference and all four policy outputs. Speed summarizes complete one-second intervals and omits the final partial second; it is not the native quarter-second signal.

A.7 Historical evidence and reproduction scope

The earlier exploratory evaluation is preserved in the repository, including its unfavorable feedback and augmentation findings. It began with 270 held-out participant IDs, of which 53 satisfied the strict moving-window generation eligibility rules. Its 512-point task, reference population and evaluation protocol differ from the present complete-journey experiment. The 53 eligible requests constitute a separate evaluation subset of the original 270 and are reported under that study's protocol. The original report remains at *the accompanying reproducibility archive*.

The earlier paired-representation evaluation is also documented in the accompanying reproducibility archive. It trained spatial and speed classifiers for matching participant labels and tested latent supervision in an earlier eight-class simulator corpus. Its planned comparisons did not establish stable independent physical gains from paired spatial–speed supervision. That result characterizes the representation interface on its own corpus and remains separate from the complete-journey experiments. The present thesis distinguishes classification information, the mechanism used to transmit it, and physical generation quality across these stages.

The accompanying reproducibility archive contains the experimental protocols, complete numerical reports, model configurations and fixed-example selection rules. Separate training, generation, evaluation and presentation programs reproduce the temporal-resolution and fixed-duration studies using versioned scenario data and saved model artifacts. This organization links the supplementary results to their computational provenance.

The original locked-result snapshot remains separately identified in the experiment repository [13]; its test population is not reused to select the present models.

A.8 Expanded recognition on a common development population

The full-day expansion completed all four 10,000-step diffusion fits and both fixed 50-epoch GAF classifiers. The native models took 2,771.9 and 2,750.8 seconds, approximately 46 minutes each; the 257-point controls took approximately 319 seconds each. Model capacity remains 891,969 parameters. The larger corpus and longer native sequences change data exposure and computational cost, so equal update counts alone do not imply equal computation. No new 257-point generation-quality claim follows merely from completing those fits.

The paired classifier comparison is stronger evidence than comparing the pilot's small development set with the expanded set. Both old and new checkpoints see the same 7,230 complete development legs, and the original training populations remain separate from that cohort. Table A.19 and Figure A.6 show a substantial recognition improvement in both representations.

Table A.19. Fixed old and expanded classifiers on the same 7,230 development legs. Percentages; no checkpoint selection.

Reader	Accuracy	Macro F1	Taxi F1
Old Spatial	60.65	50.10	19.07
New Spatial	86.17	76.47	45.85
Old Speed	63.61	61.82	30.61
New Speed	89.13	90.34	81.07

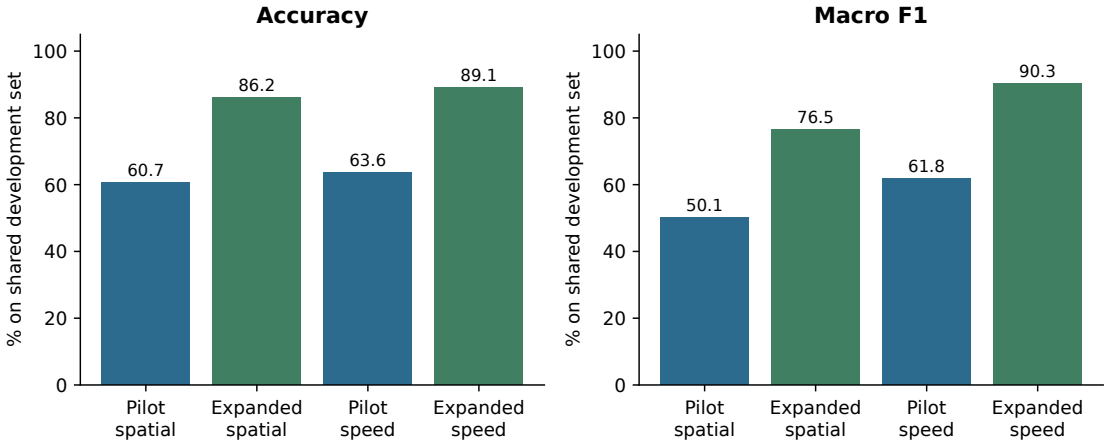

Figure A.6. Pilot and expanded classifiers evaluated on identical development inputs. Old/new refers to training corpus and optimization exposure, not different evaluation sets. Higher is better.

Spatial macro F1 increases by 26.37 percentage points, with a paired source vehicle-bootstrap 95% interval of 24.54–28.16 points. Velocity macro F1 increases by 28.52 points, with an interval of 27.27–29.73 points. The intervals characterize variability within the available public-demand dataset and fixed model fits. The improved recognition supports the decision to preserve native demand and expand complete-trip coverage. The comparison combines changes in sample count, temporal coverage and CNN optimization exposure.

The confusion matrices explain why overall accuracy alone would be incomplete. Spatial private-car F1 is 95.20%, but taxi/ride-hailing F1 is 45.85%. The velocity values are 92.23% and 81.07%, respectively. Spatial training accuracy is 97.22%, compared with 86.17% on development; the corresponding velocity figures are 92.39% and 89.13%. Generalization gaps, calibration error and shared route templates remain relevant when transferring these classifiers from simulator observations to generated candidates.

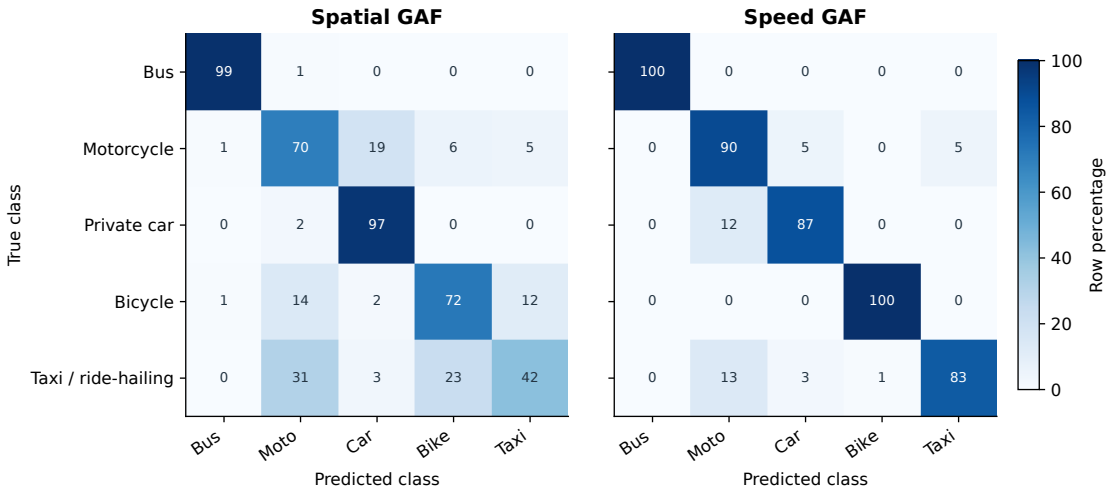

Figure A.7. Pre-adaptation spatial and speed classifiers on the same expanded development cohort. Each row is normalized within its true class. These are the pre-trained classifier checkpoints, not new evaluations of jointly adapted encoders.

A.9 Supporting old/new generator comparison

Table A.20. Expanded versus pilot native generators, full candidate pools: class–macro native W1 changes. Each cell uses its own training-side duration prior.

Scene	Seed	Speed change	Acceleration change
Morning	42	+4.7%	+22.0%
Morning	43	+12.2%	+34.4%
Later	42	-57.2%	-15.7%
Later	43	-65.5%	-5.6%

Expansion changes the generator's applicability across the observed contexts. In the later scene, native-speed distribution error falls by 57.2% and 65.5% for the two model seeds, while acceleration error falls by 15.7% and 5.6%. In the morning scene, speed error instead rises by 4.7% and 12.2%, and acceleration error rises by 22.0% and 34.4%. The signs agree across seeds within each scene, but there are only two simulator realizations and the underlying public plans are shared.

This contrast is consistent with sensitivity to demand and temporal coverage, but it does not isolate that cause. Expanded and pilot models differ in training data and duration prior, and neither receives the instantaneous background traffic state. Departure context and simulator random seed change together between scenes. The result establishes a scene-dependent change in the delivered system, not that the diffusion network has learned an explicit time-of-day controller.

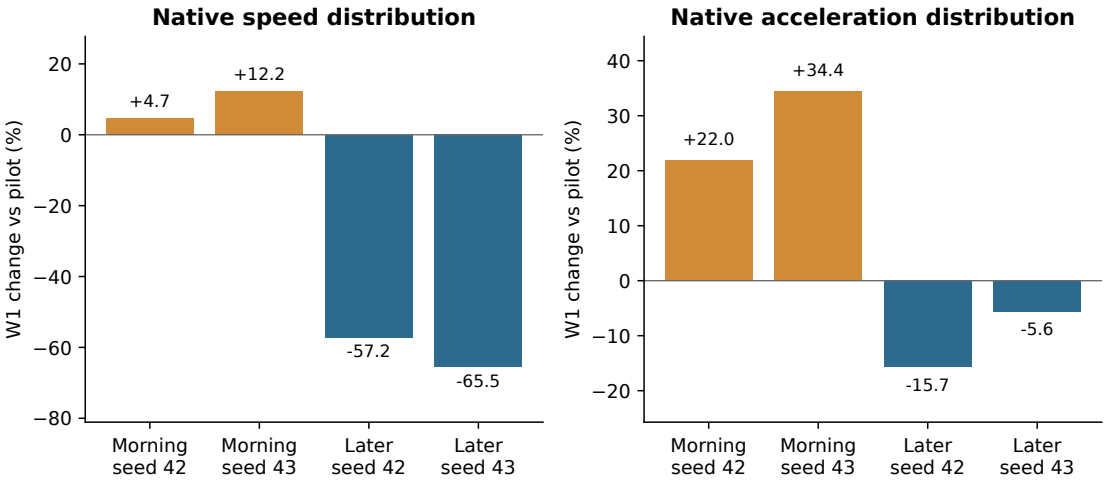

Figure A.8. Expanded versus pilot native-time generation on two new SUMO scenes. Bars show changes in full-pool five-class macro distances; negative is better. Each model uses its own training-side duration prior. The reversal across scenes prevents a universal expansion claim.

A.10 Does a more accurate GAF classifier improve selection?

The central comparison applies old and expanded classifiers to the same expanded-generator candidates. Duration, geometry, candidate noise and the requested class are unchanged by the teacher contrast. Uniform controls are verified identical array by array. The experiment therefore distinguishes improving the generator from improving the criterion used to select its outputs.

The physical ranking varies across metrics despite stronger recognition. For example, the expanded spatial classifier reduces longest-stop request error by 6.4% relative to the pilot spatial classifier on these same pools, but raises stopped-fraction error by 1.1% and stop-frequency error by 2.8%. The expanded velocity classifier reduces mean-absolute-acceleration request error by 2.8% but increases longest-stop error by 4.1% relative to the pilot velocity classifier. These comparisons evaluate a fixed generator and its selectors through independently measured physical properties.

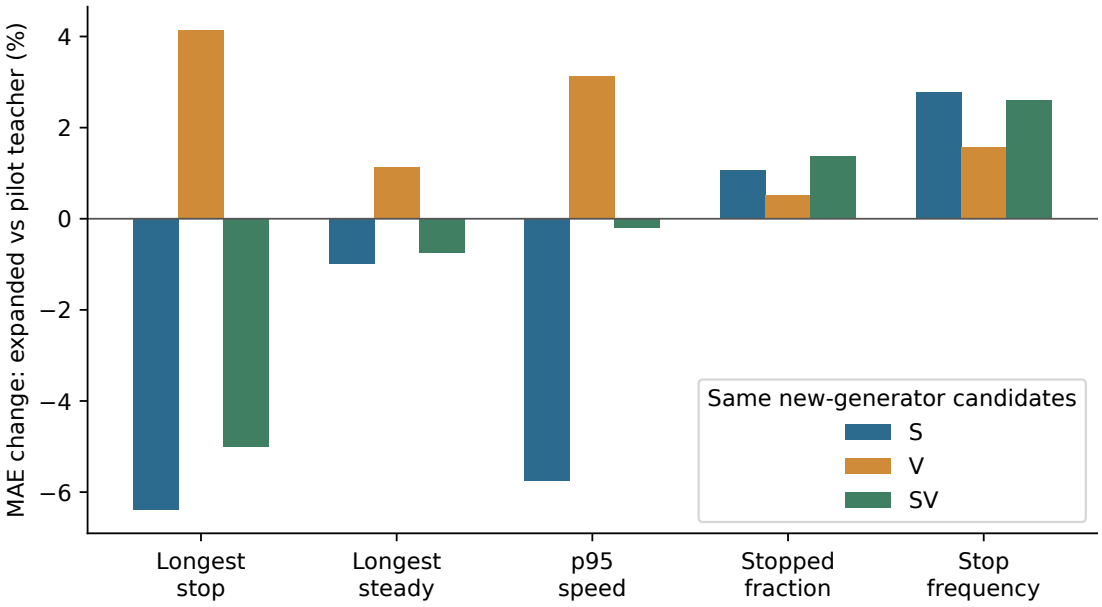

Figure A.9. Expanded versus pilot teachers on identical expanded-generator pools: changes in request-level error for five motion summaries. Negative values favour the expanded classifier. These post-hoc equal-cell summaries expose gains and costs; they are not significance tests or a replacement for the full distributional matrix.

A.11 Exploratory evidence for class-specific stopping feedback

The post-hoc request-level analysis suggests a more precise mechanism to test. Spatial feedback reduces longest-stop MAE in all four scene/seed cells, with an aggregate reduction of 7.6%. This differs from its class-marginal longest-stop W1, which improves in three cells and decreases by 4.6% in the aggregate. The distinction matters: the first statement concerns correspondence to the same request's reference, while the second concerns the distribution of that scalar over requests.

For buses, the spatial longest-stop request error falls by 20.7%, with all four cells improving. For motorcycles it rises by 5.3%, with all four cells worsening. The dual policy's corresponding bus reduction is 21.1%; that small difference from spatial-only feedback is not evidence of representation synergy. These observations motivate class-sensitive stopping analysis, but the metric does not determine whether the longest stop occurs at the correct bus station. It also does not distinguish a service stop from waiting in traffic.

A mechanism claim therefore requires a fresh, fixed comparison of stop location, service-stop coverage and non-service waiting, alongside the complete class matrix.

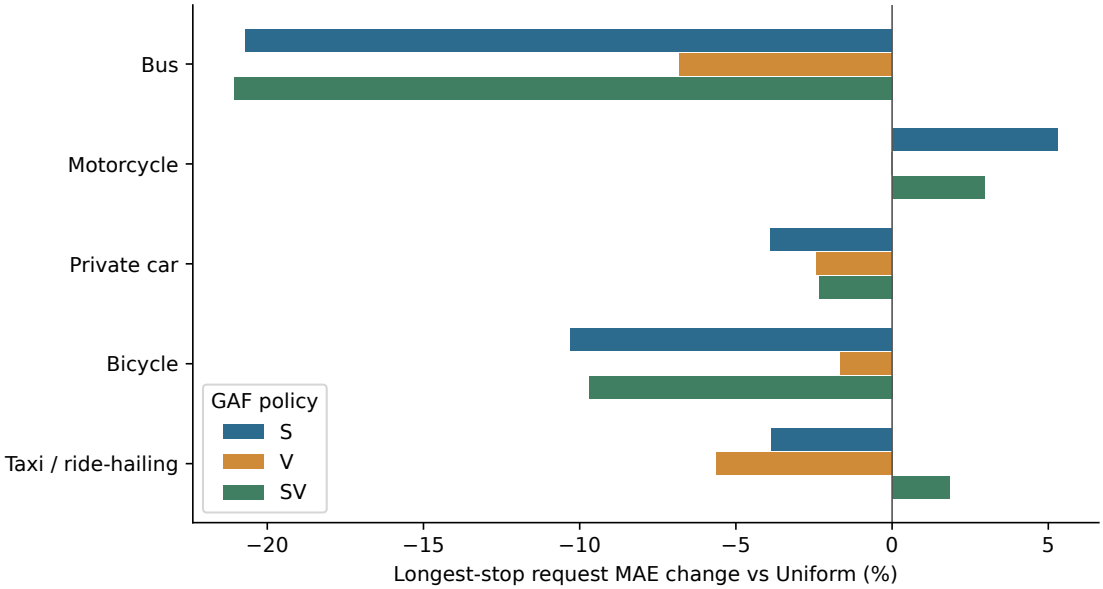

Figure A.10. Class-specific longest-stop request error after expanded GAF selection. The same five classes and all three policies are retained. The bus improvement coexists with a motorcycle cost; percentage reductions across classes are not a single common-unit effect size.

The present finding was identified after inspecting results, and its consistency over two generator seeds cannot supply an independent confirmatory test. It is valuable as a specific link between the spatial trajectory representation and an output motion property, not as proof that the spatial encoding has become a universal realism objective.

Full-pool pair diversity describes the variation available before selection. Conditional pair diversity is evaluated on the full candidate pool, while the selected output contains one trajectory per request. The common 97-request map coverage is unchanged by selection and belongs to the geometric interface.

A.12 Frozen-Feature Supervision and Candidate Selection

The frozen-feature alignment study includes seven model variants, two fitted seeds and eight new reference episodes. Its 800 declared episode requests yield 782 route-supported requests and 754 physical pairs; 470 realised reference routes are present in their catalogs. The dual-view SV variant gives K1 speed/acceleration distances of 1.29337/0.18761 compared with 1.28383/0.18674 for B: increases of 0.743% and 0.466%. The speed-only variant slightly improves primary acceleration,

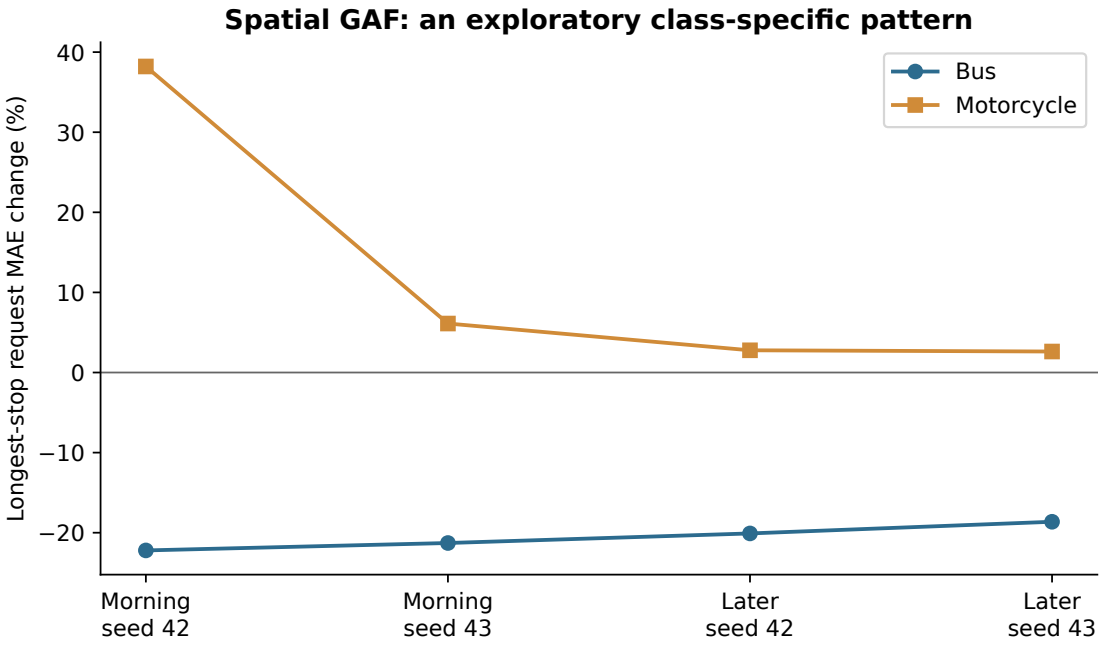

Figure A.11. Expanded spatial feedback versus Uniform for buses and motorcycles, shown separately in every scene/seed cell. The two classes illustrate opposing directions found in the complete class table. Connecting lines identify the paired scene and seed results.

but the same-class shuffled control changes it by a similar amount and the four-draw weighted comparison reverses the apparent speed-only advantage. Frozen-feature supervision produces mixed movement-quality responses across the reported endpoints.

Longer-timescale changes are also mixed. SV reduces longest-stop distance by 1.44%, while mean absolute acceleration on the one-second clock worsens by 4.86% and four-second speed-change distance worsens by 5.23%. Bicycle and motorcycle gains coexist with adverse bus and private-car results. The complete endpoints remain in the experiment records rather than being condensed into a favourable class score.

A separate same-pool diagnostic on the already exposed the route-preference evaluation scenes distinguishes global GAF weighting from weighting with fixed route and route–duration marginals. It uses four scene/seed cells and 10,784 candidates. Under the fixed joint marginals, speed-GAF weighting changes native speed/acceleration distances by only $-0.12\%/-0.13\%$, with improvements in one and two of the four cells respectively. Spatial and dual-view weighting worsen both macro endpoints. The preserved marginals are distributional constraints; an individual coupled random draw need not retain the same route and duration.

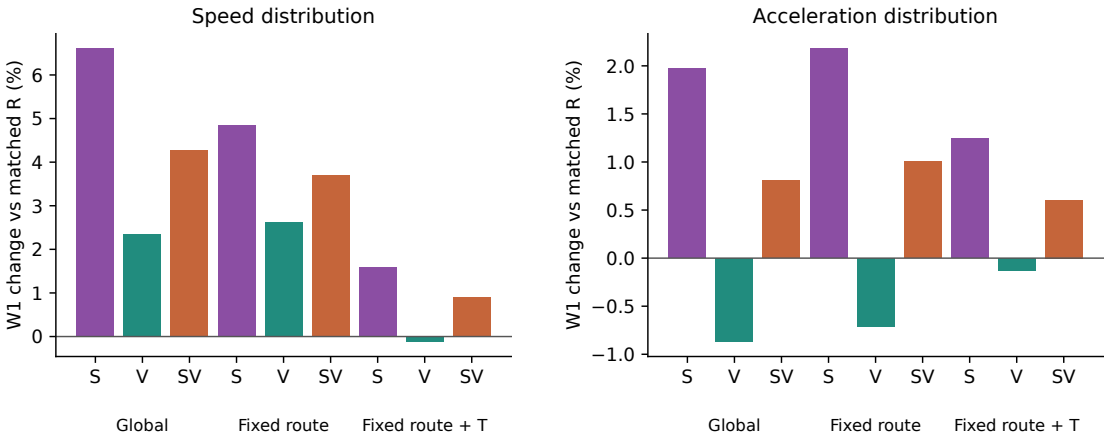

Figure A.12. Closed mechanism diagnostic: changes under global, fixed-route and fixed-route–duration weighting of identical candidate pools. The last comparison preserves the joint route/time marginal rather than asserting that each selected sample is identical. These reused scenes are development evidence; the small residual speed-view effects are not independent confirmation of guidance.

Independent augmentation nevertheless gives a useful, narrower finding. Across two raw-series learner seeds, real-only training reaches 91.25% macro F1, real plus baseline diffusion data 93.17%, real plus speed-aligned data 94.10%, and real plus RAW-aligned data 94.06%. Real plus dual-view data reaches 93.57%; synthetic-only regimes remain below real-only. Thus diffusion samples can help this downstream task, while the almost equal speed-GAF and RAW results do not demonstrate an image-specific advantage. The generator has access to the full labelled corpus, so this is not a strict low-label learning claim.

The main experiment extends these fixed-classifier controls to jointly trained generative conditions.

A.13 Implementation and Computational Performance

The prior pilot also matters. With 1,000 prior updates, sampled absolute codes reached approximately 119–178. The single declared-budget recovery used 5,000 updates and reduced the sampled pilot range substantially without refitting motion. The formal experiment uses that previously declared budget. Its largest recorded absolute code is approximately 6.50; some out-of-sample samples still exceed their variant's observed training-code maximum. Counts are 31, 35, 14, 70, 18 and 86 out of 6,272 for F-SV, J-S, J-V, J-SV, J-RAW and J-SH. All samples are retained when reporting empirical code-range exceedance. Code-range statistics complement the

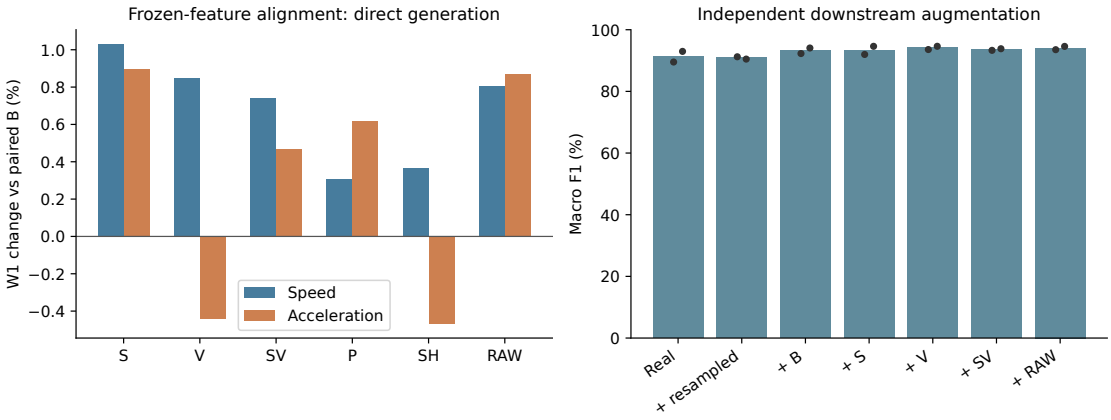

Figure A.13. Frozen-feature alignment (left) and independent augmentation (right), with both downstream seeds shown. The evaluation model reads raw time series and is independent of the GAF classifiers. Task performance and physical fidelity show different responses to feature alignment.

motion-quality evaluation.

A separate completed performance study reduces a repeated-archive evaluation unit from 104.168 to 10.017 seconds while reproducing its metrics and diagnostic JSON exactly. That I/O change is adopted. Proposed training caching gives no end-to-end speedup in the bounded benchmark and fails strict state equality, including non-identical repeats of the original implementation; it is not adopted.

Reproducibility checks link the final results to saved model configurations, generated trajectories and numerical summaries. The archive retains pilot diagnostics and implementation corrections, while presentation verification checks that figure layout changes preserve the underlying coordinate arrays, image values and numerical matrices.

A.14 Supplementary dataset and demand comparison

A.14.1 Archived trajectory exports and spatial alignment

The archived reference dataset (CNR export), *the primary archived MoST export*, contains 4,840,789 observations from 2,685 identifiers and six participant classes. A second export, *the secondary single-class export*, contains 2,189,591 observations from 12,061 identifiers but only one recorded class. Their spatial support, durations and label composition differ. The secondary table is not pooled into the labelled complete-journey experiment.

The canonical record is `time,id,x,y,type`, with seconds and local planar metres. The principal export additionally contains simulator speed and the label `class2`; its class is mapped to the common type field. Speed for physical comparisons is derived from positions and timestamps on both observed and generated data, avoiding comparison between an internal simulator state and a numerical coordinate derivative.

No original network, route file or simulation configuration accompanied the tables. A check against the public Monaco SUMO Traffic network supports alignment of the principal source [3]: all 4,841 systematically sampled principal positions lie within five metres of a lane centreline, compared with 11.2% of the 2,190-point secondary sample. This identifies a useful projected spatial frame without recovering the original run settings. The positions are not treated as longitude and latitude, and map proximity alone is not a directed-road feasibility proof.

A.14.2 Demand structure and participant recognition

A controlled comparison evaluates class-permitted random routes against activity-based demand and the archived reference dataset. Road permissions determine feasible paths, while public services and activity plans introduce class-related spatial patterns.

The comparison matched six labels and class counts, with 1,983 training and 257 validation entities per source. Original spatial encoding, ResNet18 architecture, fitting budget and final-checkpoint rule were fixed; two classifier seeds were used. Table A.21 separates an information readout from the image classifier. The readouts evaluate recognition on observed trajectories.

Table A.21. Matched source diagnosis. Values are validation macro F1 in percent; the Hilbert–GAF row averages two fixed classifier seeds.

Readout	Archived data	Random demand	Native MoST
Spatial geometry	79.12	34.85	76.95
Physical motion	88.08	80.57	90.35
Hilbert–GAF	59.88	30.15	58.69

Spatial information and GAF recognition both recover under native demand, while random demand still retains substantial motion-related class information. The source comparison identifies domain mismatch as a contributor to low spatial recognition in the initial expansion. Service routes, stops, density and activity chains change together in this comparison.

The native source is fixed to public revision `b29b2f65`, using the unchanged map, 24 bus services and 181 stops together with activities, parking and background traffic. These simulations use a separately specified configuration. In the

matched source diagnosis, 204 of 257 validation entities share source identity with the archived dataset and native validation bus lines are represented in training. That study concerns recovery of domain structure, not unseen-line validation.

Participant labels and SUMO permission classes remain separate. Source vehicle types retain their actual physical and behavioural parameters [51]. A taxi/ride-hailing label does not change a passenger-permission vehicle into a taxi-permission vehicle. Such distinctions affect route construction and prevent manufactured label differences from being presented as learned transport behaviour.

Bibliography

[1] E. Salvadori, *Unraveling Urban Mobility: Gramian Angular Fields for Trajectory Classification*. Master's thesis, Politecnico di Torino, 2024. PoliTo WebThesis 30378.

[2] G. O. Ferreira, F. Dabbene, F. Malandrino and C. Ravazzi. Unraveling Urban Mobility: A Domain Knowledge-Free Trajectory Classification Using Gramian Angular Fields. *IEEE Transactions on Intelligent Transportation Systems*, 27, 531–543, 2026. doi:10.1109/TITS.2025.3631275.

[3] L. Codecá and J. Härri. Monaco SUMO Traffic (MoST) Scenario: A 3D Mobility Scenario for Cooperative ITS. *SUMO User Conference*, 2018. Scenario repository.

[4] J. Ho, A. Jain and P. Abbeel. Denoising Diffusion Probabilistic Models. *Advances in Neural Information Processing Systems*, 33, 2020. arXiv:2006.11239.

[5] J. Song, C. Meng and S. Ermon. Denoising Diffusion Implicit Models. *International Conference on Learning Representations*, 2021. arXiv:2010.02502.

[6] Y. Zhu, Y. Ye, S. Zhang, X. Zhao and J. Yu. DiffTraj: Generating GPS Trajectory with Diffusion Probabilistic Model. *Advances in Neural Information Processing Systems*, 36, 2023. Conference proceedings.

[7] T. Wei, Y. Lin, S. Guo, Y. Lin, Y. Huang, C. Xiang, Y. Bai and H. Wan. Diff-RNTraj: A Structure-aware Diffusion Model for Road Network-constrained Trajectory Generation. 2024. arXiv:2402.07369.

[8] Z. Lv, Z. Fan, Z. Zhang and X. Song. Two-Stage Fine-Grained Trajectory Generation Constrained by Road Networks. *IJCAI*, 2915–2923, 2026. doi:10.24963/ijcai.2026/324.

[9] E. Pronovost, M. R. Ganesina, N. Hendy, Z. Wang, A. Morales, K. Wang and N. Roy. Scenario Diffusion: Controllable Driving Scenario Generation With Diffusion. *Advances in Neural Information Processing Systems*, 36, 2023. arXiv:2311.02738.

[10] Z. Wang and T. Oates. Encoding Time Series as Images for Visual Inspection and Classification Using Tiled Convolutional Neural Networks. *AAAI workshop*, 2015. Workshop paper.

[11] K. He, X. Zhang, S. Ren and J. Sun. Deep Residual Learning for Image Recognition. *IEEE Conference on Computer Vision and Pattern Recognition*, 2016.

CVPR proceedings.

[12] Historical trajectory project. *Deep Learning Based Generative Models for Traffic Trajectory Synthesis*, revision `4773ad65`. Encoding source.

[13] Thesis experiment repository. *Road-Conditioned Trajectory Diffusion: Code and Experiments*, results snapshot `6031e3f`. Archived repository snapshot and frozen evaluation report.

[14] L. Wu, B. Trippe, C. Naesseth, D. Blei and J. Cunningham. Practical and Asymptotically Exact Conditional Sampling in Diffusion Models. *Advances in Neural Information Processing Systems*, 2023. arXiv:2306.17775.

[15] Choi, Seongjin, Kim, Jiwon, Yeo, Hwasoo. TrajGAIL: Generating Urban Vehicle Trajectories using Generative Adversarial Imitation Learning. 2020. arXiv:2007.14189.

[16] Gao, Jiyang et al.. VectorNet: Encoding HD Maps and Agent Dynamics from Vectorized Representation. 2020. arXiv:2005.04259.

[17] Liang, Ming et al.. Learning Lane Graph Representations for Motion Forecasting. 2020. arXiv:2007.13732.

[18] Salzmann, Tim, Ivanovic, Boris, Chakravarty, Punarjay, Pavone, Marco. Trajectron++: Dynamically-Feasible Trajectory Forecasting With Heterogeneous Data. 2020. arXiv:2001.03093.

[19] Ngiam, Jiquan et al.. Scene Transformer: A unified architecture for predicting multiple agent trajectories. 2021. arXiv:2106.08417.

[20] Gupta, Agrim, Johnson, Justin, Fei-Fei, Li, Savarese, Silvio, Alahi, Alexandre. Social GAN: Socially Acceptable Trajectories with Generative Adversarial Networks. 2018. arXiv:1803.10892.

[21] Shi, Shaoshuai, Jiang, Li, Dai, Dengxin, Schiele, Bernt. Motion Transformer with Global Intention Localization and Local Movement Refinement. 2022. arXiv:2209.13508.

[22] Janner, Michael, Du, Yilun, Tenenbaum, Joshua B., Levine, Sergey. Planning with Diffusion for Flexible Behavior Synthesis. 2022. arXiv:2205.09991.

[23] Chi, Cheng et al.. Diffusion Policy: Visuomotor Policy Learning via Action Diffusion. 2023. arXiv:2303.04137.

[24] Jiang, Chiyu Max, Cornman, Andre, Park, Cheolho, Sapp, Ben, Zhou, Yin, Anguelov, Dragomir. MotionDiffuser: Controllable Multi-Agent Motion Prediction using Diffusion. 2023. arXiv:2306.03083.

[25] Dhariwal, Prafulla, Nichol, Alex. Diffusion Models Beat GANs on Image Synthesis. 2021. arXiv:2105.05233.

[26] Ho, Jonathan, Salimans, Tim. Classifier-Free Diffusion Guidance. 2022. arXiv:2207.12598.

[27] Bansal, Arpit et al.. Universal Guidance for Diffusion Models. 2023. arXiv:2302.07121.

[28] Kawar, Bahjat, Elad, Michael, Ermon, Stefano, Song, Jiaming. Denoising Diffusion Restoration Models. 2022. arXiv:2201.11793.

[29] Chung, Hyungjin, Kim, Jeongsol, Mccann, Michael T., Klasky, Marc L., Ye, Jong Chul. Diffusion Posterior Sampling for General Noisy Inverse Problems. 2022. arXiv:2209.14687.

[30] Singhal, Raghav et al.. A General Framework for Inference-time Scaling and Steering of Diffusion Models. 2025. arXiv:2501.06848.

[31] Nichol, Alex, Dhariwal, Prafulla. Improved Denoising Diffusion Probabilistic Models. 2021. arXiv:2102.09672.

[32] Karras, Tero, Aittala, Miika, Aila, Timo, Laine, Samuli. Elucidating the Design Space of Diffusion-Based Generative Models. 2022. arXiv:2206.00364.

[33] Lu, Cheng, Zhou, Yuhao, Bao, Fan, Chen, Jianfei, Li, Chongxuan, Zhu, Jun. DPM-Solver: A Fast ODE Solver for Diffusion Probabilistic Model Sampling in Around 10 Steps. 2022. arXiv:2206.00927.

[34] Rasul, Kashif, Seward, Calvin, Schuster, Ingmar, Vollgraf, Roland. Autoregressive Denoising Diffusion Models for Multivariate Probabilistic Time Series Forecasting. 2021. arXiv:2101.12072.

[35] Tashiro, Yusuke, Song, Jiaming, Song, Yang, Ermon, Stefano. CSDI: Conditional Score-based Diffusion Models for Probabilistic Time Series Imputation. 2021. arXiv:2107.03502.

[36] Kong, Zhifeng, Ping, Wei, Huang, Jiaji, Zhao, Kexin, Catanzaro, Bryan. DiffWave: A Versatile Diffusion Model for Audio Synthesis. 2020. arXiv:2009.09761.

[37] Bai, Shaojie, Kolter, J. Zico, Koltun, Vladlen. An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling. 2018. arXiv:1803.01271.

[38] Cho, Kyunghyun et al.. Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation. 2014. arXiv:1406.1078.

[39] Fawaz, Hassan Ismail et al.. InceptionTime: Finding AlexNet for Time Series Classification. 2019. arXiv:1909.04939.

[40] Dempster, Angus, Petitjean, François, Webb, Geoffrey I.. ROCKET: Exceptionally fast and accurate time series classification using random convolutional kernels. 2019. arXiv:1910.13051.

[41] Dempster, Angus, Schmidt, Daniel F., Webb, Geoffrey I.. MINIROCKET: A Very Fast (Almost) Deterministic Transform for Time Series Classification. 2020. arXiv:2012.08791.

[42] Franceschi, Jean-Yves, Dieuleveut, Aymeric, Jaggi, Martin. Unsupervised Scalable Representation Learning for Multivariate Time Series. 2019. arXiv:1901.10738.

[43] Lopez-Paz, David, Oquab, Maxime. Revisiting Classifier Two-Sample Tests. 2016. arXiv:1610.06545.

[44] Sajjadi, Mehdi S. M., Bachem, Olivier, Lucic, Mario, Bousquet, Olivier, Gelly, Sylvain. Assessing Generative Models via Precision and Recall. 2018. arXiv:1806.00035.

[45] Kynkäänniemi, Tuomas, Karras, Tero, Laine, Samuli, Lehtinen, Jaakko, Aila, Timo. Improved Precision and Recall Metric for Assessing Generative Models. 2019. arXiv:1904.06991.

[46] Esteban, Cristóbal, Hyland, Stephanie L., Rätsch, Gunnar. Real-valued (Medical) Time Series Generation with Recurrent Conditional GANs. 2017. arXiv:1706.02633.

[47] Eclipse SUMO project. FCD Output and Randomness documentation. Accessed 5 November 2026. Floating-car output; random seeds.

[48] C. Guo, G. Pleiss, Y. Sun and K. Q. Weinberger. On Calibration of Modern Neural Networks. *ICML*, PMLR 70, 1321–1330, 2017. Conference proceedings.

[49] S. Kornblith, M. Norouzi, H. Lee and G. Hinton, Similarity of Neural Network Representations Revisited, *Proceedings of ICML*, PMLR 97, 2019, pp. 3519–3529. Proceedings page.

[50] S. Yu, S. Kwak, H. Jang, J. Jeong, J. Huang, J. Shin and S. Xie. Representation Alignment for Generation: Training Diffusion Transformers Is Easier Than You Think. *ICLR*, 2025. arXiv:2410.06940.

[51] Eclipse SUMO project. Vehicle Type Parameter Defaults. Accessed 6 November 2026. Official documentation.

[52] T. Gneiting and A. E. Raftery. Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association*, 102(477), 359–378, 2007. doi:10.1198/016214506000001437.

[53] K. Preechakul, N. Chatthee, S. Wizadwongsa and S. Suwajanakorn. Diffusion Autoencoders: Toward a Meaningful and Decodable Representation. *Proceedings of CVPR*, 2022. arXiv:2111.15640.

[54] M. Zimmermann and E. Frejinger. A tutorial on recursive models for analyzing and predicting path choice behavior. *EURO Journal on Transportation and Logistics*, 9(2), 100004, 2020. doi:10.1016/j.ejtl.2020.100004.